SA-AI (Spalart–Allmaras with Autogenous Inception): Technical Summary

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Technical summary of the SA-AI one-equation transition model: the complete formulation with every constant and its determination, the protocol needed to run it, and the results. Derivations, discussion, and the reference list are in the full paper (`sa-ai.tex`); references below are given by author/report name only. Every number is copied from the paper or its committed data tables and is reproducible from `paper/repro/` in the repository: the analytic figures from `repro/analytic/` alone; the coupled-RANS results additionally require the Flow360 solver binary (`repro/driver/` re-runs the cases, `repro/cfd/` regenerates the figures); the independent cross-solver replication runs from `openfoam/`. The appendix collects the main paper's appendix figure sets (meshes, wall-anchored contour sheets, Daedalus surface and section sheets). The 6:1 prolate spheroid (crossflow) study is under revision and is omitted here.

1 Model

1.1 Transport equation and blended sources

The model is the Spalart–Allmaras (SA) transport equation for the working variable $\tilde{\nu}$, with the laminar sub-$O(1)$ range of $\chi = \tilde{\nu}/\nu$ repurposed as an e^N-style envelope amplification: $\ln(\tilde{\nu}/\tilde{\nu}_\infty)$ accumulates as an amplification factor, and the working variable gives rise to its own transition at $\chi \sim 1$. The single equation is

$$\frac{D\tilde{\nu}}{Dt} = P - D + \frac{1}{\sigma}\nabla\cdot\left[(c_{\nu,\mathrm{ai}}\,\nu + \tilde{\nu})\,\nabla\tilde{\nu}\right] + \frac{c_{b2}}{\sigma}\,|\nabla\tilde{\nu}|^2, \tag{1}$$

$$P = \max\left[\,(1-\sigma_P)\,a\,S\,\omega\,\tilde{\nu},\;\; \sigma_P\,P_{\mathrm{SA}}\,\right], \qquad D = \sigma_D\,D_{\mathrm{SA}}, \tag{2}$$

identical in form to standard SA except in four places: the blended production P (its amplification branch built on the rate coordinate $\hat{\Omega}\hat{I}$ below), the tied destruction D, the factor $c_{\nu,\mathrm{ai}}$ on the molecular diffusion, and the lifted-layer bypass weight b in the eddy-viscosity assembly. $\sigma_P \equiv 1$ (and with it $\sigma_D = 1$), $c_{\nu,\mathrm{ai}} = 1$, and $b \equiv 0$ recover baseline SA exactly. The baseline SA production, destruction, and closure functions are carried over unchanged:

$$P_{\mathrm{SA}} = c_{b1}\tilde{S}\,\tilde{\nu}, \qquad D_{\mathrm{SA}} = c_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2, \qquad \tilde{S} = \omega + \frac{\tilde{\nu}}{\kappa^2 d^2}\,f_{v2}, \qquad f_{v2} = 1 - \frac{\chi}{1+\chi f_{v1}},$$

$$f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}, \qquad f_w = g\left(\frac{1+c_{w3}^6}{g^6 + c_{w3}^6}\right)^{1/6}, \qquad g = r + c_{w2}(r^6 - r), \qquad r = \min\left(\frac{\tilde{\nu}}{\tilde{S}\,\kappa^2 d^2},\, 10\right), \tag{3}$$

with ω the vorticity magnitude and d the wall distance. The non-negative $\tilde{S}$ modification and the negative-$\tilde{\nu}$ safeguards of Allmaras et al. (2012) are retained, with one consistency requirement: the negative-branch diffusion factor f_n is formed with the scaled coefficient $c_{\nu,\mathrm{ai}}\,\nu$, which keeps the total diffusivity positive, $c_{\nu,\mathrm{ai}}\nu + f_n\tilde{\nu} > 0$. The handover weights are

$$\sigma_P = \sigma_t(\chi;\tau), \qquad \sigma_t(\chi;\tau) = \max\left[1 - e^{-(\chi-1)/\tau},\, 0\right], \qquad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}}\,(1-\sigma_P). \tag{4}$$

1.2 Local indicators

In a thin quasi-two-dimensional layer the profile is captured to second order about any wall-normal point by three velocity scales formed from the wall distance d: the velocity u, the shear $d\,u'$, and the curvature $\frac{1}{2}d^2u''$. Normalized by their common magnitude $R = \sqrt{u^2 + (du')^2 + (\frac{1}{2}d^2u'')^2}$, they place the local state on the unit sphere of directions

$$(X, Y, Z) = \big(u,\ d\,u',\ \tfrac{1}{2}d^2u''\big)/R, \tag{5}$$

a direction defined only up to sign (a layer and its mirror are equally unstable): a point of the real projective plane $\mathbb{RP}^2$, on which only even products of odd sphere functions are single-valued. Two odd functions carry the model:

$$\hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}} \quad \text{(vorticity fraction)}, \qquad \hat{I} = Y - X - Z \quad \text{(accumulated inflection)}. \tag{6}$$

A near-wall parabola $u = By + Cy^2$ satisfies $\hat{I} = 0$ identically, so the parabola great circle $\hat{I} = 0$ is the neutral locus; the unnormalized combination integrates the third derivative,

$$I(d) \equiv R\,\hat{I} = d\,u' - u - \tfrac{1}{2}d^2u'' = -\int_0^d \tfrac{1}{2}\,s^2\,u'''(s)\,\mathrm{d}s, \tag{7}$$

the accumulated curvature of the vorticity profile—the inflectional content Rayleigh's theorem ties to inviscid instability; $\hat{I} > 0$ marks the inflectional side. Every profile leaves the wall with $\hat{I}$ third-order small ($u'''_w = 0$ on a steady wall), so the kernel produces nothing at the wall by construction. The fourth local number, the vorticity Reynolds number $Re_\Omega = d^2\omega/\nu$, does not enter the rate (the envelope rate is Reynolds-independent beyond its viscous cutoff) and is reserved for the onset gate. Fig. 1 shows the Falkner–Skan family, its linear-stability amplifying bands, and the contours of $\hat{\Omega}\hat{I}$ on the indicator sphere.

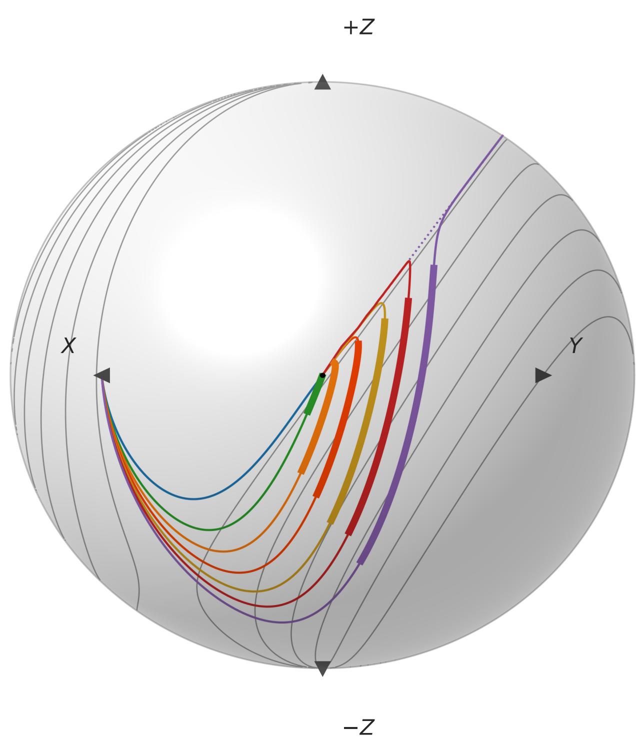


Figure 1: Indicator sphere.

In the solver the triple is assembled from four local vector-calculus objects: the velocity $\mathbf{u}$ (relative to the nearest wall point, which makes the indicators Galilean invariant), the vorticity $\boldsymbol{\omega}$, the wall-distance vector $\mathbf{d} = d\,\nabla d$, and the curvature vector $\boldsymbol{\ell} = \frac{1}{2}\langle \mathbf{d}, \mathbf{d}\rangle\, \nabla^2\mathbf{u}$, with $\nabla^2\mathbf{u} = -\nabla\times\boldsymbol{\omega}$ supplying the second derivative. The division-free Gram form is

$$(X,\, Y,\, Z) \;=\; \Big(\langle \mathbf{u},\mathbf{u}\rangle,\; \sqrt{\langle \mathbf{u},\mathbf{u}\rangle\,\langle \mathbf{d},\mathbf{d}\rangle\,\langle \boldsymbol{\omega},\boldsymbol{\omega}\rangle},\; \langle \boldsymbol{\ell},\mathbf{u}\rangle\Big), \qquad \hat\Omega = \frac{Y}{\sqrt{X^2+Y^2}}, \quad \hat I = \frac{Y-X-Z}{\sqrt{X^2+Y^2+Z^2}}, \tag{8}$$

homogeneous of degree zero, with no unit vector, no division by a vanishing magnitude, and the only radical over a non-negative product; the solver's equivalent ratio realization ($|\mathbf{u}|$, $d|\boldsymbol{\omega}|$, $\frac{1}{2}d^2(\nabla^2\mathbf{u})\cdot\hat{\mathbf{u}}$) coincides with it wherever $\mathbf{u}\neq 0$. Because the rate depends on the frozen mean flow and never on $\tilde\nu$, the production's $\tilde\nu$-derivative is analytic and is treated implicitly with the same clipping as the baseline SA production.

1.3 Amplification rate and onset threshold

The complete amplification source is

$$P_{\mathrm{AI}} = a\,S\,\omega\,\tilde\nu, \qquad a = a_{\max}\ \mathrm{clip}\big\langle\, \hat\Omega\,\hat I\,\big\rangle_0^1, \qquad S = S\big(Re_\Omega/Re_\Omega^{\mathrm{c}}(\hat\Omega\hat I)\big), \qquad S(\zeta) = \tfrac{1}{2}\big[1+\tanh\big((\zeta-1)/w\big)\big], \tag{9}$$

$$Re_\Omega^{\mathrm{c}}(\hat\Omega\hat I) = k\ \mathrm{softmin}_2\Big(C,\ A + B\,\langle\hat\Omega\hat I\rangle_+^{-2}\Big), \qquad \mathrm{softmin}_2(x_1,x_2) = \frac{x_1x_2}{\sqrt{x_1^2+x_2^2}}, \qquad \langle x\rangle_+ = \max(x,0). \tag{10}$$

Two functions of the one even coordinate $\hat\Omega\hat I$—the rate and the onset threshold—carry the whole model. A favorable gradient keeps the profile's curvature single-signed, so $\hat\Omega\hat I\to 0$ and both shut off geometrically, with no pressure-gradient parameter anywhere.

The ceiling $a_{\max} = 0.19$ is an eigenvalue, not a fit: the hyperbolic-tangent free shear layer's most-amplified temporal Kelvin–Helmholtz mode grows at $\omega_{i,\max}/\omega_{\mathrm{peak}} = 0.1897$ (Michalke), reached where $\hat\Omega\hat I \geq 1$ at the vorticity pole, the free-shear limit of the profile continuum. The onset threshold's shape $(C, A, B) = (2600,\ 175,\ 2)$ is the common outer envelope that the attached Falkner–Skan profiles, each evaluated at its Drela–Giles critical Reynolds number, graze in the $(\hat\Omega\hat I, Re_\Omega)$ plane (graze ratios 0.93–1.14, root-mean-square $\sim 6\%$, from incipient separation through $\beta = +0.15$; Fig. 2); the shape constants are never refit. The ramp width $w = 0.35$ is not fitted either: the marched $N = 1$ stations move by under 1% when w is halved and under 2.5% when doubled. The single scale $k = 0.712$ compensates the retained laminar diffusion: it is chosen so that the model's own disturbance transport, marched on the Blasius layer at the committed $c_{\nu,\mathrm{ai}} = \frac{1}{6}$, crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$. Everything after that anchor is prediction: the marched Blasius $N = 9$ crossing lands at $Re_\theta = 1181$, 7% past Drela's 1108; across the attached family the $N = 1$ stations land on the Drela–Giles stations to 6% root-mean-square (adverse wedges within $\pm 4\%$, the favorable pair 11–12% early). The strongly favorable branch degrades: by $\beta = +0.35$ the rate is $\approx 0.1\times$ Drela and the onset $\approx 3\times$ its critical Re_θ; by $\beta \approx 0.45$ transition has retreated beyond any station of practical relevance, and stronger favorable wedges are suppressed outright (Figs. 3–4).

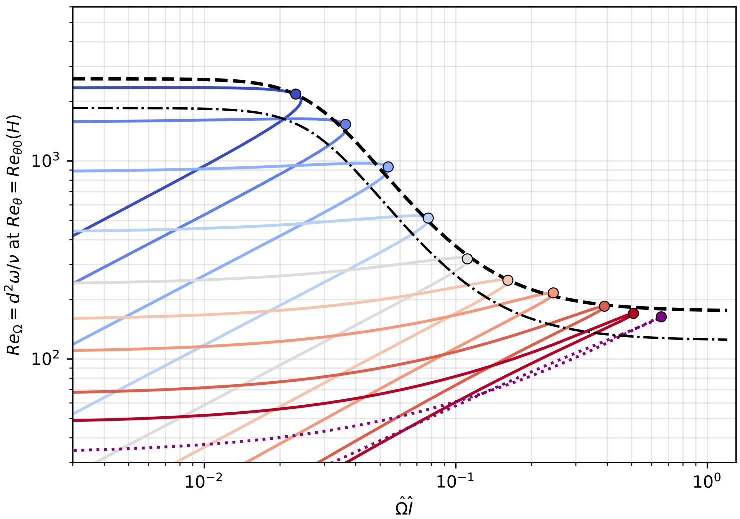


Figure 2: Onset-threshold graze.

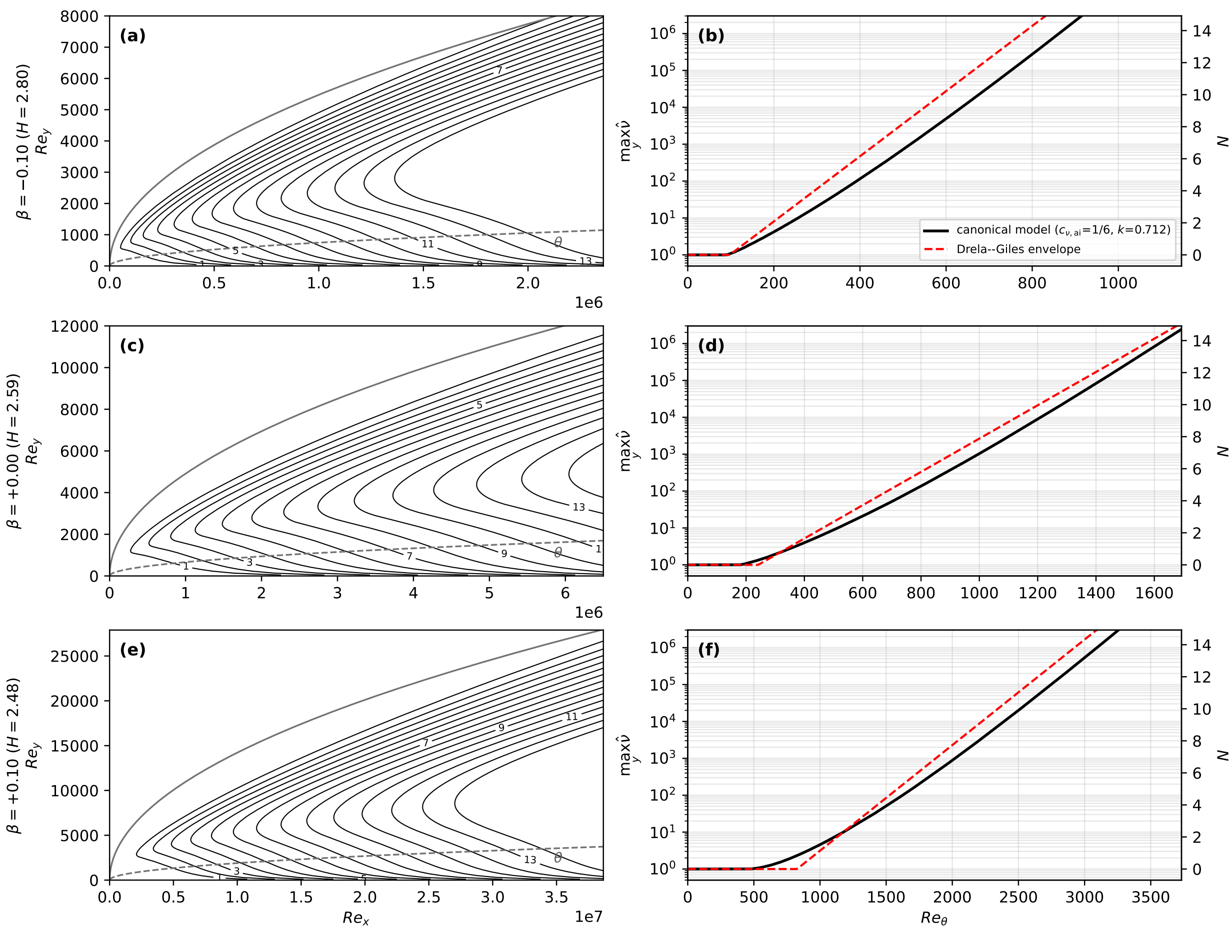


Figure 3: Disturbance transport, three wedges.

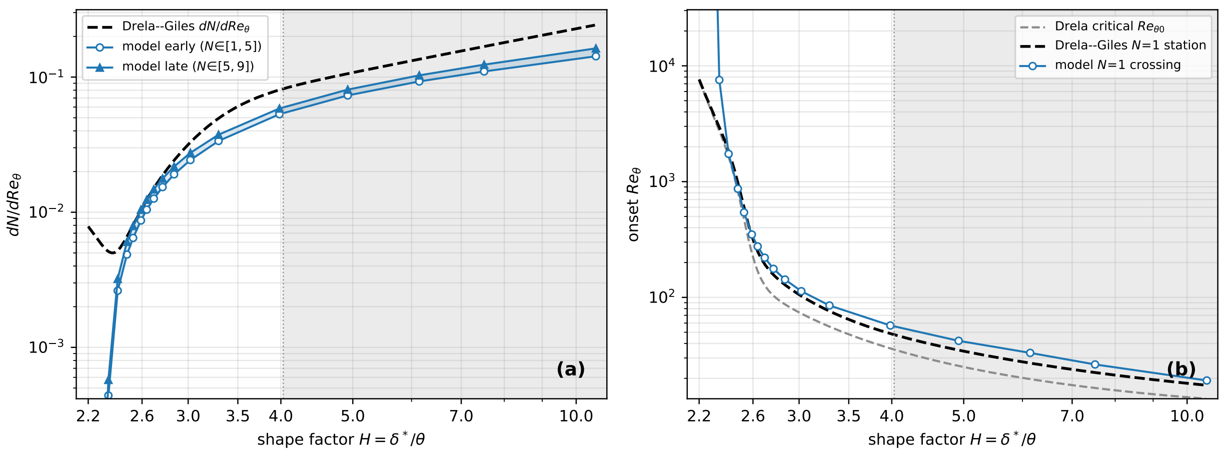


Figure 4: Calibrated model vs Drela–Giles.

1.4 Reduced laminar diffusion

The molecular coefficient in the diffusion term of Eq. (1) only is scaled by $c_{\nu,\mathrm{ai}} < 1$, leaving the mean-flow viscosity, the SA production and destruction, and $\chi = \tilde{\nu}/\nu$ (formed with the true ν) untouched. At full molecular diffusion the drain overwhelms the gentle amplification: the frozen-profile growth eigenvalue on Blasius at $Re_\theta = 400$ delivers only 0.49 of the Drela–Giles rate, and the marched inception fails outright (the Blasius $N = 1$ crossing sits at $Re_\theta = 392$ even with the threshold removed entirely). Down the ladder $c_{\nu,\mathrm{ai}} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}$ that station's eigenvalue ratio climbs 0.49, 0.75, 0.87, 1.02, 1.13: the value

$$c_{\nu,\mathrm{ai}} = \tfrac{1}{6}$$

puts the just-past-critical stations on the correlation (the separation branch holds 0.82–0.91). Pressing further costs mesh: the equilibrium disturbance-band thickness scales as $\sqrt{c_{\nu,\mathrm{ai}}}$, so every halving thins the band the grid must resolve by $\sqrt{2}$.

1.5 Handover blend and destruction tie

The max in Eq. (2) hands production from amplification to SA at the crossover: for $\chi \le 1$ ($\sigma_P = 0$) it returns P_{AI}, and once $\sigma_P P_{\mathrm{SA}}$ overtakes it just past $\chi = 1$ the source reverts to standard SA with no laminar residue. The production width $\tau = 4$ is a numerics bracket: too wide throttles SA production past $\chi = c_{v1}$, too narrow compresses the transition zone into a front steeper than the streamwise mesh can resolve; at $\tau = 4$ the handover is 78% complete where f_{v1} is half active ($\chi = c_{v1}$) and 97% complete where f_{v1} reaches 90% ($\chi \approx 14.8$). The destruction is not gated off but tied: σ_D tracks σ_P with slope $c_{b1}/(\kappa^2 c_{w1}) \approx 0.249$ from the floor $(1 + c_{b2})/(\sigma c_{w1}) \approx 0.751$, the two summing to one by the definition of c_{w1}. The tie is derived from the constant-stress wall layer, on which SA's solution is the linear profile $\tilde{\nu} = \kappa u_\tau y$ with height-independent terms $P_{\mathrm{SA}} = c_{b1} u_\tau^2$, $D_{\mathrm{SA}} = c_{w1}\kappa^2 u_\tau^2$, and self-diffusion $(1 + c_{b2})\kappa^2 u_\tau^2/\sigma$; requiring the gated balance to hold pointwise gives Eq. (4), under which the linear profile is an exact solution of the gated equation at every height and for any τ—continuously and discretely (verified to below 10^{-10} in profile and log-law intercept with two unrelated discretizations). The linear profile has zero curvature, so the reduced molecular diffusion does not act on the inner layer at all: the inner-layer eddy viscosity, and with it the log law, is standard SA's exactly. In the laminar range the destruction floor is inert at the disturbance's quadratic scale: its ratio to the amplification production stays below 10^{-3} up to the $\chi = 1$ crossing for every wedge and seed tested, and at $\chi = c_{v1}$ is at most 1.4% of the σ_P-weighted SA production it is tied against; marching the calibration instrument with the floor included moves the anchor and prediction stations by less than 1%.

1.6 Eddy-viscosity assembly in lifted transitional layers

The near-wall damping f_{v1} enforces the attached-wall-layer identity $\chi = \kappa y^+$; in a lifted transitional shear layer (the free shear layer over a laminar separation bubble in mid-handover) that identity is void and the damping is an unearned sink on the young turbulence that must close the bubble. Whether a point is in an attached wall layer is decidable from the state itself, with no new calibrated constants: writing $y^+ := \chi/\kappa$, the reconstruction

$$q = \frac{|\mathbf{u}|\,d/\nu}{y^+\,U^+(y^+)}, \qquad U^+(y^+) = \tfrac{1}{\kappa}\ln(1+\kappa y^+) + 7.8\big(1 - e^{-y^+/11} - \tfrac{y^+}{11}e^{-y^+/3}\big) \ \text{(Reichardt)}, \tag{11}$$

equals unity identically in an attached equilibrium layer and is many times larger in a lifted layer. The assembly is blended,

$$\mu_t = \big[(1-b)\,f_{v1}(\chi) + b\big]\,\rho\tilde{\nu}, \qquad b = s(\chi)\,G(q), \qquad s = \operatorname{clip}(\chi - 1, 0, 1), \qquad G = \operatorname{clip}((q-2)/2, 0, 1), \tag{12}$$

inert for $\chi < 1$ (negligible μ_t either way), for $\chi \gtrsim 30$ ($f_{v1} \to 1$), and in the attached inner layer ($q = 1$, G shut). The s ramp is deliberately faster than the σ_t handover: σ_t encodes turbulence maturity and is already applied to the production, while f_{v1} encodes wall proximity. Controlled on/off batteries bound the net effect on attached-layer outer edges below 1.5 drag counts (flat plate and NLF at their cold-budget protocol); every result below is from the bypass-active model.

1.7 Freestream seed

A higher freestream turbulence intensity is a larger seed χ_∞ and thus fewer e-folds to the handover: a case seeded with χ_∞ reaches the handover where its envelope crosses $N_{\text{crit}} = \ln(c_{v1}/\chi_\infty)$. The map is Mack's e^N critical-N-factor correlation, adopted unchanged:

$$N_{\text{crit}}(Tu) = -8.43 - 2.4\,\ln(Tu_{\text{frac}}), \qquad \chi_\infty = c_{v1}\,e^{-N_{\text{crit}}}, \tag{13}$$

with Tu_{frac} the rms turbulence intensity as a fraction. The correspondence is not exact: e^N methods declare transition at a sharp N_{crit}, while the model's threshold is smeared over the $\chi = 1 \to O(c_{v1})$ handover—several e-folds of the envelope. Sensitivity to the seed is that of any e^N method: one e-fold in χ_∞ is one unit of N_{crit}, which on the Blasius envelope ($dN/dRe_\theta \approx 10^{-2}$) shifts onset by $\Delta Re_\theta \approx 100$. Strongly bypass-dominated conditions ($Tu \gtrsim 1\%$) are out of scope: at $Tu = 1\%$ the map leaves only $N_{\text{crit}} \approx 2.6$, and by $Tu \approx 3\%$ the seed exceeds c_{v1} outright.

1.8 Constants

Baseline SA, unchanged:

$$c_{b1} = 0.1355, \quad \sigma = \tfrac{2}{3}, \quad c_{b2} = 0.622, \quad \kappa = 0.41, \quad c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma}, \quad c_{w2} = 0.3, \quad c_{w3} = 2, \quad c_{v1} = 7.1.$$

SA-AI:

constant	value	determination
$a_{\max}$	0.19	tanh-layer Kelvin–Helmholtz temporal eigenvalue, $\omega_{i,\max}/\omega_{\text{peak}} = 0.1897$
(C, A, B)	(2600, 175, 2)	graze envelope of the attached Falkner–Skan family at its critical Re_θ (rms $\sim 6\%$)
k	0.712	Blasius march crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$ at $c_{\nu,\text{ai}} = \frac{1}{6}$
w	0.35	onset ramp width; $N = 1$ stations move $< 1\%$ halved, $< 2.5\%$ doubled
$c_{\nu,\text{ai}}$	1/6	frozen-profile growth eigenvalue reaches the Drela–Giles rate just past critical (1.02 at $Re_\theta = 400$)
τ	4	numerics bracket: handover 78% complete at $\chi = c_{v1}$, 97% at $\chi \approx 14.8$
σ_D tie	Eq. (4)	derived: constant-stress wall-layer balance; linear wall solution exact
bypass (s, G)	Eq. (12)	parameter-free law-of-the-wall discriminator; G ramps over $q \in (2, 4)$
$(A, B)_{\text{Mack}}$	$(-8.43, 2.4)$	Mack 1977 e^N correlation, wholesale

1.9 Running the model: branch selection and seed handling

Every laminar solution may coexist with a second, spurious fixed point of the discretized equations: a turbulent wedge anchored at the leading-edge attachment point, sustained by *standard* SA production. On the frozen Hiemenz stagnation field with the freestream seed exactly zero, standard SA sustains the wedge above a critical nose Reynolds number bisected to $Re_r = 4.66$–4.74×10^5 (Fig. 5); physically the stagnation zone relaminarizes, so the branch is spurious, and any transition model that leaves SA's fully turbulent behavior unmodified is bistable at high Reynolds number. An implementer needs a branch-selection protocol; the printed transport equation alone does not determine which fixed point an initialization finds.

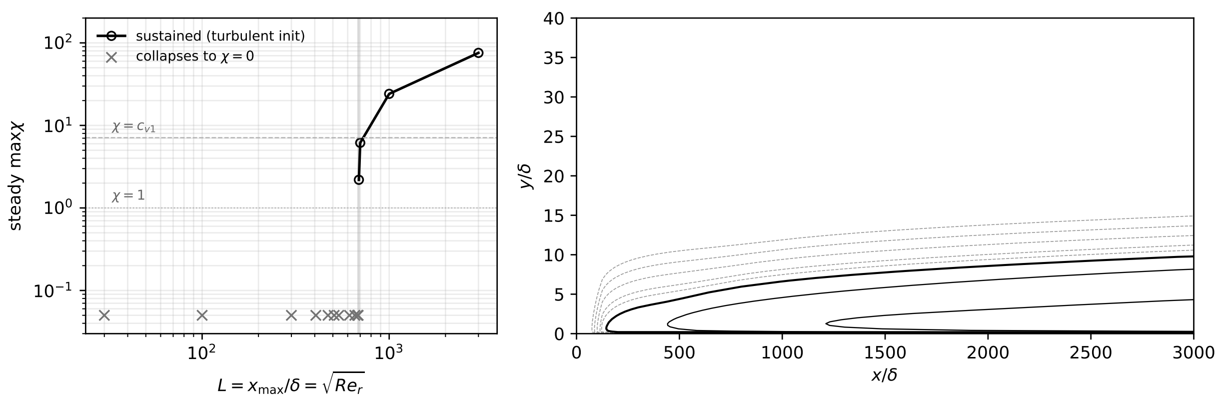


Figure 5: Attachment-anchored turbulent branch.

The protocol used for every result in this paper: the $\tilde{\nu}$-equation residual is scaled by $f^{\,1-g}$, where g rises smoothly from 0 (laminar) to 1 (turbulent) with χ and $f = 10^{-2}$ (the solver uses $g = 1 - e^{-(\chi - c_{v1})/4}$ above $\chi = c_{v1}$, zero below; the gate's exact shape is immaterial because the scaling preserves fixed points). The scaling multiplies the residual, so every converged solution is a fixed point of the unmodified equations; it lets the pressure field and the slow laminar amplification equilibrate before the fast turbulent branch can act on the startup transient—a branch selector, not an accelerator. Because the freestream boundary condition flows through the same scaling, the freestream viscosity ratio supplied to the solver is $\chi_\infty f$; the seeds quoted throughout are the physical χ_∞, recovered at the boundary-layer edge. (In the reference implementation the scaling is `AI_LAMINAR_SLOWDOWN` $= f$, and the case files carry the pre-multiplied boundary seed $\chi_\infty f$; running a committed case at $f = 1$ without rescaling the seed mis-seeds it by $1/f$.) Equivalent protocols for solvers without residual scaling: advance the turbulence equation at a reduced pseudo-time rate (CFL) relative to the mean flow during initial convergence, or—the route the OpenFOAM replication takes—converge the flow, reset $\tilde{\nu}$ to the seed, and continue to convergence, at the cost of one extra partial solve.

2 Results

Coupled RANS: Flow360, unstructured node-centered second-order finite volume, Roe flux, $M = 0.1$; the same constants and the convergence protocol above everywhere. Airfoil cases: $\chi_\infty = c_{v1}e^{-9} \approx 8.76 \times 10^{-4}$ (the $N_{\text{crit}} = 9$ anchor; via Mack's map, $Tu \approx 0.07\%$); two independently generated mesh families—all-triangular unstructured cavity meshes, constrained-Delaunay through the boundary layer, and all-quadrilateral structured O-grids—at three refinement levels, 1.5×10^4 to 2.6×10^5 nodes (meshes in the appendix). Transition locations are the solver's near-wall $\chi = 1$ crossing (the $\chi = c_{v1}$ crossing sits up to $0.05\,c$ aft where the front grows slowly); bubble stations are signed-C_f zero crossings. The cross-solver replication is a cell-centered, pressure-based, incompressible OpenFOAM implementation of the model on the same flat-plate grid specification and the same structured airfoil grids (the port omits the f_{v1} bypass); thirty airfoil cases plus the flat-plate sweep.

2.1 Flat plate, $Tu = 0.04$–0.60%

Structured 320×80 grid, $Re_x \leq 6 \times 10^6$, $y^+ \lesssim 0.3$; five seeds from Mack's map ($\chi_\infty \approx 2.3 \times 10^{-4}$, 1.2×10^{-3}, 6.3×10^{-3}, 2.9×10^{-2}, 1.5×10^{-1}). Figure 6: the Flow360 sweep—at the $\chi = c_{v1}$ crossing the onset brackets the Abu-Ghannam & Shaw correlation within 10% across the range. Figure 7: the OpenFOAM replication—$\chi = 1$ crossings within 7% in Re_θ, consistently slightly early. Table 1: both solvers' crossings.

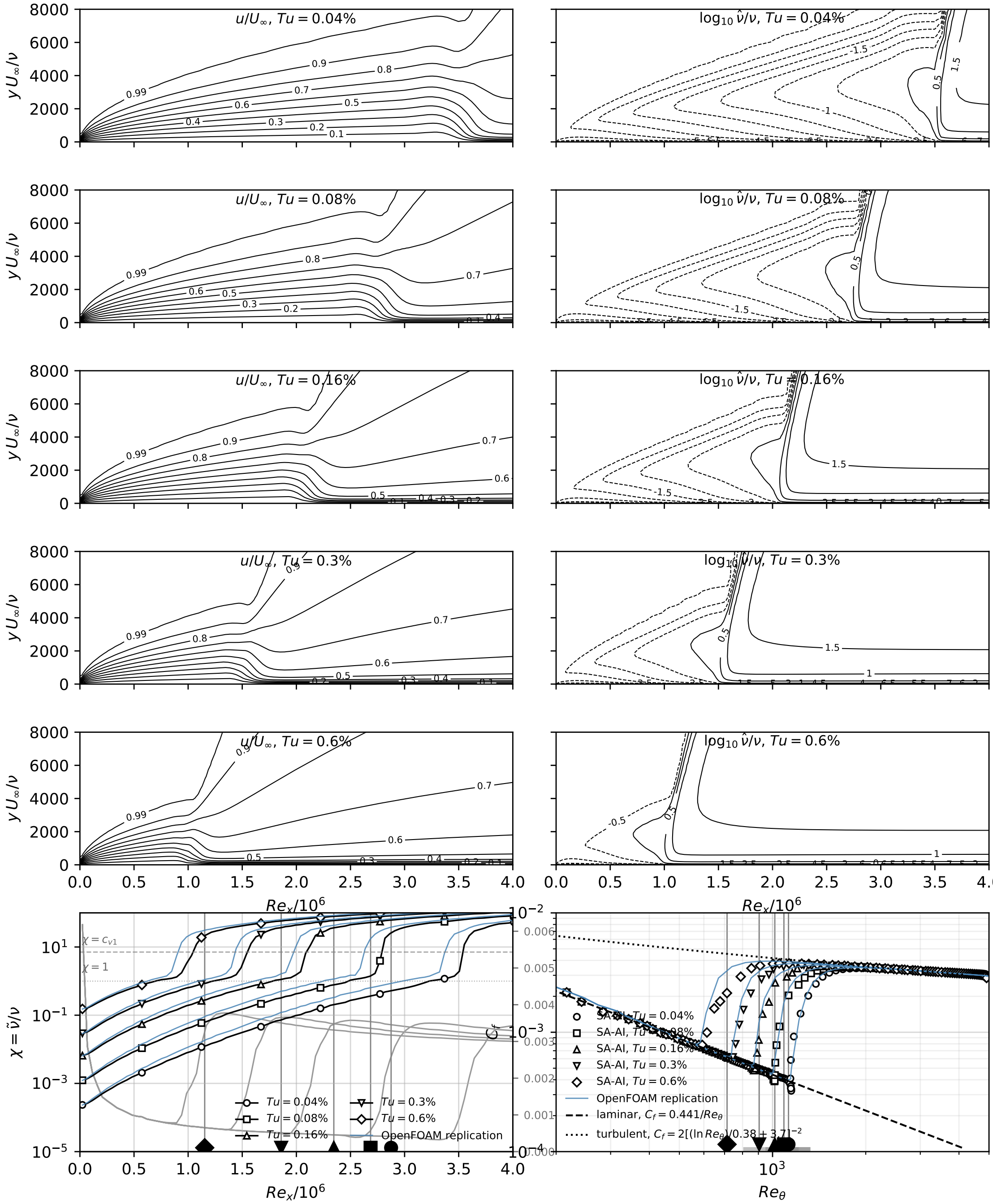


Figure 6: Flat plate, Flow360.

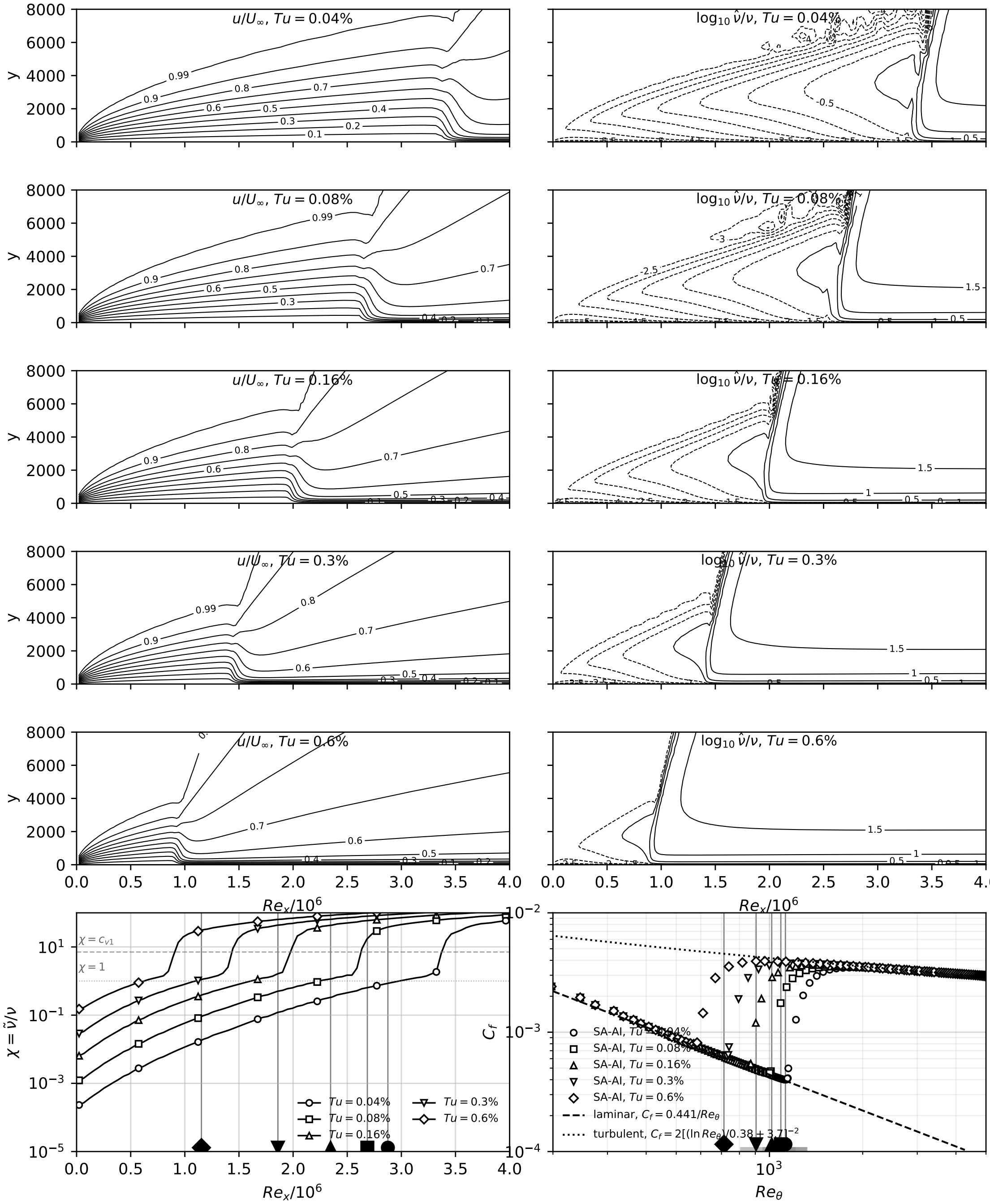


Figure 7: Flat plate, OpenFOAM.

Table 1: Flat-plate transition-onset Re_θ (integrated from the computed profiles) at the $\chi = 1$ and $\chi = c_{v1}$ crossings, Flow360 and the OpenFOAM replication (OF), against the Abu-Ghannam & Shaw correlation.

Tu [%]	AGS Re_θ	$\chi = 1$	$\chi = c_{v1}$	OF $\chi = 1$	OF $\chi = c_{v1}$
0.04	1126	1149	1203	1098	1154
0.08	1088	1002	1069	949	1004
0.16	1017	845	934	798	887
0.3	905	700	815	656	739
0.6	713	524	685	487	589

2.2 NLF(1)-0416 at $Re = 4 \times 10^6$

$\alpha \in \{0°, 4°, 9°, 15°\}$ on both families at L0–L2 plus $\alpha = -8°, -4°$ on all six grids (thirty-six cases). Figure 8: transition fronts against the LTPT microphone measurements, Coder's AFT (OVERFLOW) at $N_{\text{crit}} = 10.07$ and recalibrated 7.18, the fourteen First AIAA Transition Modeling Workshop submittals, SA-LM2015, Langtry–Menter, and XFOIL—the untuned $N = 9$-class fronts land on the measurements where the recalibrated AFT lands. Figure 9: drag polar against Somers TP-1861, the e^9 reference, and fully turbulent SA. Figures 10–12: surface distributions (five-row suites). Table 2: forces and fronts. Table 3: the OpenFOAM replication—at L2 every front within $0.016\,c$ and the lift within 0.9% of the SA-AI values.

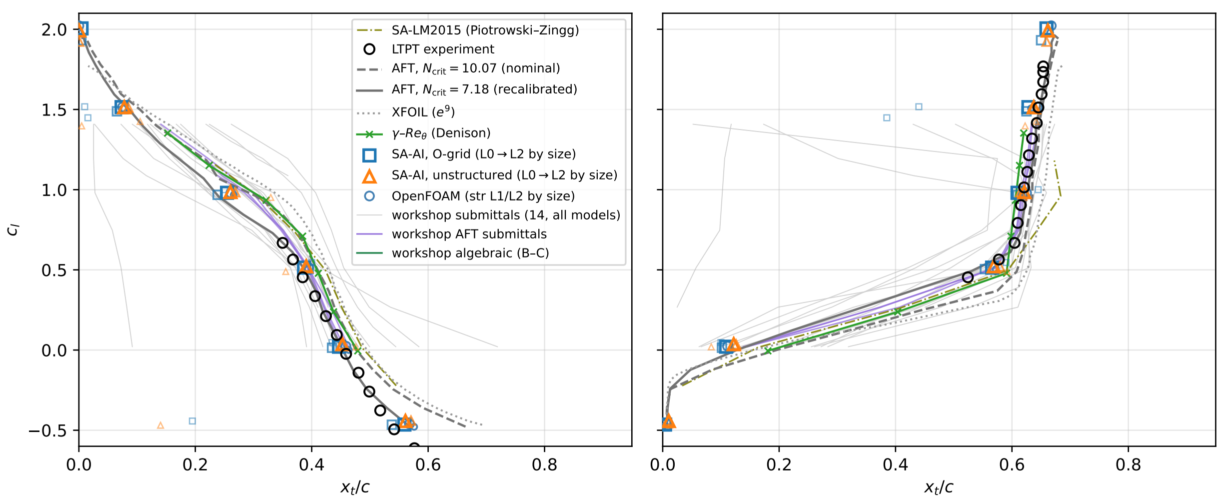


Figure 8: NLF(1)-0416 transition fronts.

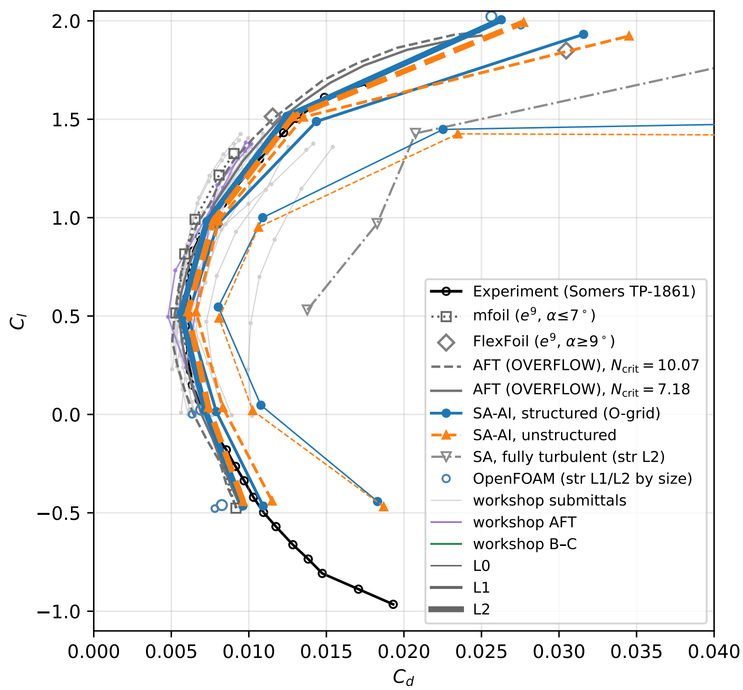


Figure 9: NLF(1)-0416 drag polar.

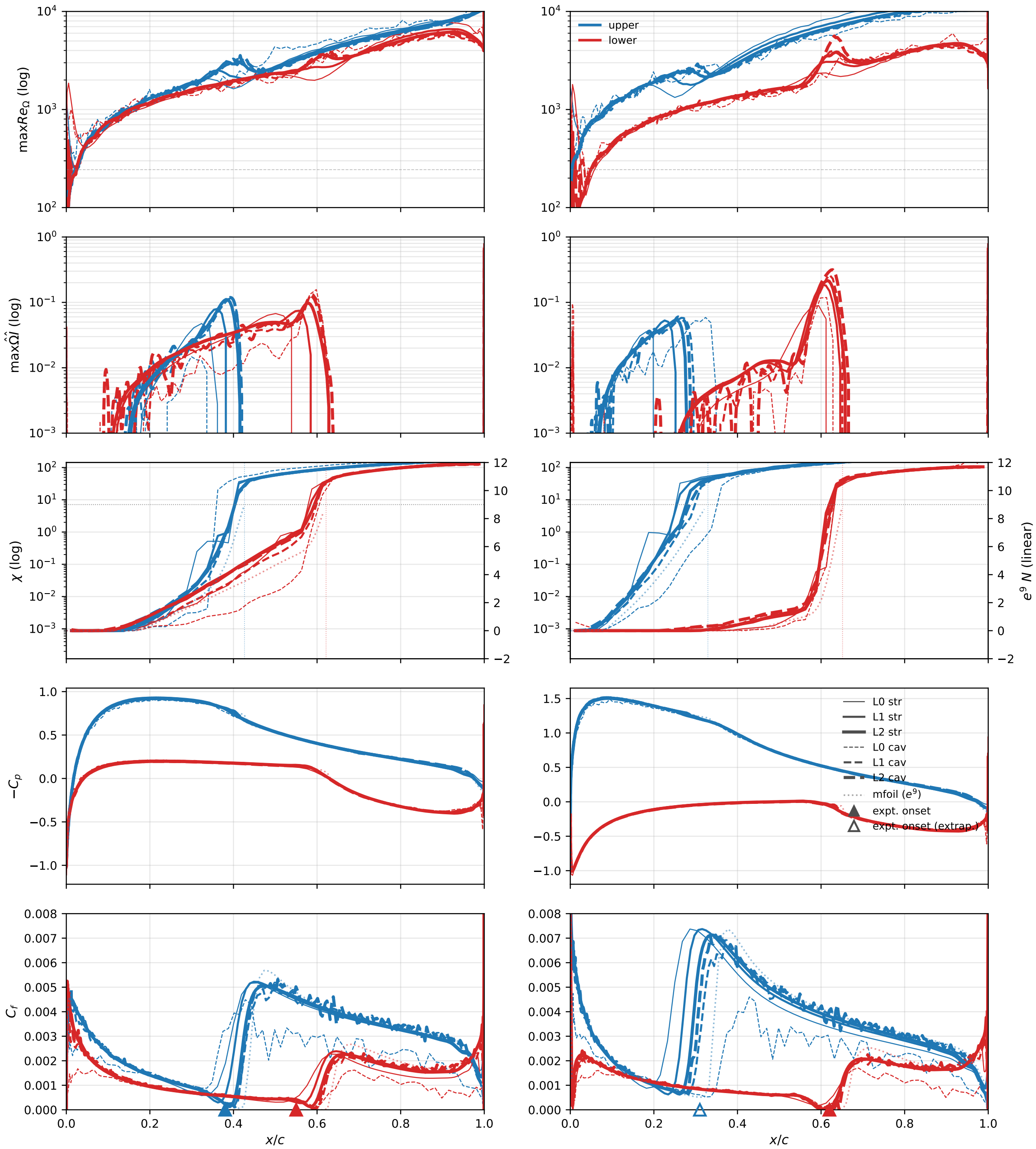


Figure 10: NLF(1)-0416, $\alpha = 0^\circ, 4^\circ$.

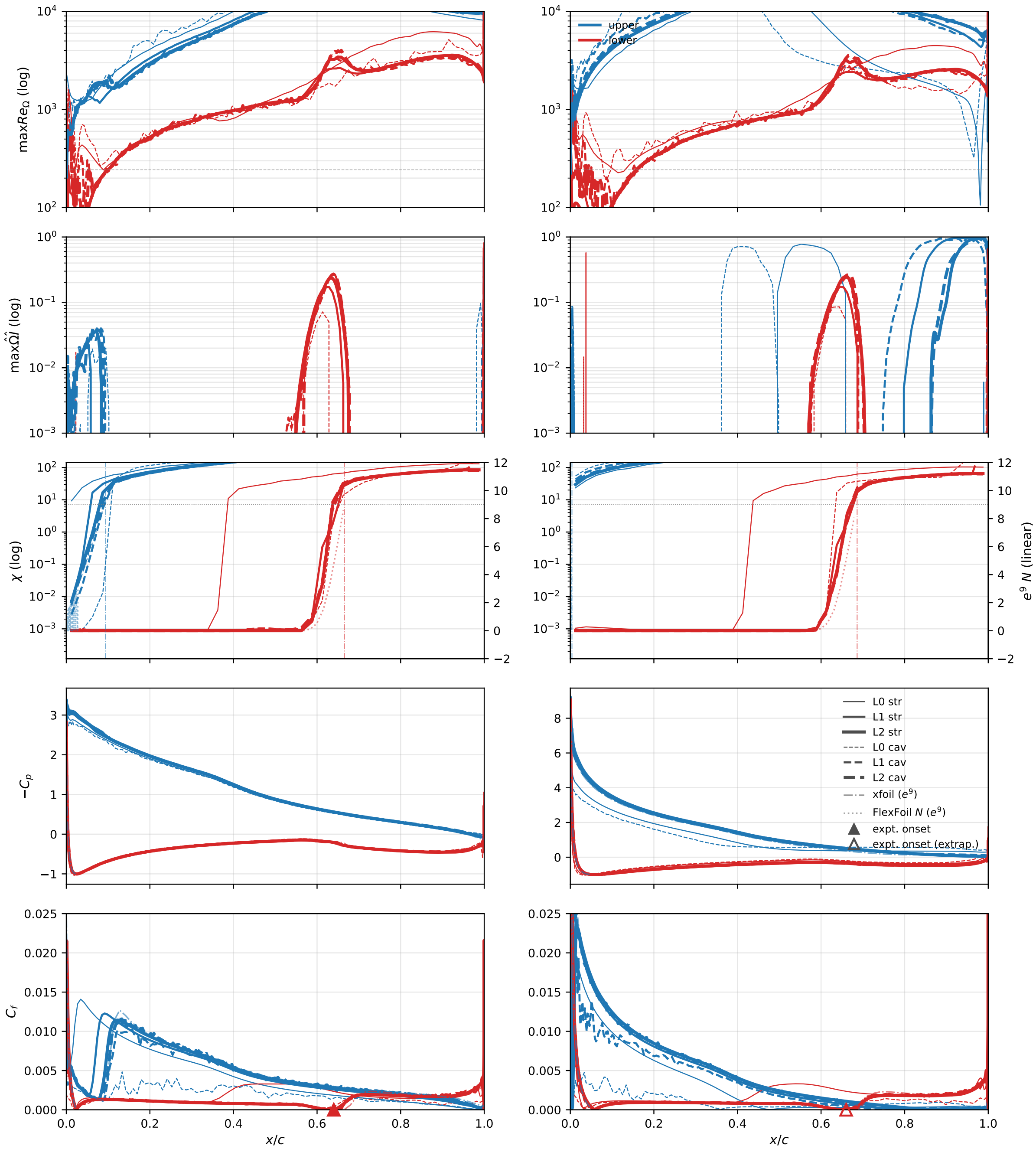


Figure 11: NLF(1)-0416, $\alpha = 9°, 15°$.

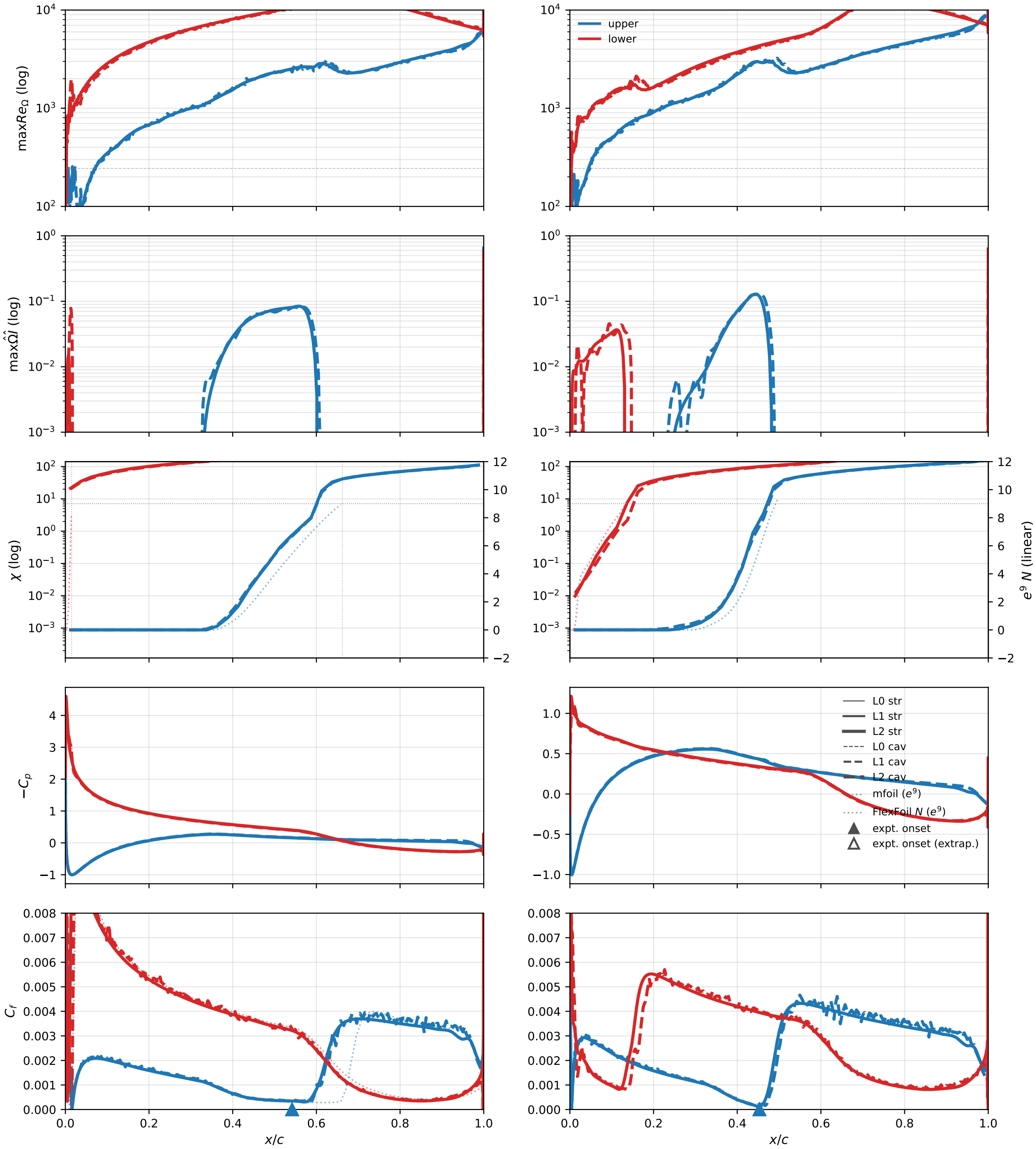


Figure 12: NLF(1)-0416, $\alpha = -8°, -4°$.

Table 2: NLF(1)-0416, $Re = 4\times10^6$: computed forces and transition locations (near-wall $\chi = 1$ front stations).

α	grid	structured O-grid				unstructured cavity			
		C_L	C_D	$x_{\mathrm{tr}}^{\mathrm{up}}$	$x_{\mathrm{tr}}^{\mathrm{lo}}$	C_L	C_D	$x_{\mathrm{tr}}^{\mathrm{up}}$	$x_{\mathrm{tr}}^{\mathrm{lo}}$
−8	L0	-0.4426	0.01830	0.195	0.011	-0.4689	0.01869	0.140	0.012
−8	L1	-0.4656	0.01093	0.538	0.006	-0.4397	0.01150	0.571	0.011
−8	L2	-0.4646	0.00962	0.559	0.004	-0.4426	0.00964	0.561	0.010
−4	L0	0.0469	0.01078	0.435	0.100	0.0197	0.01025	0.476	0.083
−4	L1	0.0142	0.00790	0.435	0.102	0.0357	0.00832	0.452	0.126
−4	L2	0.0208	0.00718	0.447	0.109	0.0363	0.00739	0.453	0.122
0	L0	0.5458	0.00803	0.400	0.590	0.4896	0.00808	0.355	0.592
0	L1	0.5048	0.00607	0.395	0.553	0.5209	0.00660	0.392	0.579
0	L2	0.5139	0.00564	0.387	0.566	0.5219	0.00606	0.391	0.569
4	L0	1.0001	0.01090	0.250	0.645	0.9513	0.01062	0.330	0.621
4	L1	0.9678	0.00800	0.238	0.610	0.9839	0.00805	0.269	0.614
4	L2	0.9805	0.00730	0.255	0.610	0.9848	0.00772	0.260	0.621
9	L0	1.4483	0.02252	0.015	0.385	1.4251	0.02345	0.105	0.641
9	L1	1.4886	0.01436	0.065	0.625	1.5112	0.01353	0.085	0.635
9	L2	1.5146	0.01245	0.073	0.628	1.5139	0.01286	0.078	0.638
15	L0	1.5172	0.07109	0.010	0.440	1.3976	0.11602	0.005	0.623
15	L1	1.9317	0.03159	0.004	0.650	1.9232	0.03454	0.001	0.659
15	L2	2.0057	0.02628	0.005	0.659	1.9924	0.02772	0.000	0.662

Table 3: NLF(1)-0416: the independent OpenFOAM implementation on the structured L1 and L2 grids, fronts at the same near-wall $\chi = 1$ convention (the port omits the f_{v1} bypass).

α	L1				L2			
	C_L	C_D	$x_{\mathrm{tr}}^{\mathrm{up}}$	$x_{\mathrm{tr}}^{\mathrm{lo}}$	C_L	C_D	$x_{\mathrm{tr}}^{\mathrm{up}}$	$x_{\mathrm{tr}}^{\mathrm{lo}}$
−8	-0.4785	0.00781	0.576	0.000	-0.4606	0.00826	0.568	0.000
−4	0.0005	0.00634	0.464	0.117	0.0239	0.00693	0.459	0.111
0	0.5034	0.00603	0.393	0.572	0.5153	0.00573	0.395	0.568
4	0.9670	0.00778	0.266	0.616	0.9793	0.00749	0.261	0.625
9	1.5066	0.01273	0.076	0.633	1.5151	0.01270	0.075	0.644
15	1.9780	0.02754	0.000	0.665	2.0221	0.02564	0.000	0.667

2.3 Eppler 387 at $Re = 2 \times 10^5$

$\alpha \in \{0°, 2°, 5°, 7°\}$ on both families at L0–L2 plus six extension incidences (−2°, 1°, 3°, 4°, 6°, 8.5°) on the finest pair; same seed. Figure 13: drag polar against the LTPT measurement (McGhee et al., TM-4062), the e^9 reference, fully turbulent SA, and the digitized γ–Re_θ and SA-BC curves—the finest grids track the measured bucket at matched lift (−6.4% to +3.7% over $\alpha = 0°$–$7°$). Figure 14: bubble stations against the oil flow, Selig et al., Cole–Mueller, and the e^9 references—separation reproduced to a few percent of chord everywhere; reattachment 0.02–0.04 c late at $\alpha \leq 5°$, at the edge of bubble collapse at 7°. Figures 15–16: surface suites. Table 4: forces. Table 5: bubble stations. Table 6: the OpenFOAM replication—at L2 lift within 0.5%, separation 0.030–0.037 c ahead of and reattachment 0.033–0.045 c behind the SA-AI stations.

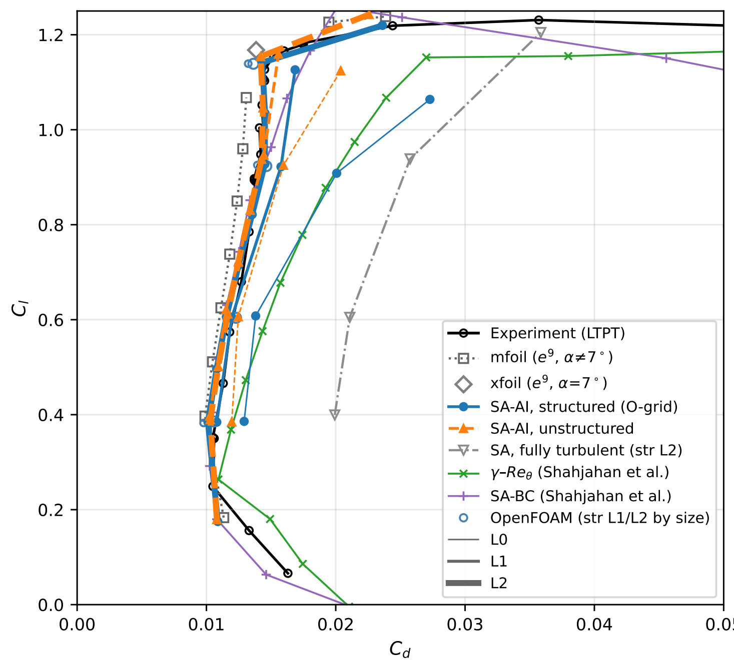


Figure 13: Eppler 387 drag polar.

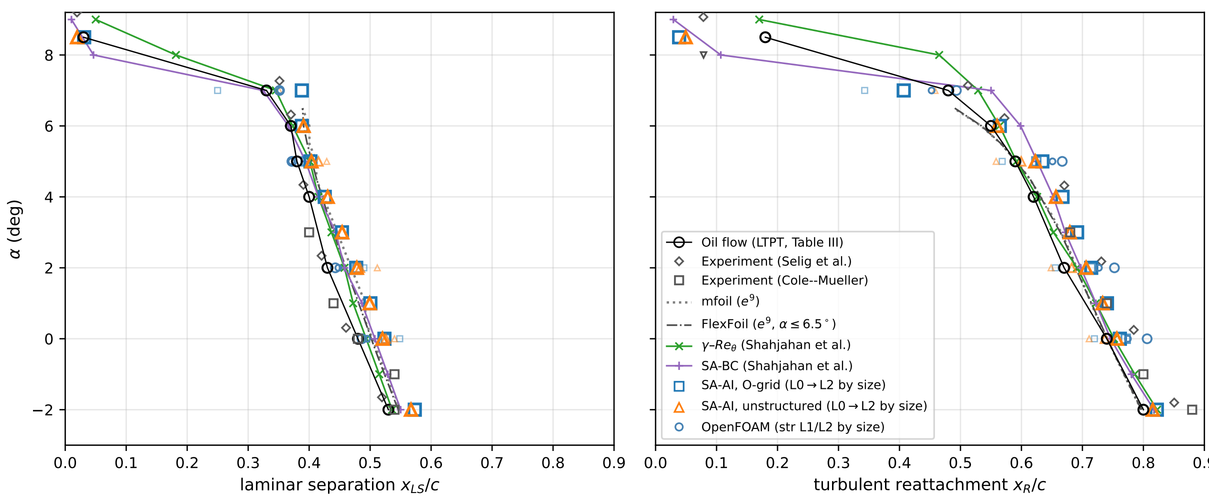


Figure 14: Eppler 387 bubble stations.

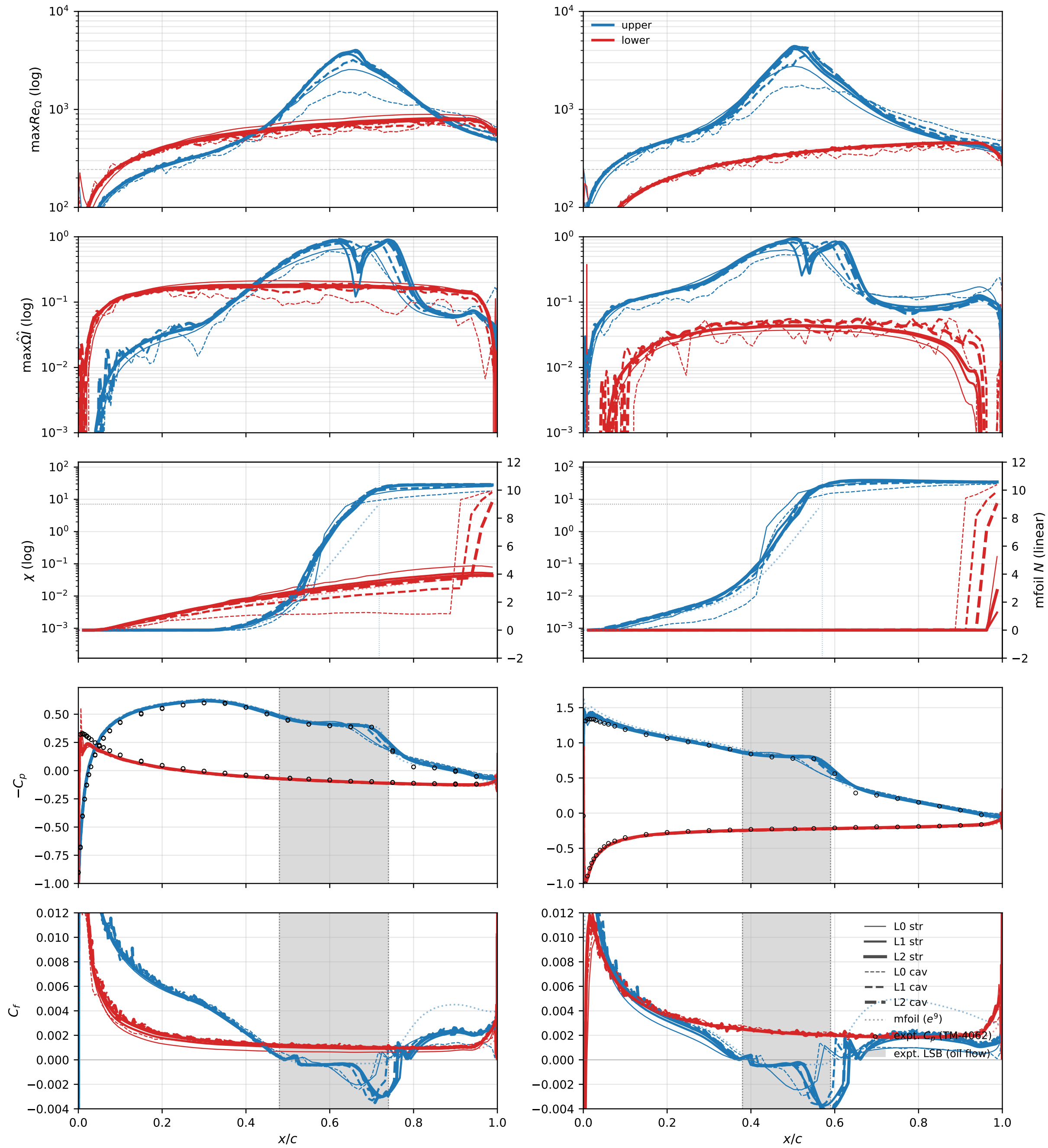


Figure 15: Eppler 387, $\alpha = 0°, 5°$.

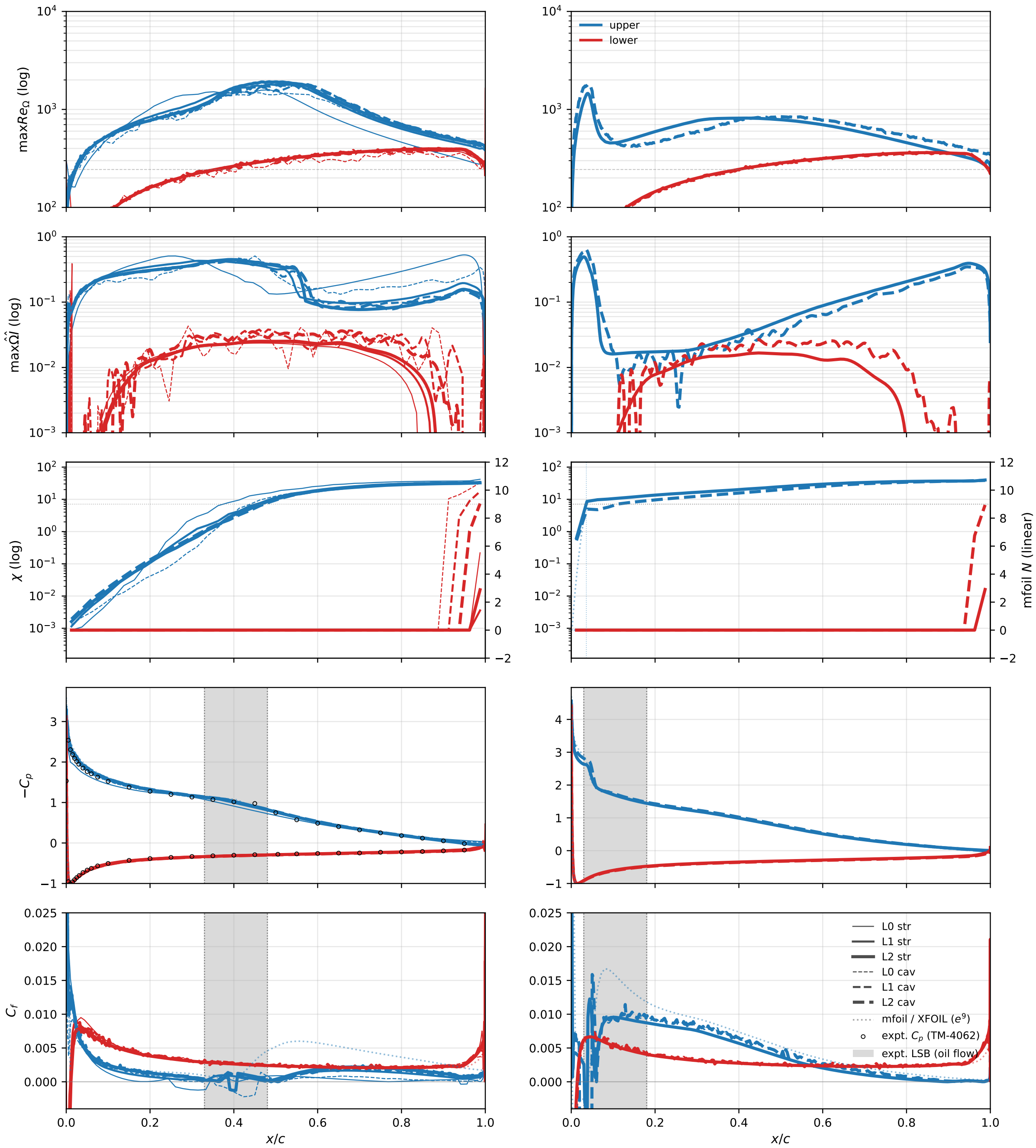


Figure 16: Eppler 387, $\alpha = 7°, 8.5°$.

Table 4: Eppler 387, $Re = 2 \times 10^5$: computed forces (the $-2°$–$8.5°$ extension incidences exist on the finest pair only).

α	grid	structured O-grid		unstructured cavity	
		C_L	C_D	C_L	C_D
−2	L2	0.1752	0.01089	0.1791	0.01080
0	L0	0.3859	0.01292	0.3847	0.01196
0	L1	0.3843	0.01080	0.3969	0.01026
0	L2	0.3844	0.01010	0.3894	0.01030
1	L2	0.4957	0.01078	0.5012	0.01092
2	L0	0.6079	0.01381	0.6061	0.01246
2	L1	0.6048	0.01196	0.6207	0.01152
2	L2	0.6059	0.01154	0.6124	0.01167
3	L2	0.7146	0.01247	0.7218	0.01245
4	L2	0.8218	0.01354	0.8304	0.01341
5	L0	0.9083	0.02007	0.9253	0.01596
5	L1	0.9218	0.01575	0.9465	0.01444
5	L2	0.9274	0.01451	0.9375	0.01428
6	L2	1.0346	0.01453	1.0456	0.01439
7	L0	1.0639	0.02728	1.1241	0.02038
7	L1	1.1262	0.01686	1.1593	0.01557
7	L2	1.1415	0.01431	1.1535	0.01419
8.5	L2	1.2190	0.02310	1.2611	0.02150

Table 5: Eppler 387, $Re = 2 \times 10^5$: computed upper-surface laminar-separation and turbulent-reattachment stations (signed-C_f zero crossings; missing entries transition ahead of separation and have no bubble). The measured oil-flow reattachments (TM-4062 Table III) at $\alpha = 0°/2°/5°$ are $0.74/0.67/0.59\,c$.

α	grid	structured O-grid		unstructured cavity	
		x_{LS}/c	x_R/c	x_{LS}/c	x_R/c
−2	L2	0.573	0.822	0.567	0.815
0	L0	0.549	0.720	0.540	0.711
0	L1	0.526	0.771	0.526	0.737
0	L2	0.523	0.761	0.520	0.757
1	L2	0.500	0.740	0.499	0.733
2	L0	0.490	0.656	0.512	0.649
2	L1	0.479	0.713	0.483	0.682
2	L2	0.477	0.714	0.478	0.706
3	L2	0.454	0.691	0.454	0.679
4	L2	0.426	0.667	0.430	0.657
5	L0	0.393	0.568	0.429	0.558
5	L1	0.394	0.624	0.415	0.600
5	L2	0.402	0.634	0.404	0.623
6	L2	0.388	0.565	0.391	0.560
7	L0	0.250	0.343	0.351	0.458
7	L1	–	–	–	–
7	L2	0.388	0.407	–	–
8.5	L2	0.031	0.039	0.019	0.049

Table 6: Eppler 387: the independent OpenFOAM implementation on the structured L1 and L2 grids (fronts at the near-wall $\chi=1$ convention, stations at the signed-C_f zero crossings; the port omits the f_{v1} bypass).

α	L1					L2				
	C_L	C_D	$x_{\rm tr}^{\rm up}$	x_{LS}	x_R	C_L	C_D	$x_{\rm tr}^{\rm up}$	x_{LS}	x_R
0	0.3826	0.00978	0.605	0.495	0.775	0.3857	0.01058	0.621	0.486	0.806
2	0.6037	0.01174	0.543	0.448	0.727	0.6039	0.01229	0.564	0.443	0.752
5	0.9256	0.01394	0.468	0.371	0.651	0.9234	0.01463	0.487	0.372	0.667
7	1.1391	0.01323	0.329	0.353	0.453	1.1388	0.01363	0.379	0.351	0.493

2.4 Eppler 387 Reynolds sweep at $\alpha = 5°$

$Re \in \{6\times10^4,\ 10^5,\ 3\times10^5,\ 4.6\times10^5\}$ on the same grids (24 further solutions). Figure 17: forces and bubble stations across the sweep—the refinement fan closes onto the measurement at 2–4.6×10^5 and opens at 10^5, the model's bursting boundary, one Reynolds step above the measurement's (6×10^4); the computed state at every swept Reynolds number is unique among protocol-admitted initializations. Figures 18–19: surface suites. Table 7: L2 coefficients.

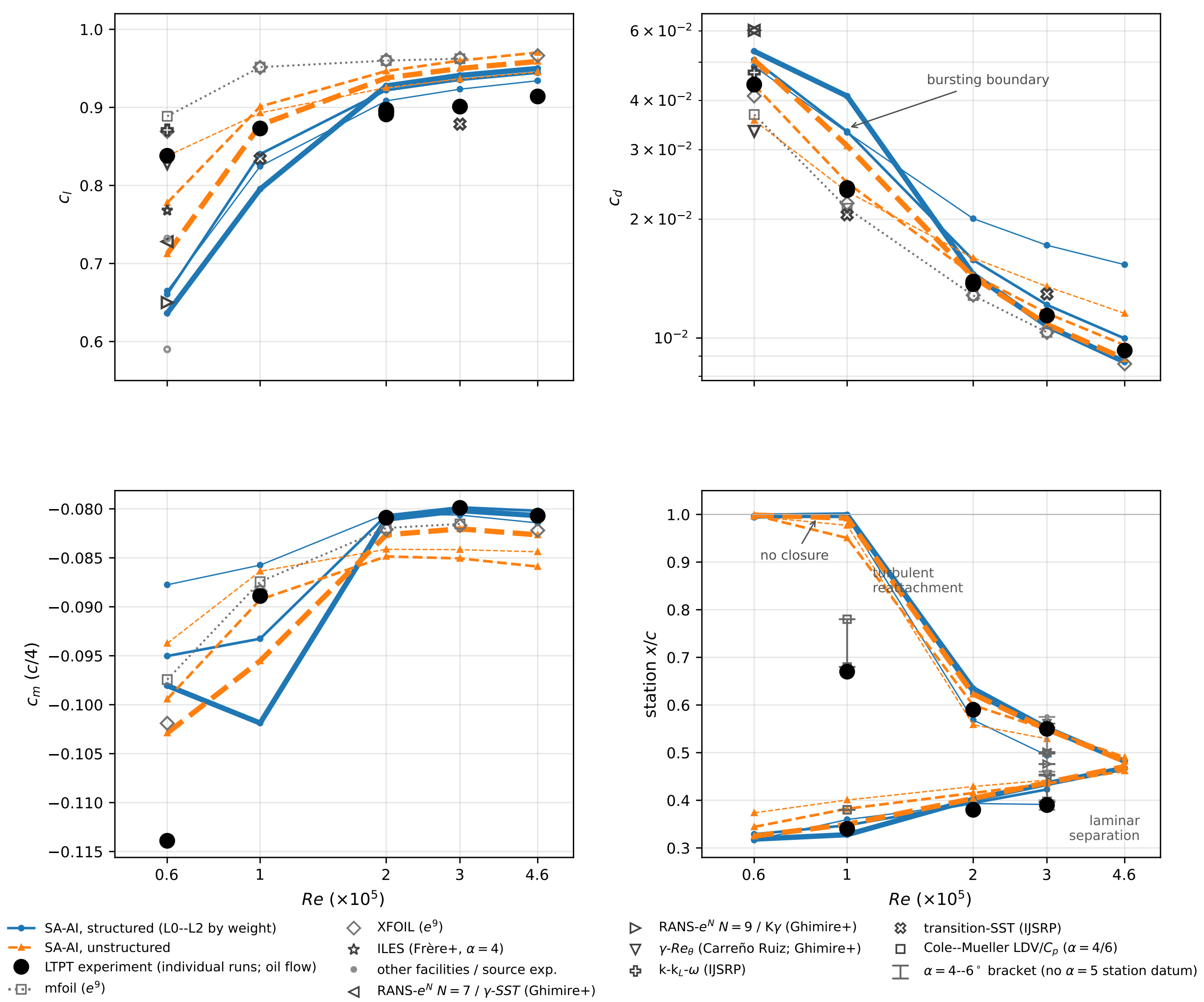


Figure 17: Reynolds-sweep forces and stations.

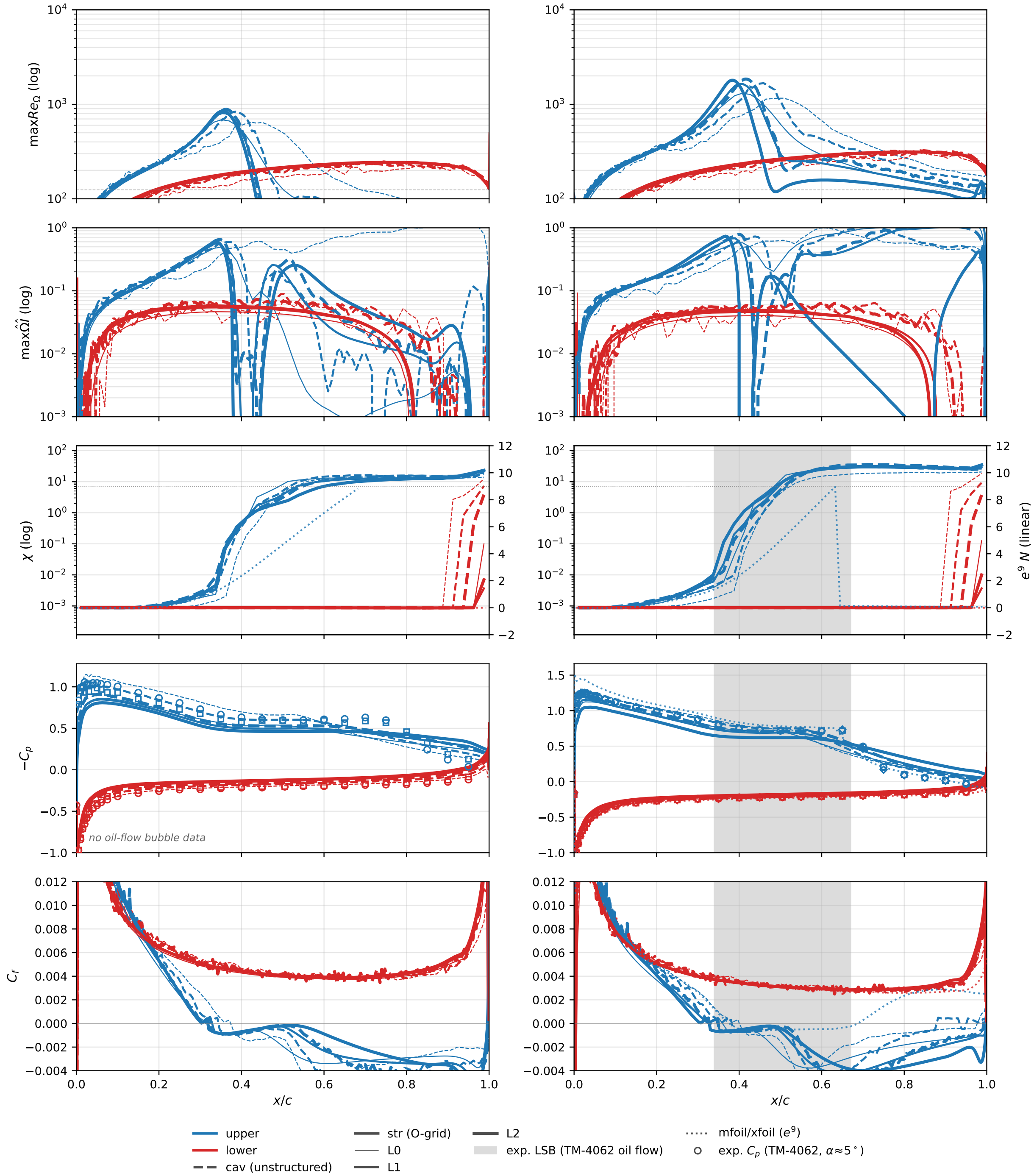


Figure 18: Reynolds sweep, $Re = 6 \times 10^4$, 10^5.

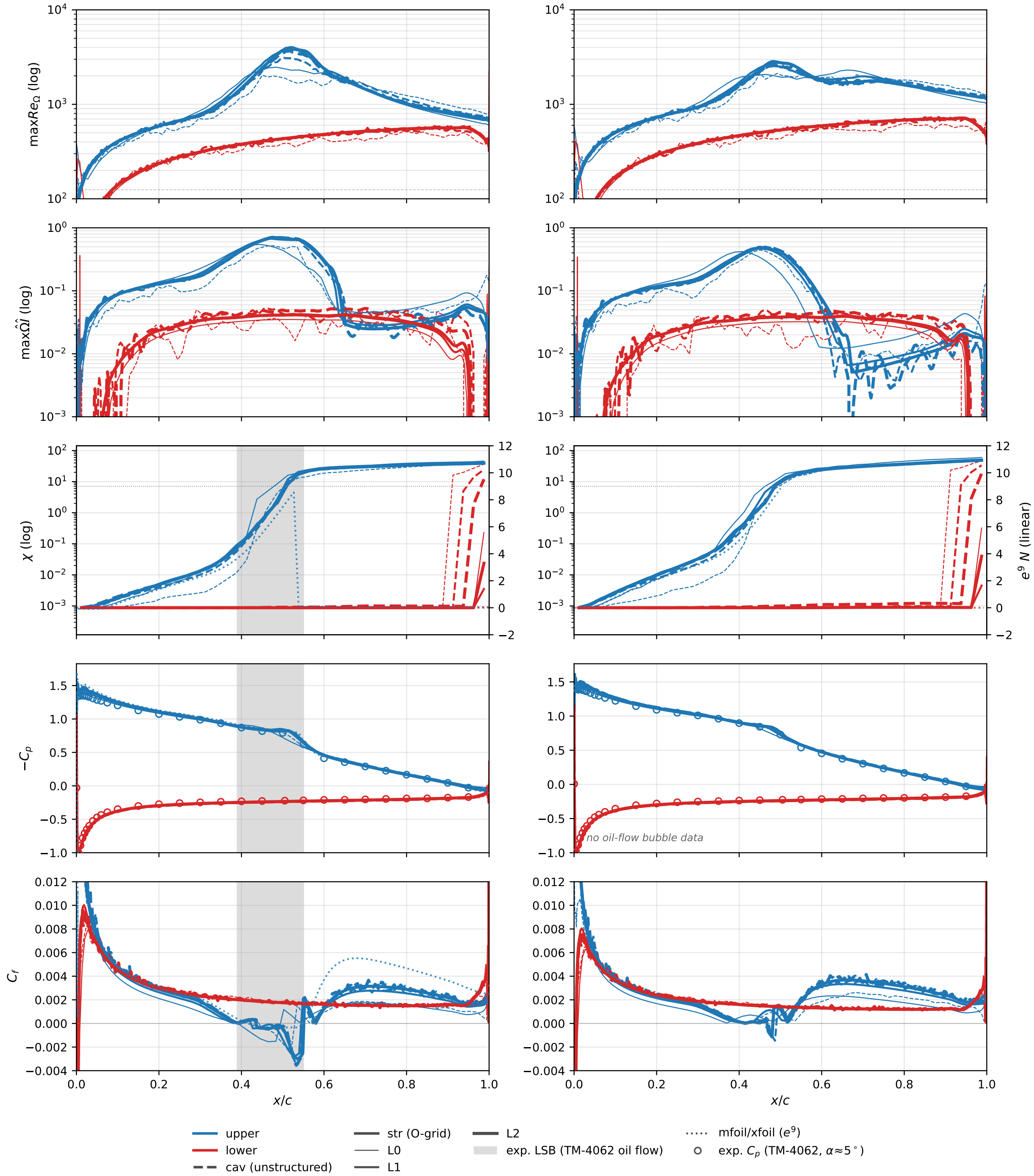


Figure 19: Reynolds sweep, $Re = 3,\ 4.6 \times 10^5$.

Table 7: Eppler 387 at $\alpha = 5°$ across the Reynolds sweep: L2 section coefficients against the e^9 panel reference and the LTPT measurement (read at the tabulated angle nearest 5°). Computed moments transferred to the quarter chord.

	SA-AI, structured L2			SA-AI, unstructured L2			e^9 panel			Exp (LTPT)		
Re	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m
6×10^4	0.636	0.0533	−0.098	0.712	0.0506	−0.103	0.869‡	0.0411‡	−0.102‡	0.838	0.0439	−0.114
1×10^5	0.795	0.0410	−0.102	0.877	0.0307	−0.096	0.952	0.0213	−0.087	0.873	0.0237	−0.089
2×10^5	0.927	0.0145	−0.081	0.938	0.0143	−0.083	0.960	0.0128	−0.082	0.891	0.0138	−0.081
3×10^5	0.941	0.0107	−0.080	0.950	0.0108	−0.082	0.962	0.0103	−0.082	0.901	0.0114	−0.080
4.6×10^5	0.950	0.0087	−0.081	0.959	0.0089	−0.083	0.966	0.0086	−0.082	0.914	0.0093	−0.081

‡XFOIL, whose forces sit closest to the measurement, is the reference at this bistable condition; mfoil gives $c_l = 0.889$, $c_d = 0.0368$, $c_m = -0.097$, and FlexFoil sides with XFOIL ($c_l = 0.868$, $c_d = 0.0406$, $c_m = -0.102$).

2.5 Cylinder drag crisis, $Re_D = 1$–2×10^7

A steady-branch traverse over 7.3 decades with one closure and one set of constants. Twelve Reynolds numbers, $Re_D = 6 \times 10^4$–2×10^6, at three seeds ($\chi_\infty = 3.9 \times 10^{-4}$, 1.1×10^{-2}, 0.22; Mack's map at $Tu = 0.05$, 0.2, 0.7%) on one quasi-two-dimensional O-grid (1200 × 148, first spacing $8 \times 10^{-6}\,D$, outer boundary $100\,D$; $y^+ \leq 0.64$ at 2×10^6), each seed swept as an upward and a downward continuation ladder plus twelve cold starts at the middle seed—84 steady cases; a large-domain laminar grid (600 × 121, outer boundary $1000\,D$) carries $Re_D = 1$–10^3 and a fine supercritical grid (1600 × 170, first spacing $10^{-6}\,D$; $y^+ \leq 0.71$ at 10^7, 1.4 at the 2×10^7 endpoint, quoted with that caveat) carries 4×10^6–2×10^7—129 cases in all. At the grid seams C_d agrees to 0.5–0.6% at $Re_D = 300$ and 10^3 and to 2.9% between cold-started pairs at 2×10^6; 37 of 129 cases are flagged steady limit cycles; every case lands on the symmetric branch, $|C_L| < 2 \times 10^{-7}$.

Figure 20: the traverse—the crisis moves from $Re_D \approx 5$–7×10^5 at the two quiet seeds to ≈ 3–4×10^5 at $Tu = 0.7\%$, and the supercritical floor ($C_d = 0.217$–0.233 at 2×10^6) rises with the seed. Figure 21: the transition and separation angles across the whole traverse, with the radial-ray χ profiles that define the transition angle. Figure 22: the favorable-pressure-gradient amplification audit—the model's frozen-profile envelope slope tracks the Drela–Giles fit through $\beta \leq 0.1$, halves at $\beta = 0.2$, and collapses three orders of magnitude by $\beta = 0.5$, while on the cylinder nose the e^N $N = \ln(1/\chi_\infty)$ crossing sits behind the computed front at all three Reynolds numbers shown. Figure 23: fields at three stations of the traverse. Table 8: the matrix C_d. Below the shedding onset ($Re_D \approx 46$) the steady solution is the physical flow: $C_d = 10.67$ at $Re_D = 1$ and 2.797 at 10, −1.7% against the steady benchmark of Dennis & Chang (+6.4% at 100); between the onset and roughly 10^3 the steady-branch points are drawn for continuity, not validation. Seed independence holds to 0.2% from 10^4 to 4×10^4, and χ first crosses unity at 4×10^4, in the base region.

Six concessions, all visible in the figures:

1. Through the crisis band the steady branch is non-unique and weakly unsteady: 27 of the 84 matrix cases end in a limit cycle (tail peak-to-peak median 0.011, maximum 0.199 in C_d), the three protocols separate by up to 0.055 in C_d there (against 0.003 elsewhere), and the downward ladder lands at higher drag—opposite to the physical hysteresis sense—so no hysteresis claim is made.

2. $C_L = 0$ in all 84 matrix cases is a statement about the symmetric steady protocol, not the physics.

3. The subcritical plateau, $C_d \approx 0.84$, sits well below measurement, as any steady symmetric (shedding-free) wake solution must.

4. The supercritical turbulent separation angles (106–118°) remain early against measurement.

5. There is no transcritical recovery: the computed floors sit flat at 0.185/0.199/0.215 (by seed) through 2×10^7 while measurement rebuilds toward $C_d \approx 0.7$; the $\chi = 1$ front marches from 101° to 82° over 2×10^6–2×10^7, so the failure is the turbulent separation, pinned near 120°, not the transition content.

6. On the fine grid the downward ladder finds a second supercritical steady state, 13–22% lower in C_d (separation 123.5° against 120.4°, base pressure −0.18 against −0.23 at 10^7).

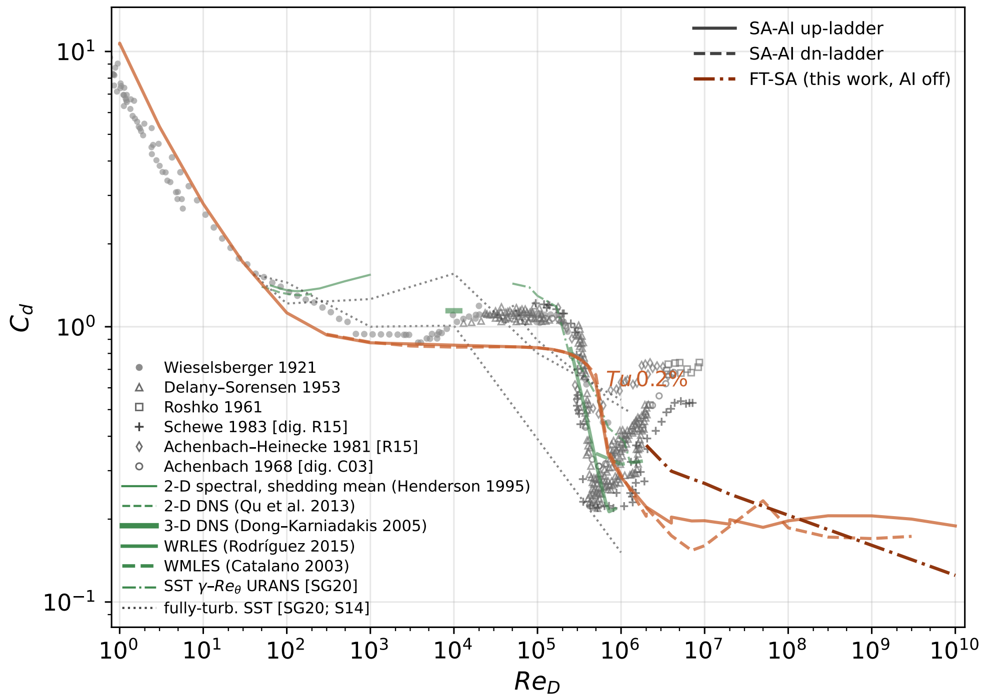


Figure 20: Cylinder drag crisis.

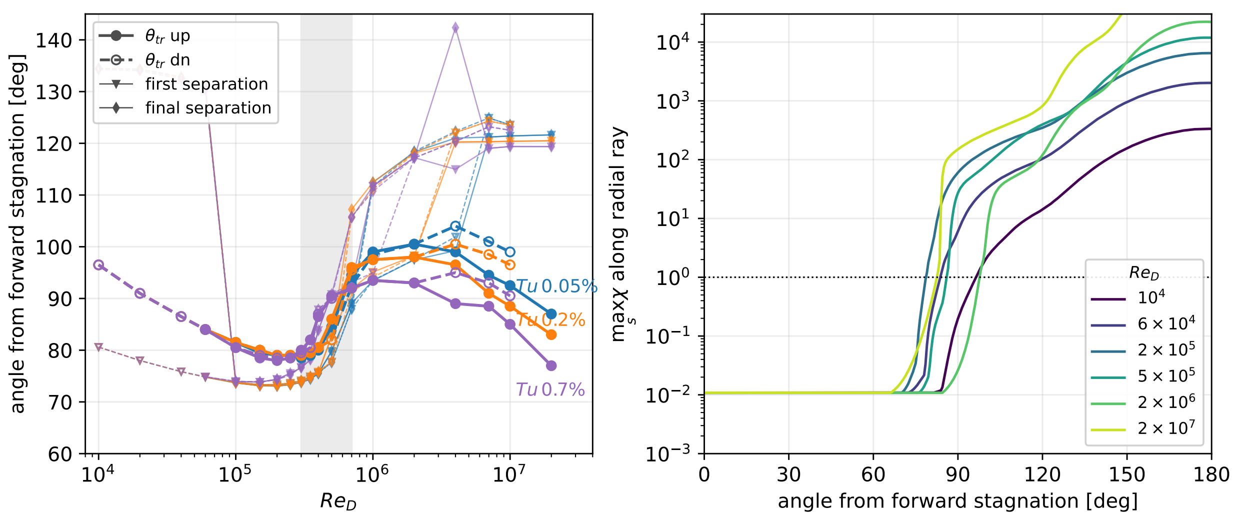


Figure 21: Transition and separation angles.

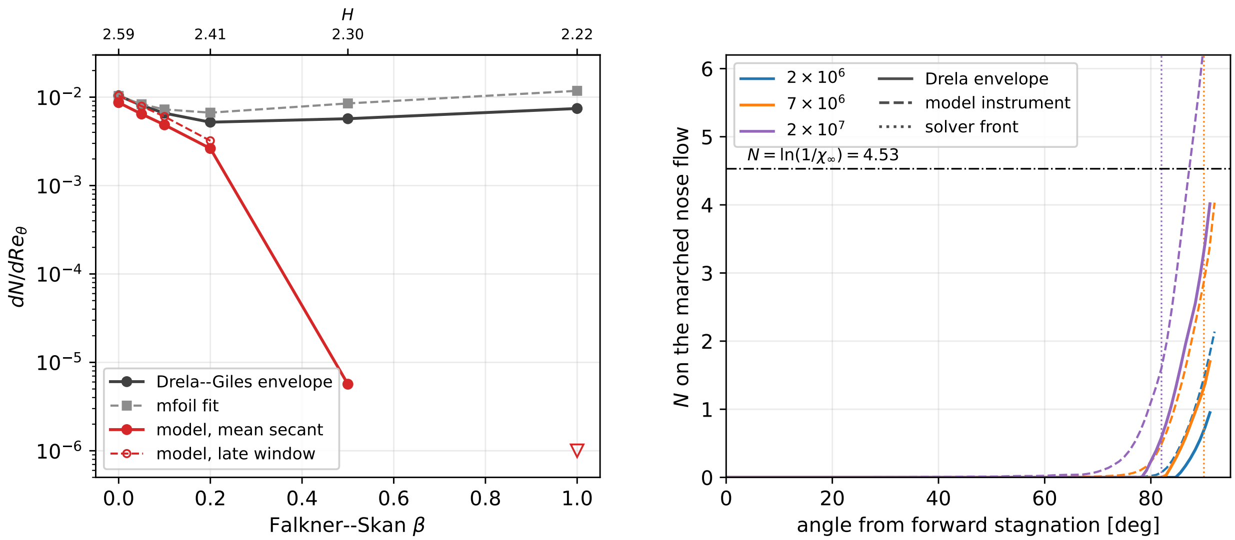


Figure 22: Favorable-gradient amplification audit.

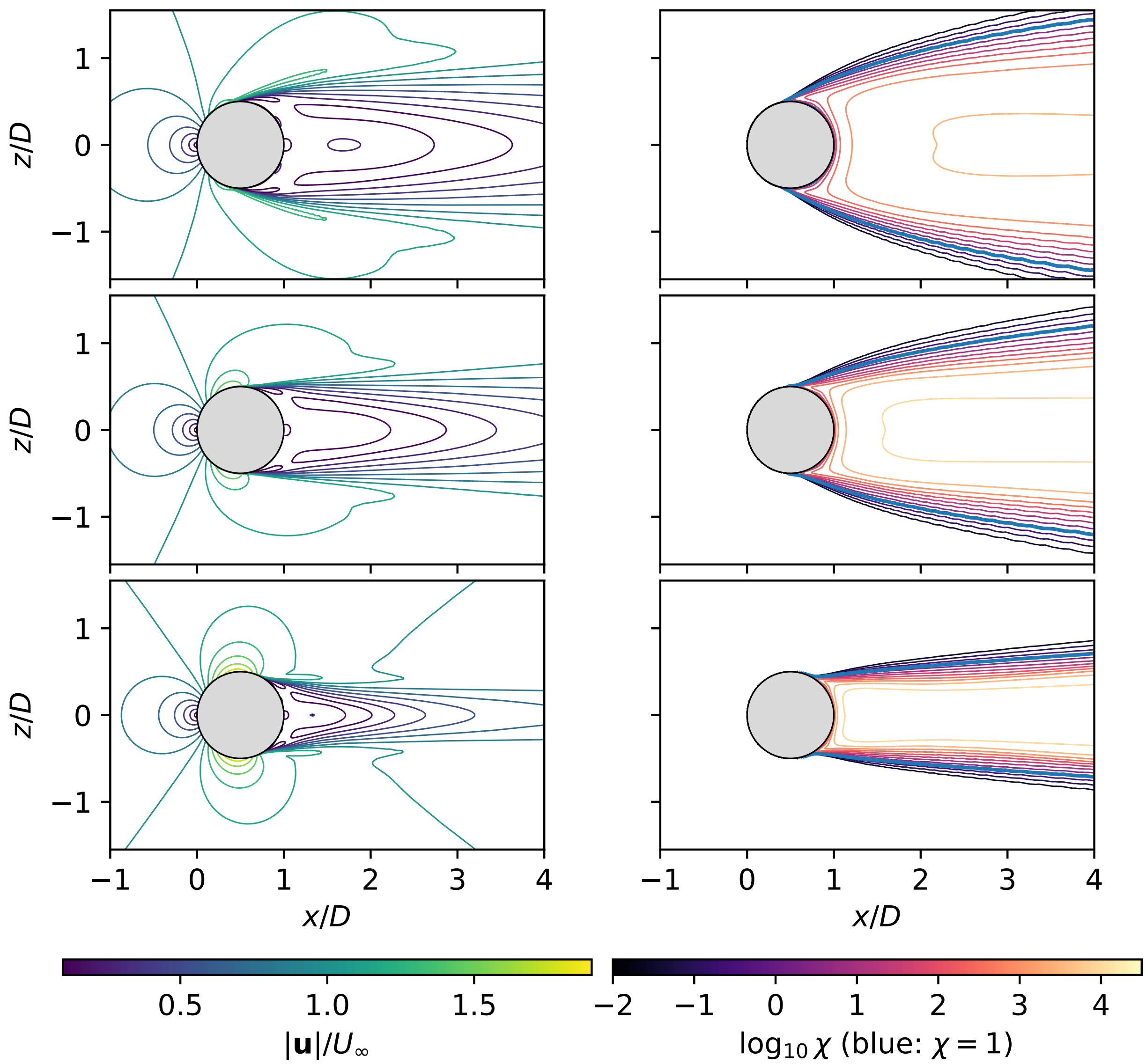


Figure 23: Drag-crisis fields.

Table 8: Cylinder steady drag-crisis matrix: C_d (median over the final-stage tail window) for the 84 matrix cases. Bracketed values: tail-window C_d peak-to-peak for cases flagged as steady limit cycles; $|C_L| < 2 \times 10^{-7}$ in all 84 cases.

Re_D	$Tu\,0.05\%$		$Tu\,0.2\%$			$Tu\,0.7\%$	
($\times 10^5$)	up	dn	up	dn	cold	up	dn
0.6	0.845	0.845	0.845	0.845	0.845	0.842	0.842
1	0.840	0.839	0.839	0.839	0.839	0.831	0.832
1.5	0.828	0.827	0.828	0.827	0.827	0.809	0.812
2	0.814	0.812	0.813	0.811	0.811	0.781	0.786
2.5	0.796	0.794	0.795	0.793	0.793	0.745	0.755
3	0.776	0.774	0.774	0.771	0.772	0.677 [0.075]	0.709
3.5	0.753	0.750	0.749	0.748	0.749	0.616 [0.199]	0.671 [0.030]
4	0.725	0.726	0.719	0.722	0.724	0.495 [0.122]	0.547 [0.086]
5	0.644 [0.076]	0.670 [0.036]	0.620 [0.129]	0.668 [0.047]	0.650 [0.136]	0.430 [0.008]	0.439
7	0.409 [0.051]	0.407 [0.024]	0.344 [0.002]	0.355	0.347 [0.003]	0.365 [0.006]	0.356
10	0.286 [0.006]	0.285	0.284 [0.005]	0.290	0.282 [0.004]	0.294 [0.007]	0.303
20	0.219 [0.013]	0.217 [0.008]	0.221 [0.011]	0.220 [0.008]	0.220 [0.008]	0.233 [0.010]	0.232 [0.005]

2.6 Daedalus wing (three-dimensional deployment)

The Daedalus human-powered-aircraft wing: half-span 17.07 m, root chord 0.917 m, aspect ratio 37.8, DAE-11/21/31 sections; $Re_{\text{root}} = 5 \times 10^5$, $\alpha \in \{4^\circ, 5^\circ, 6^\circ\}$ ($C_L \approx 1.02$–1.23), flight-quiet seed $\chi_\infty = 8.76 \times 10^{-6}$ ($N_{\text{crit}} \approx 13.6$); two mesh families (spanwise-stacked structured O-grid, 0.72–36.7M nodes; unstructured prism/tet/hex, 1.70–54.9M nodes) at three levels, eighteen solutions; the reference is AVL with strip-wise XFOIL section polars at matched local c_l and N_{crit}, trimmed to the RANS lift. Figure 24: wing polar and sectional loading—the families agree at L2 to 0.35–0.63% in C_L and 0.7–1.7 counts in C_D, and the finest-grid drag runs 15.6/10.3/5.6 counts (7.3/4.4/2.2%) below the matched-lift reference at $\alpha = 4^\circ/5^\circ/6^\circ$. Figure 25: surface maps at $\alpha = 4^\circ$ against the e^N strips—the bubble matches station by station (at $\eta = 0.31$, separation within $0.017\,c$ and reattachment within $0.009\,c$ at every incidence), the $\chi = 1$ front sits $\sim 0.09\,c$ ahead of the strips' N_{crit} contour (the seed convention), and the tip vortex trips transition to the leading edge for $\eta \gtrsim 0.992$. Table 9: totals by level ($\alpha = 5^\circ$, 6° surface maps and section sheets: appendix).

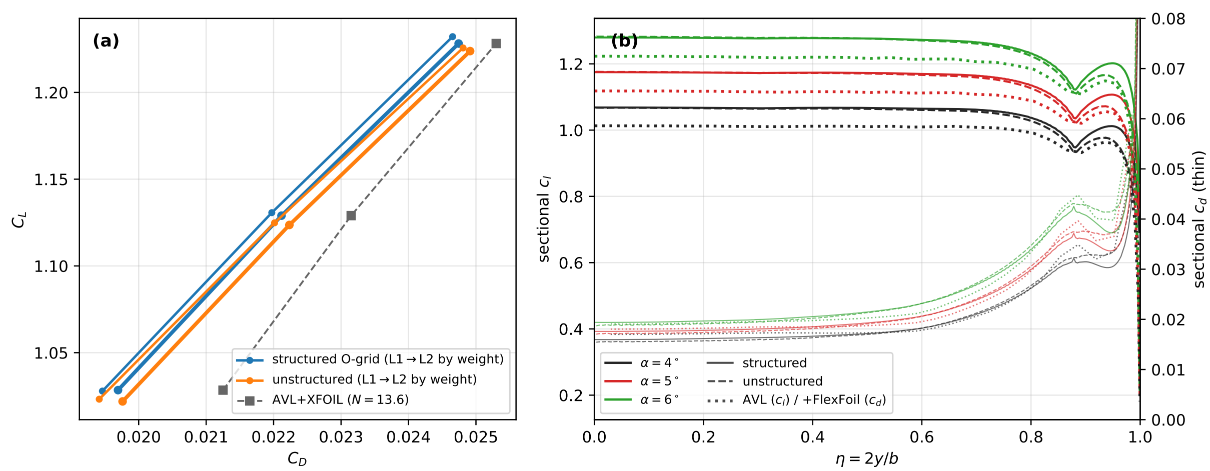


Figure 24: Daedalus polar and sections.

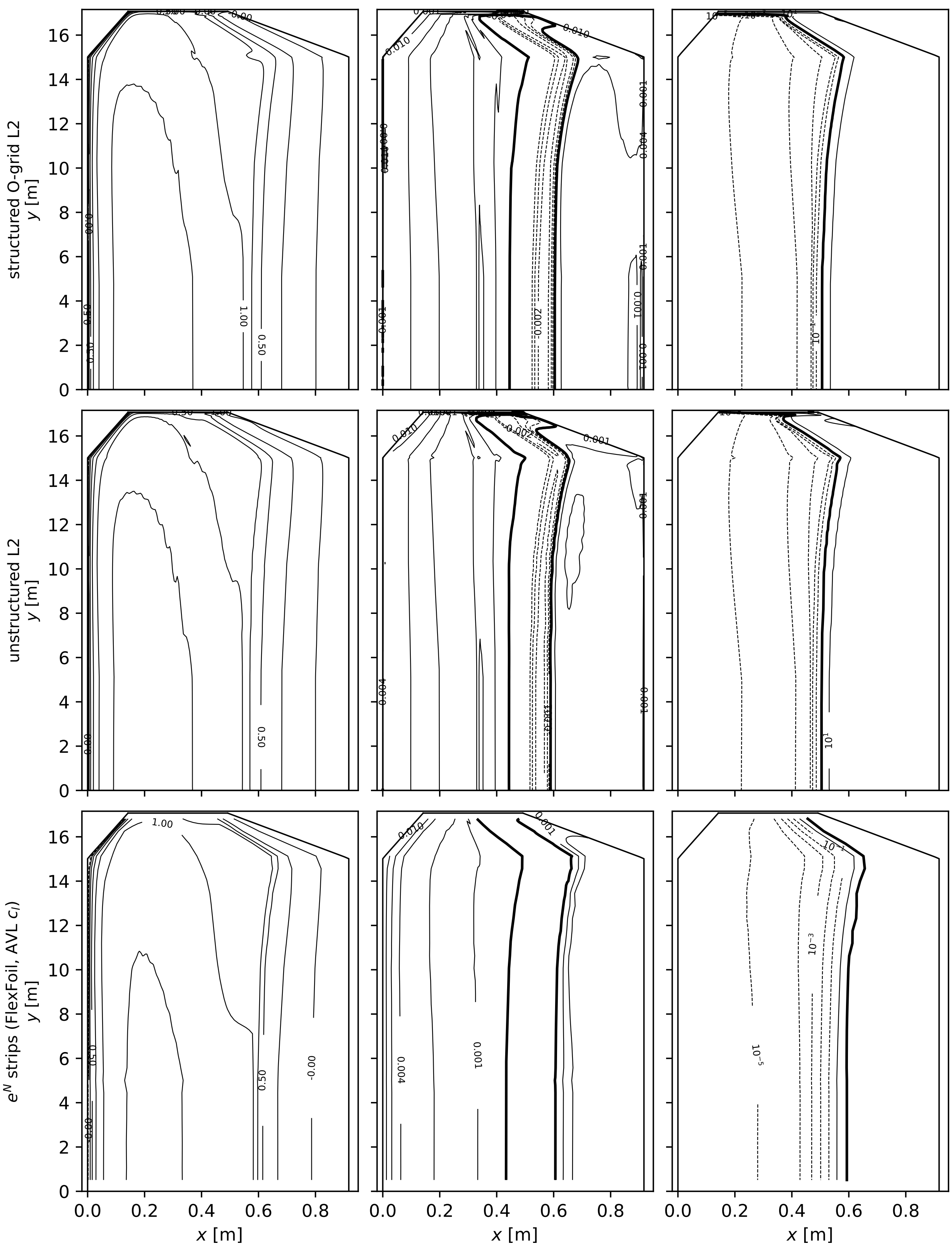
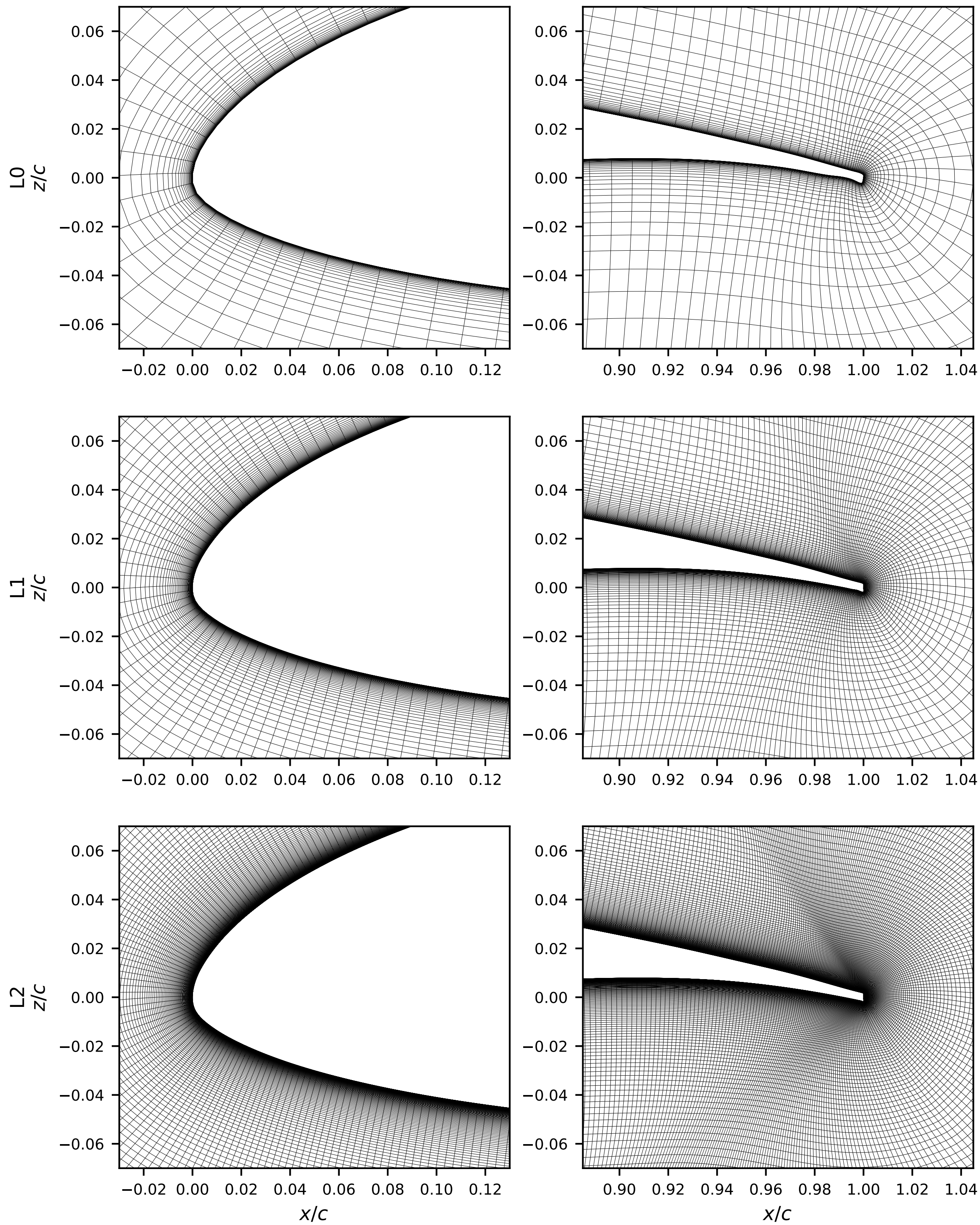


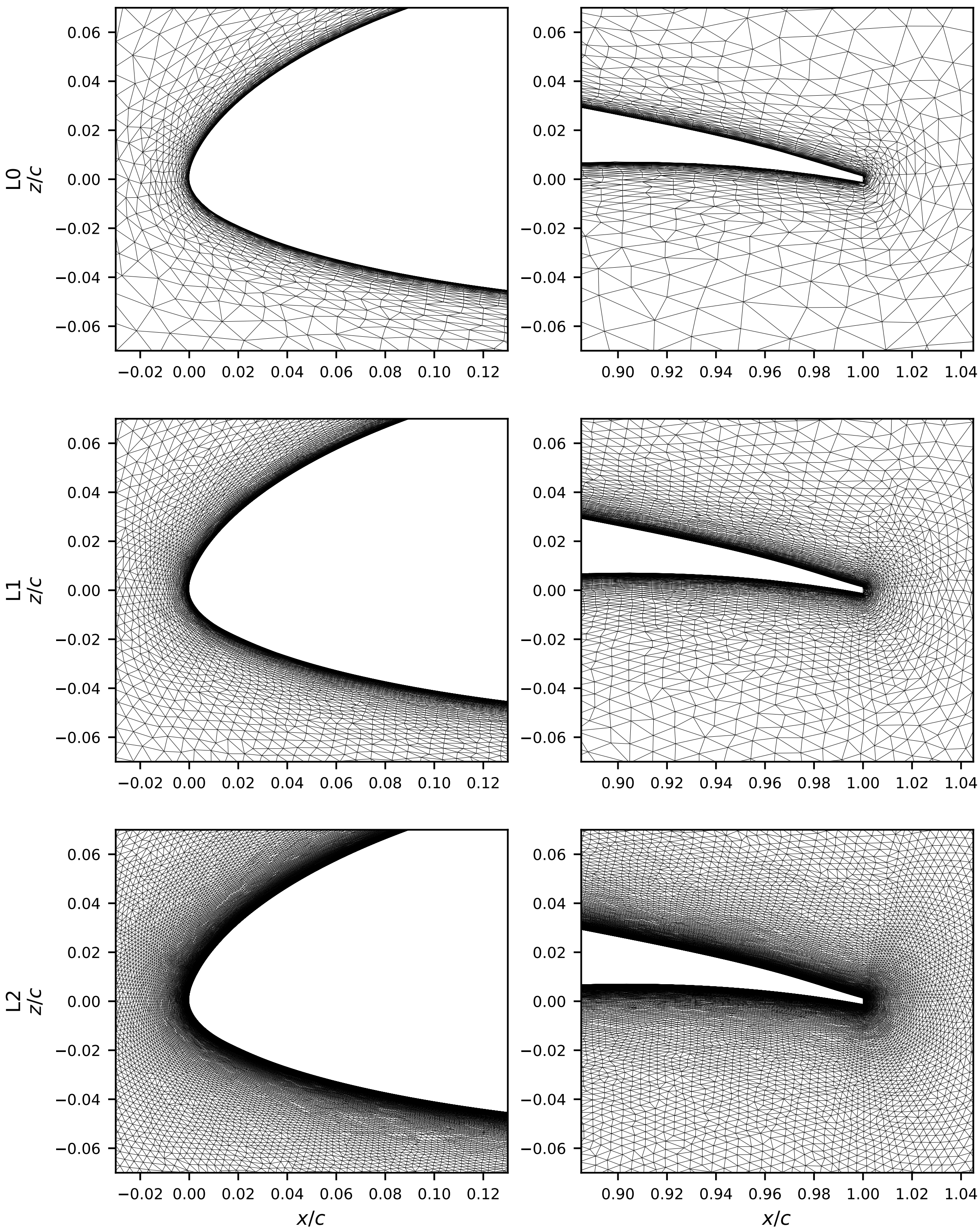
Figure 25: Daedalus surface maps, $\alpha = 4°$.

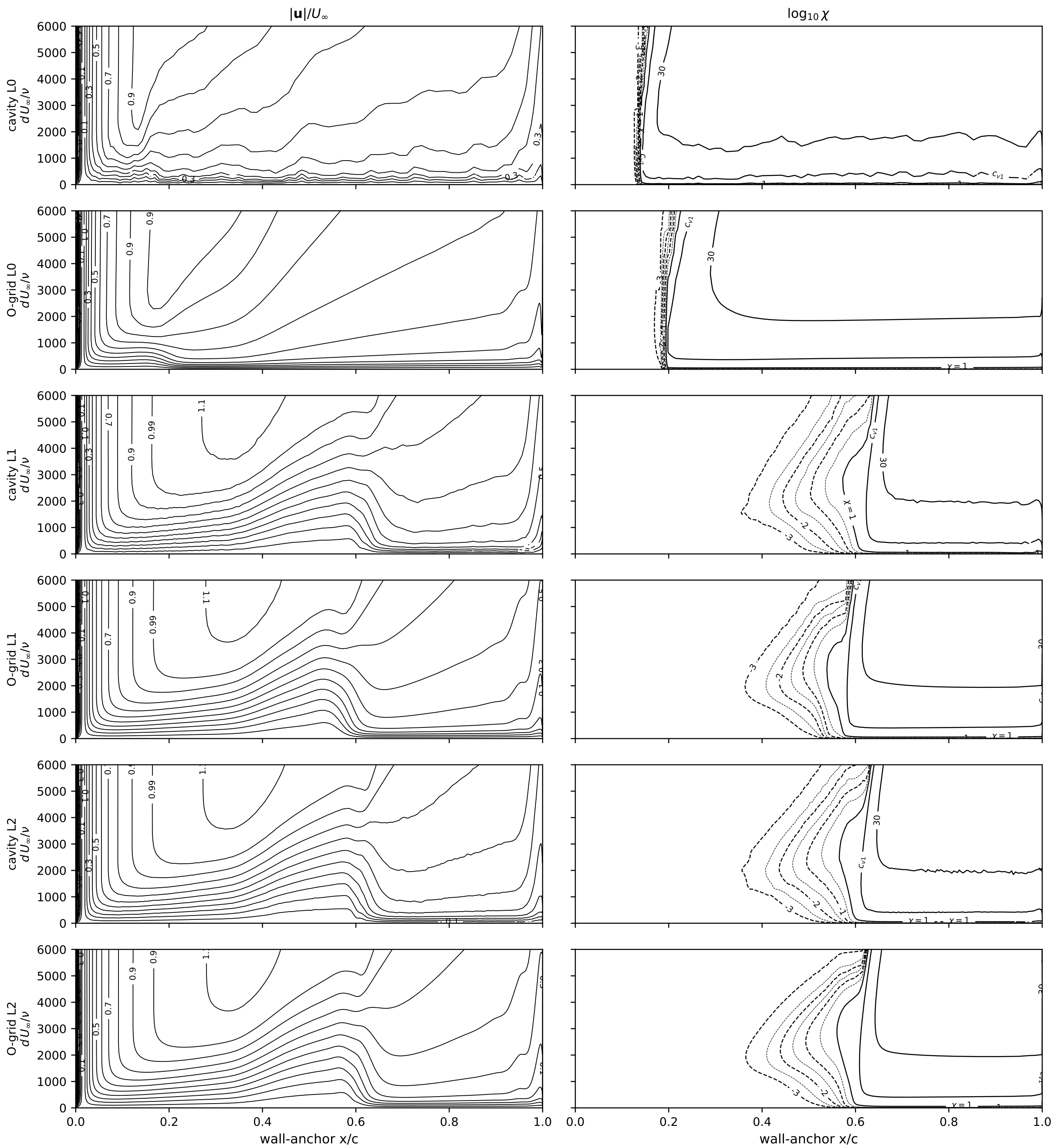
Table 9: Daedalus wing totals by grid level: SA-AI on both mesh families against the AVL+XFOIL reference trimmed to the RANS lift at each incidence.

		structured O-grid		unstructured	
α	level	C_L	C_D	C_L	C_D
4°	L0	1.0244	0.01958	1.0156	0.02134
	L1	1.0279	0.01946	1.0232	0.01941
	L2	1.0284	0.01969	1.0219	0.01976
	AVL+XFOIL	1.0284	0.02125		
5°	L0	1.1254	0.02219	1.1145	0.02427
	L1	1.1307	0.02197	1.1248	0.02201
	L2	1.1290	0.02212	1.1236	0.02223
	AVL+XFOIL	1.1290	0.02315		
6°	L0	1.2262	0.02505	1.2171	0.02713
	L1	1.2321	0.02465	1.2256	0.02481
	L2	1.2282	0.02474	1.2238	0.02491
	AVL+XFOIL	1.2282	0.02530		

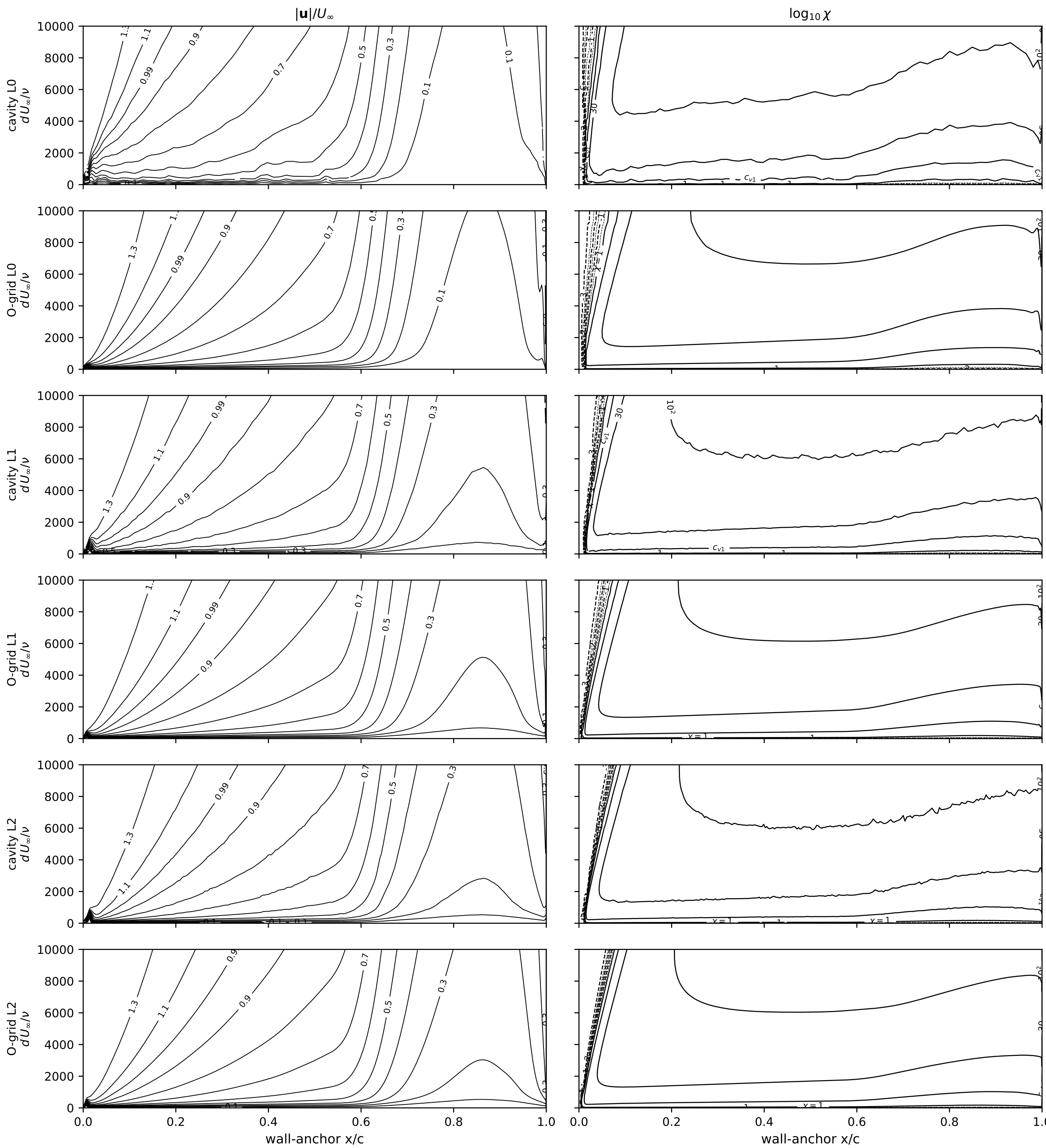
Appendix



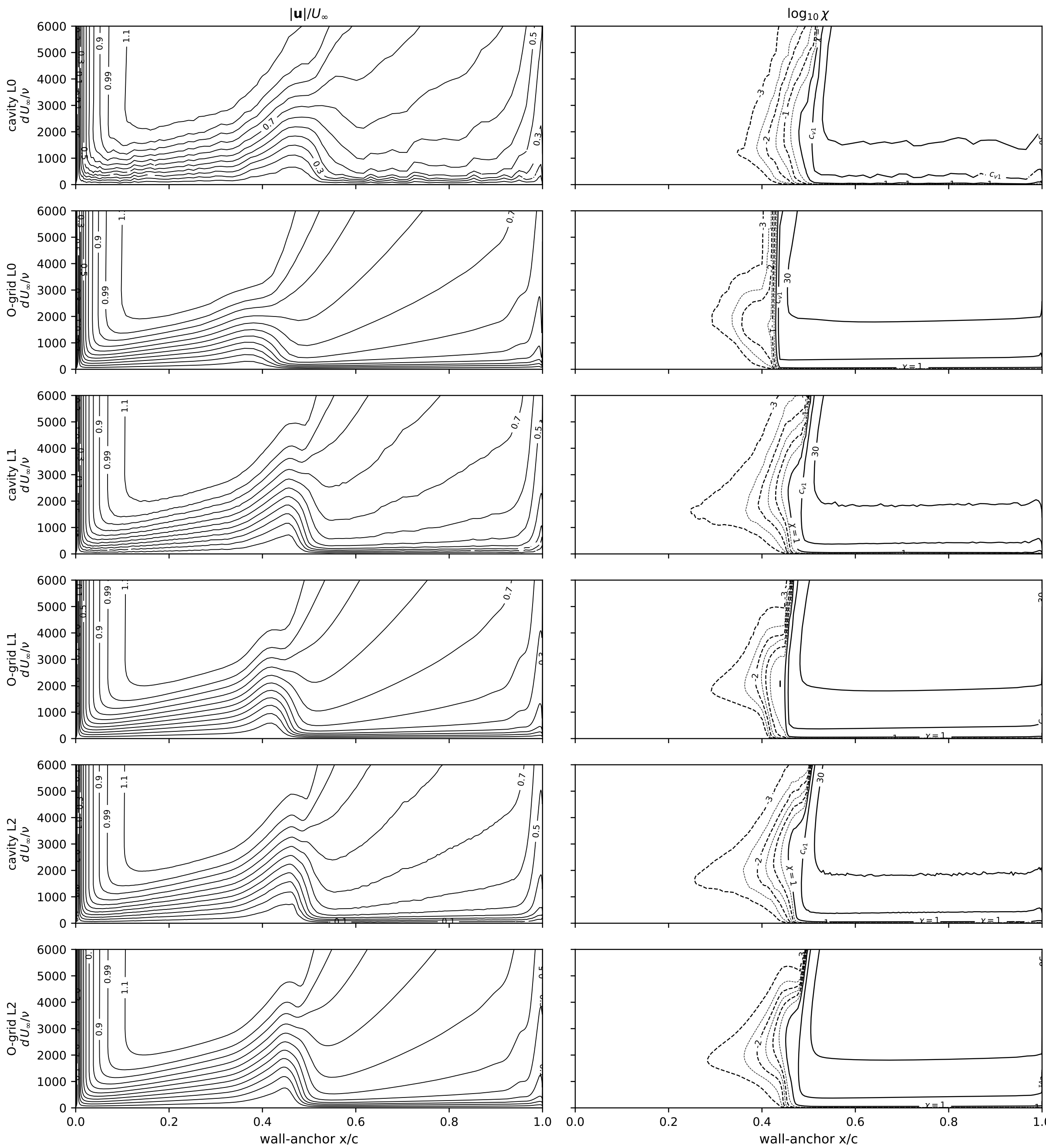

L0
L1
L2
z/c
x/c
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.02 0.00 0.02 0.04 0.06 0.08 0.10 0.12
0.90 0.92 0.94 0.96 0.98 1.00 1.02 1.04



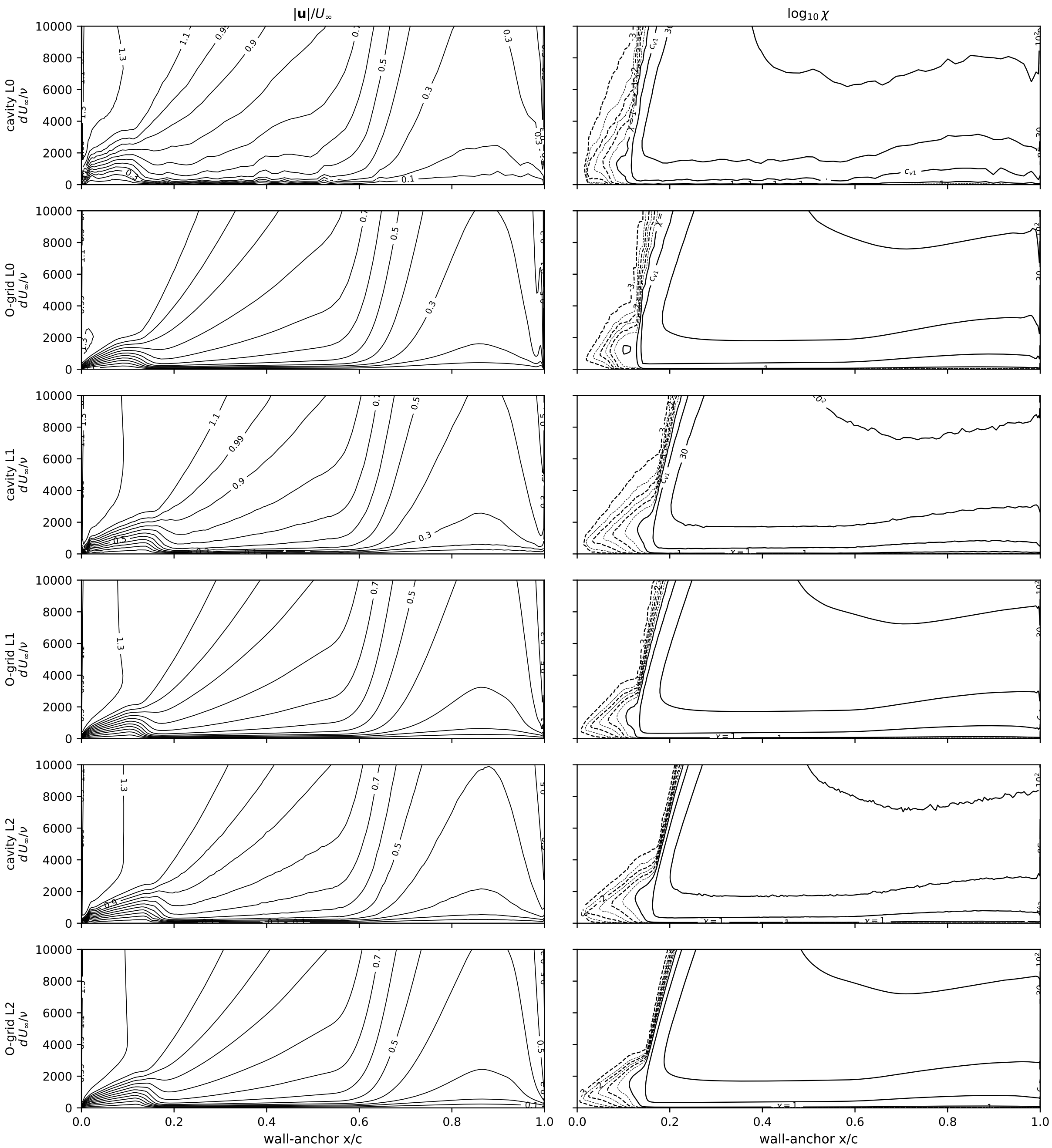
|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c

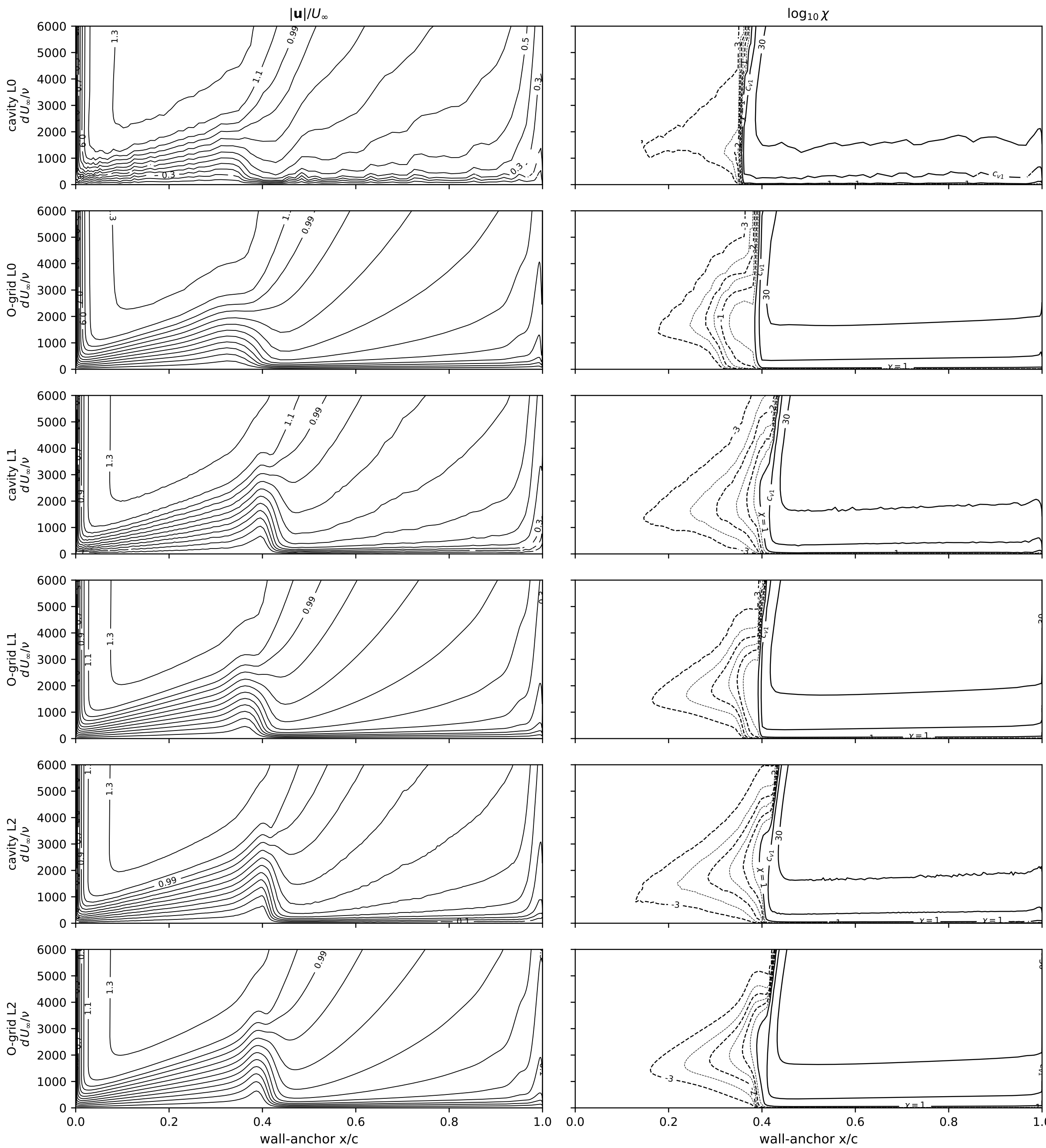


|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c

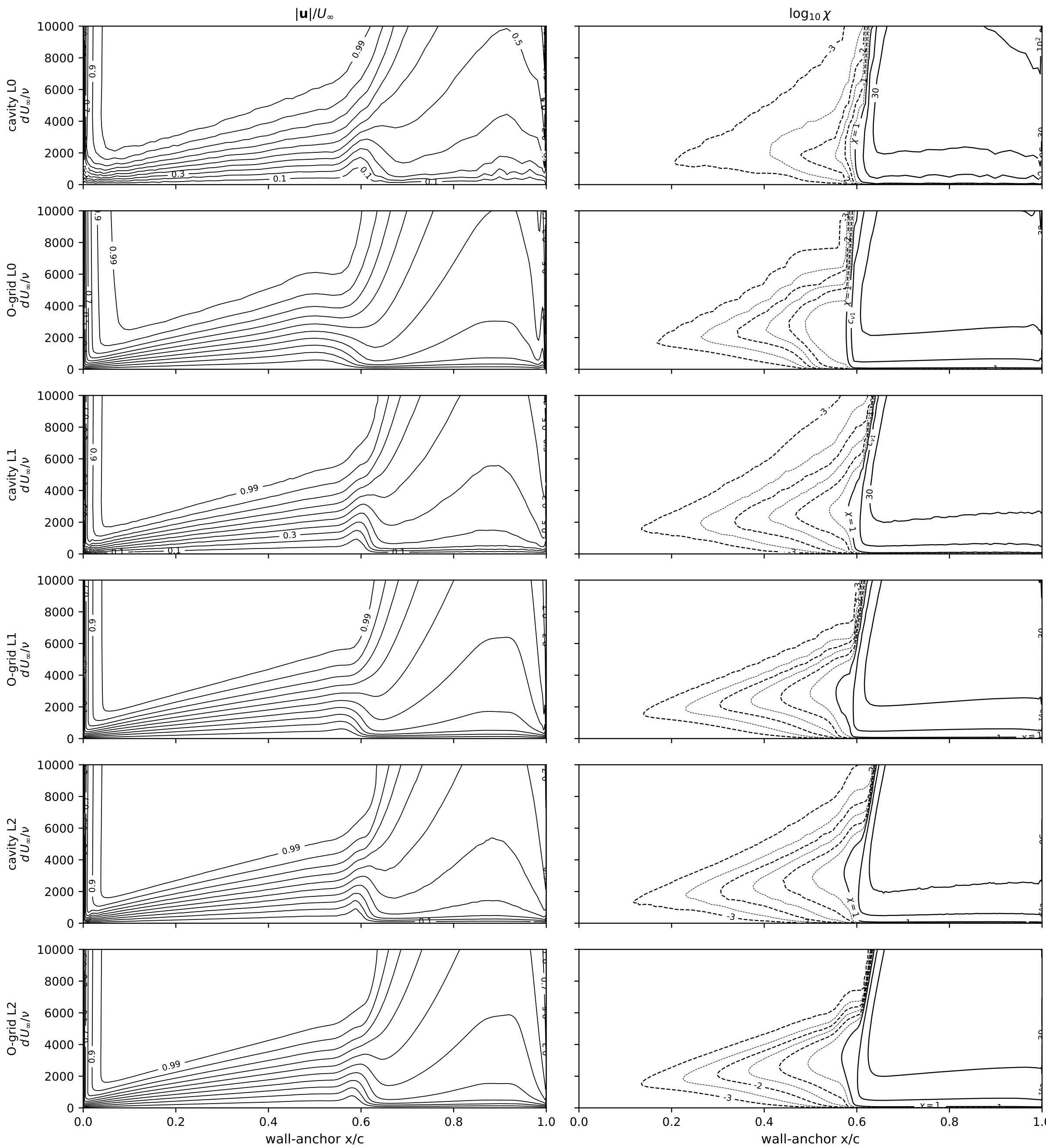


|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c

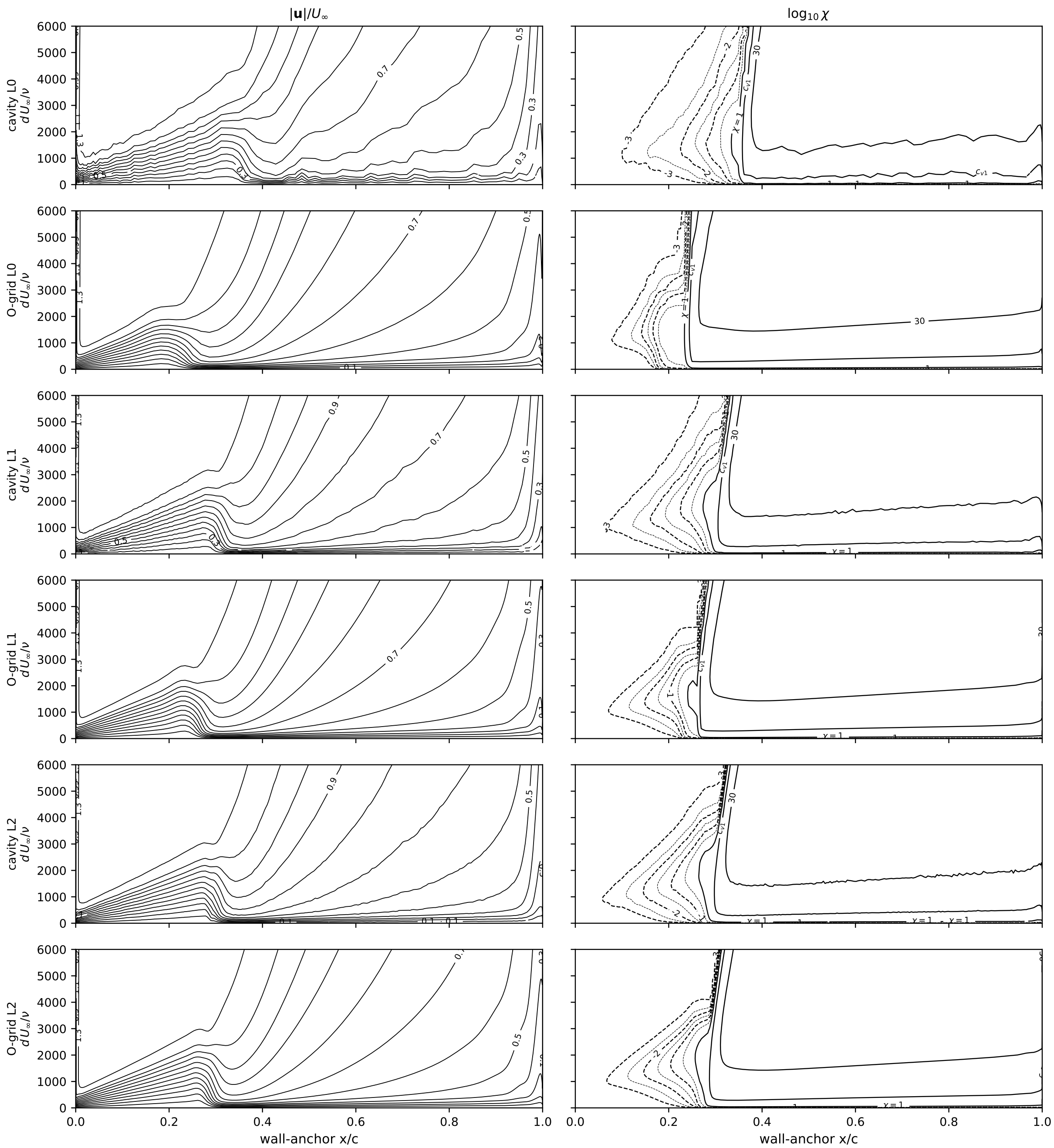

|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c



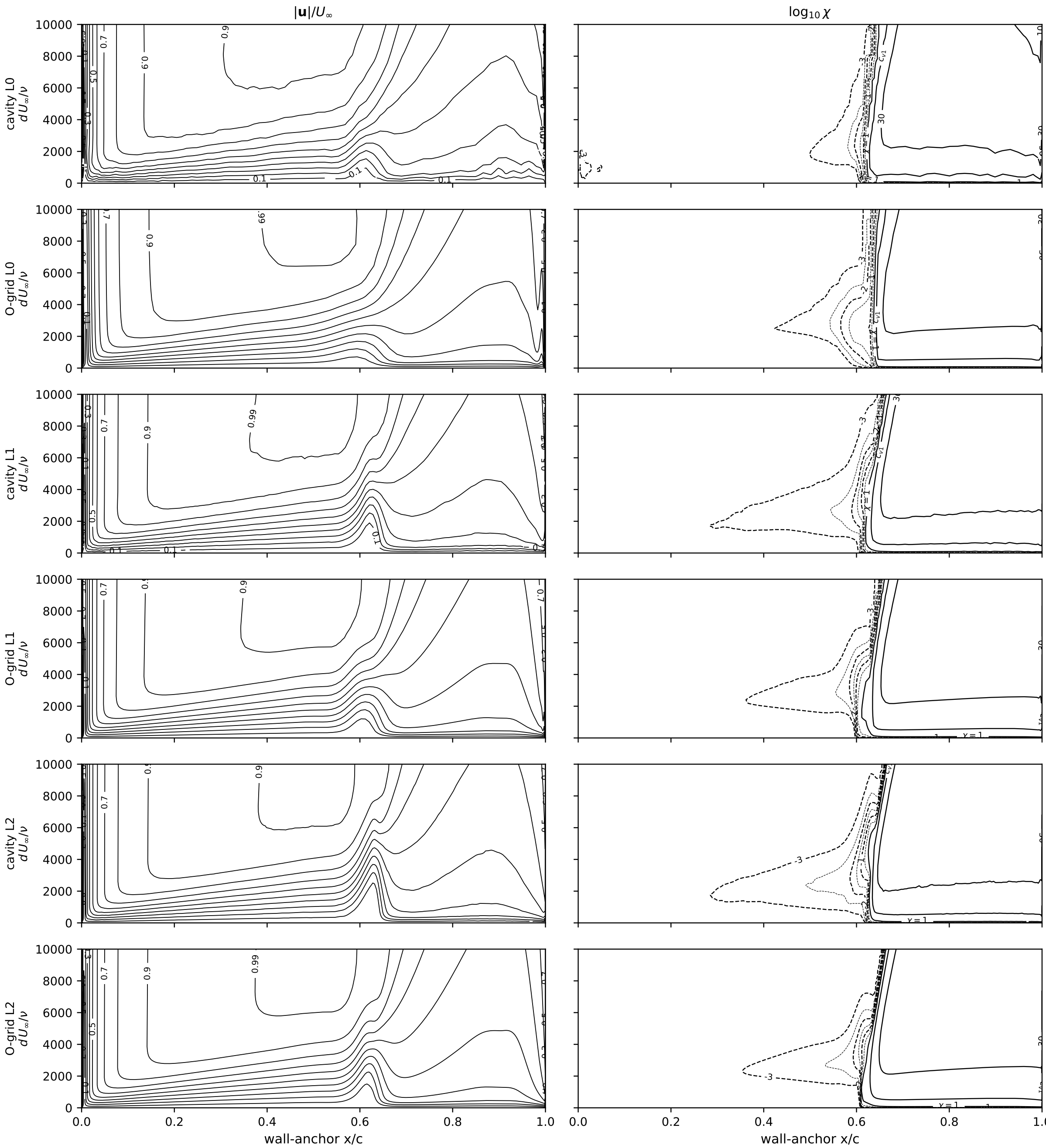
|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c



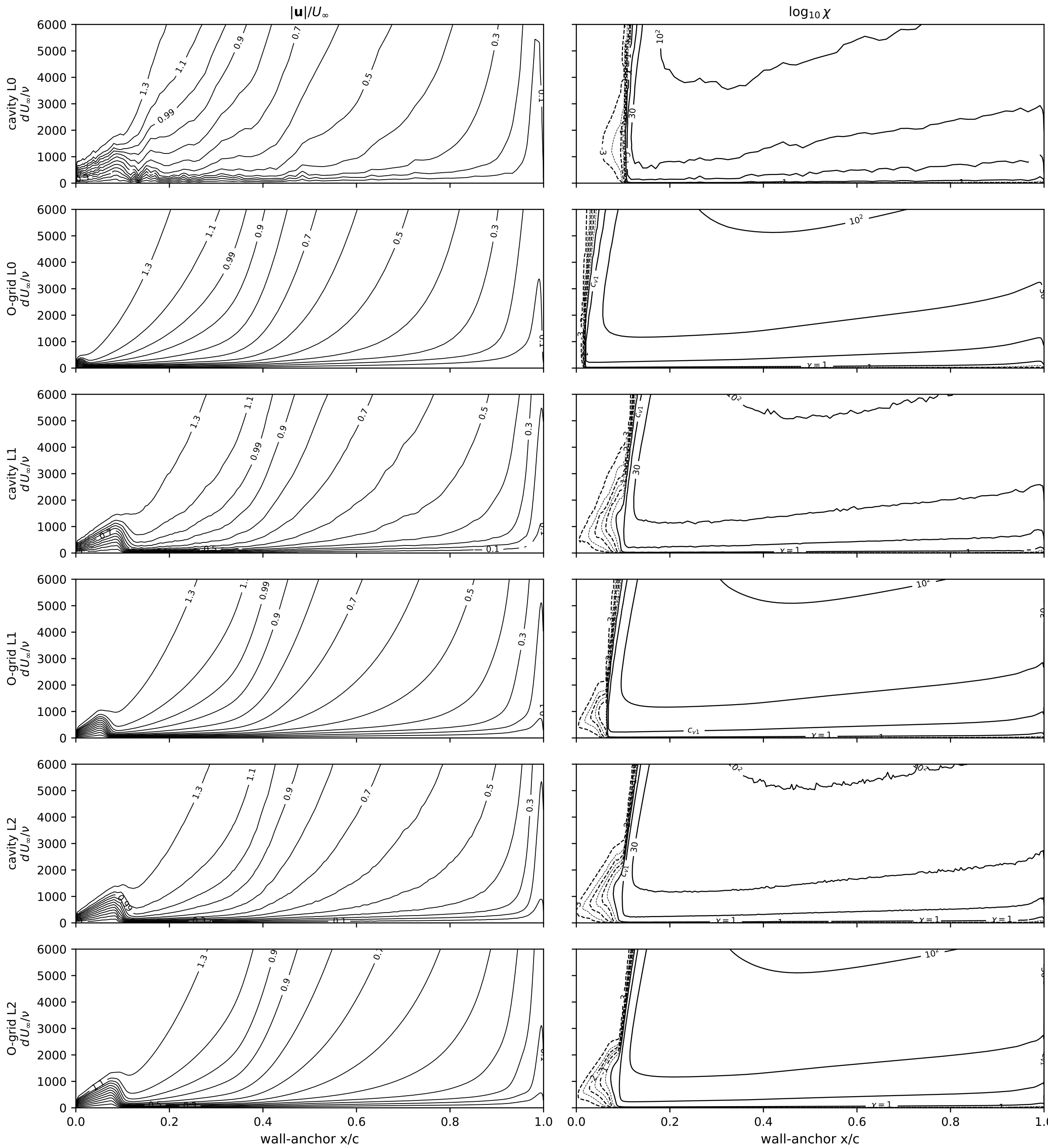
$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c



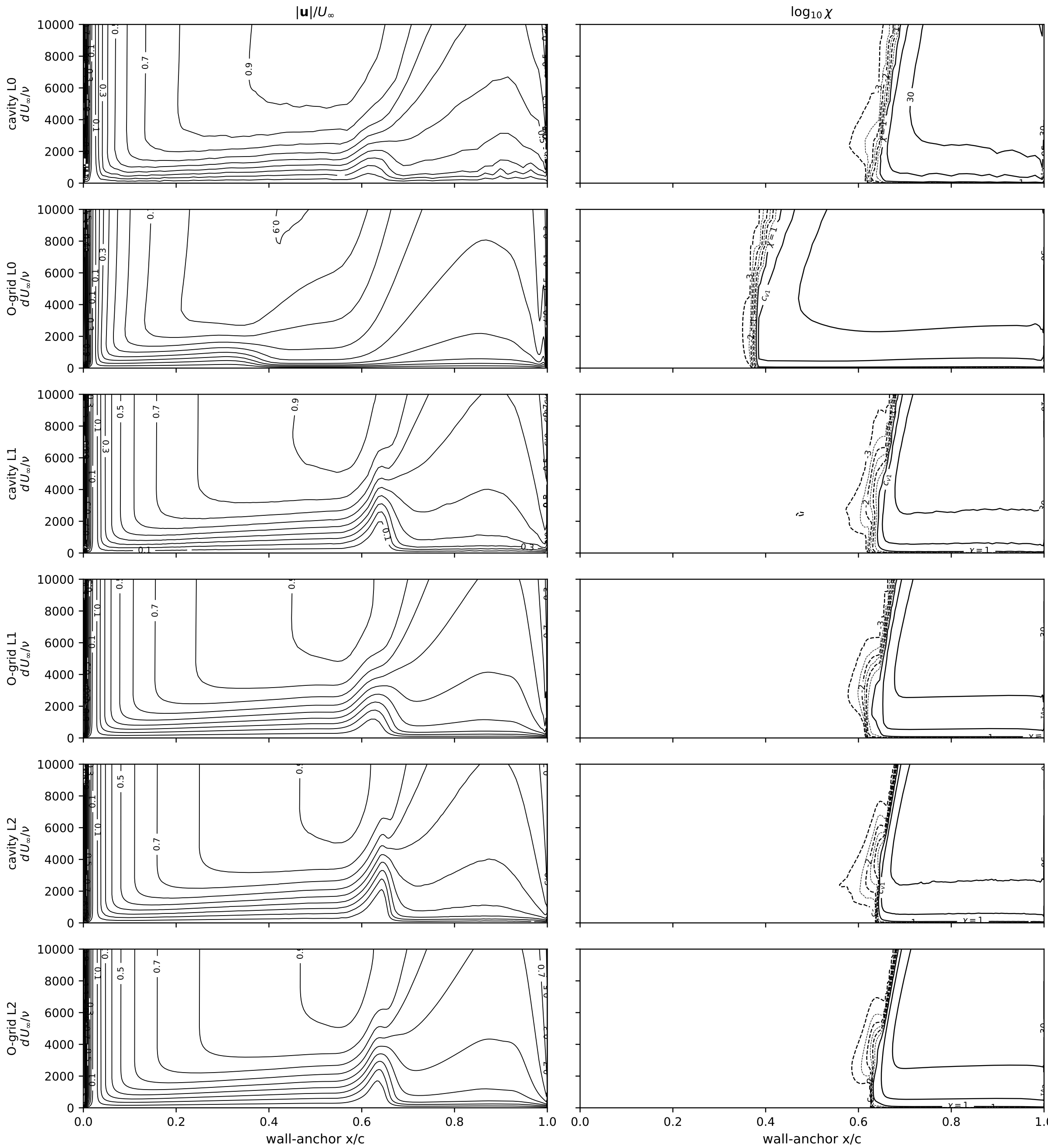
$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c

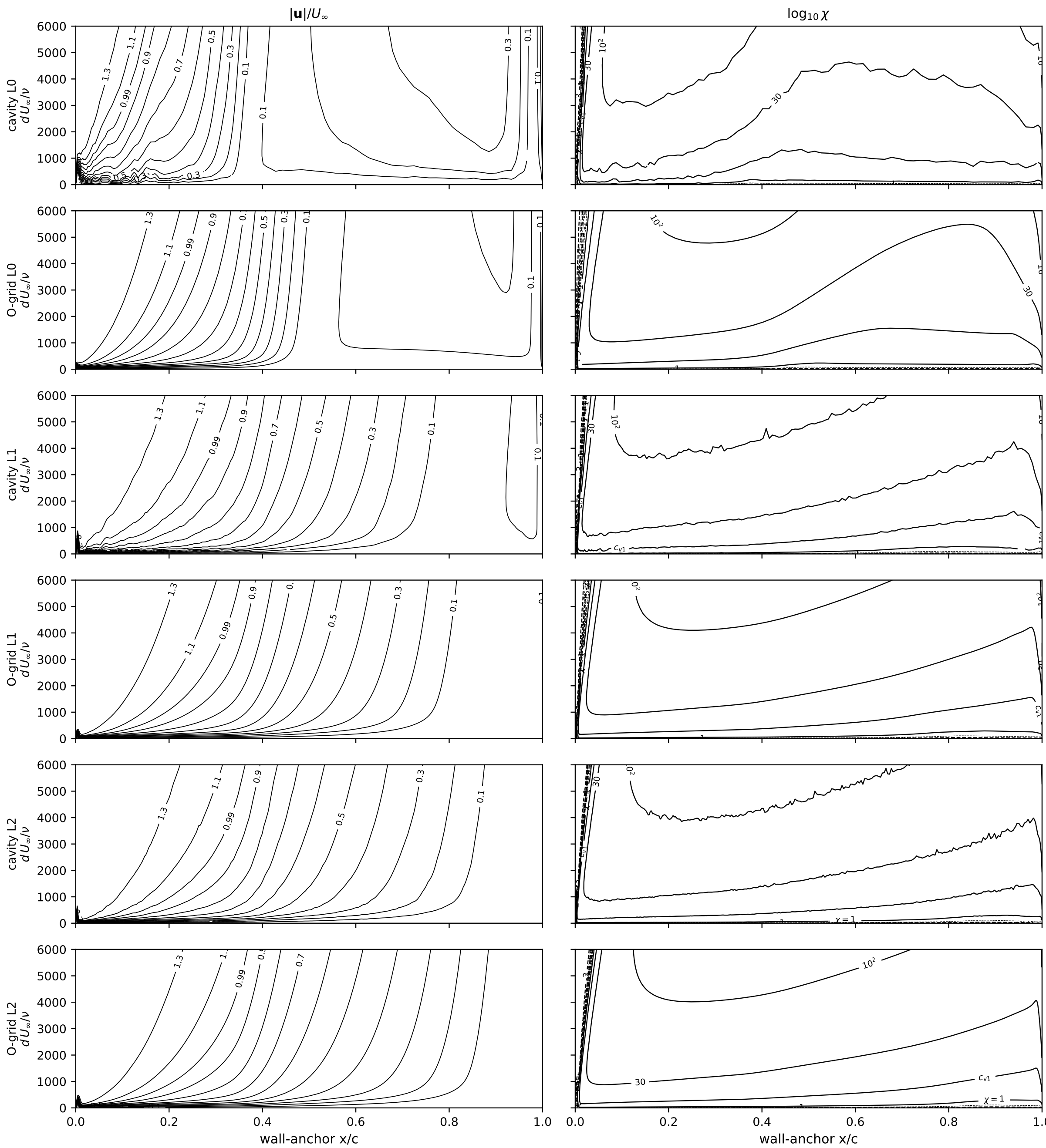


|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c

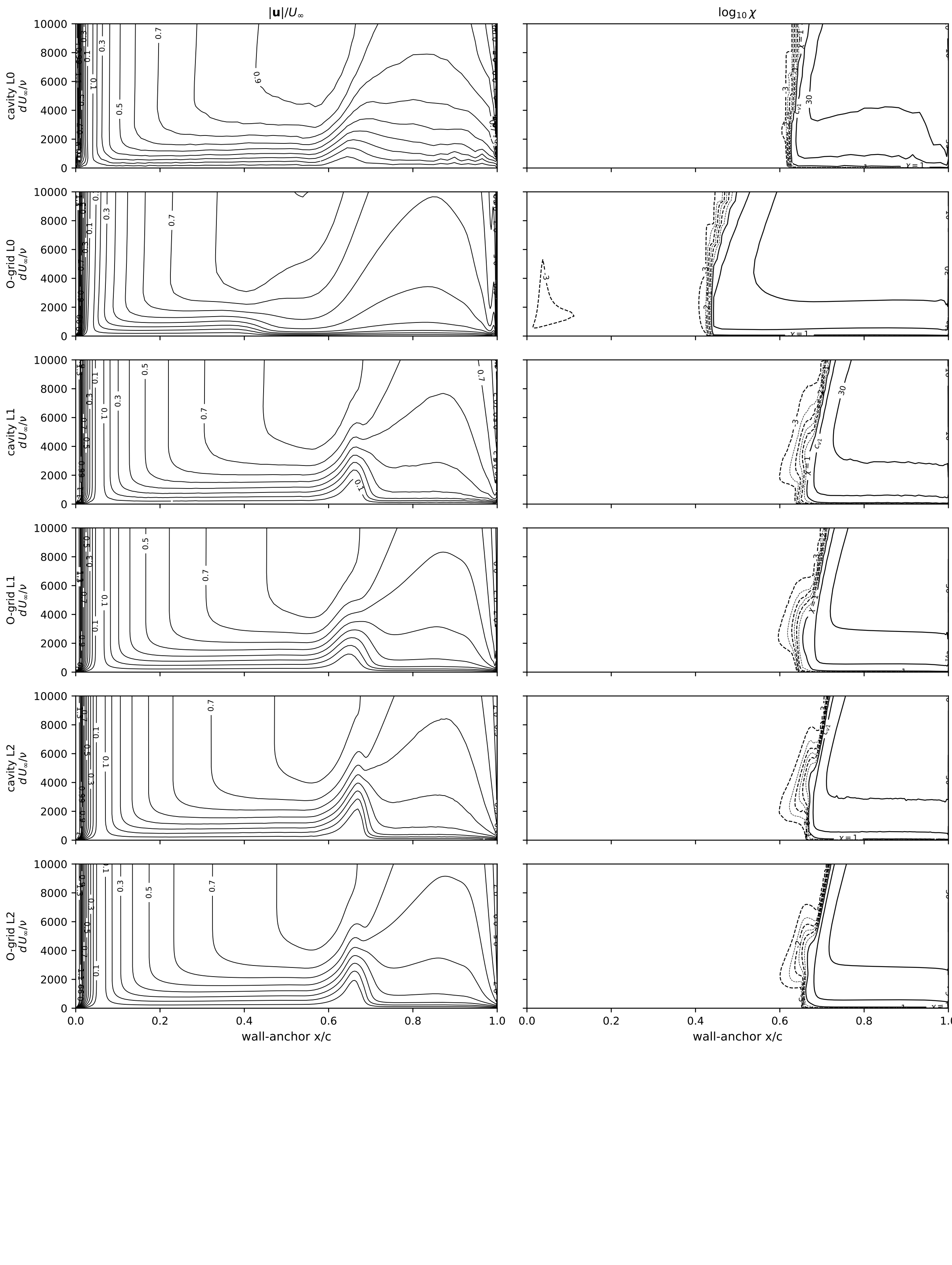


$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c

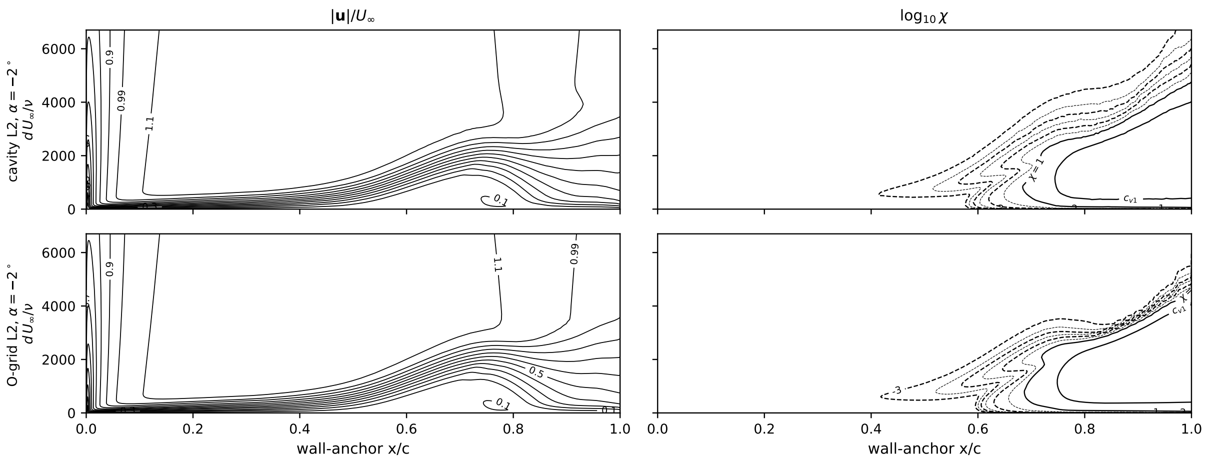

|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c



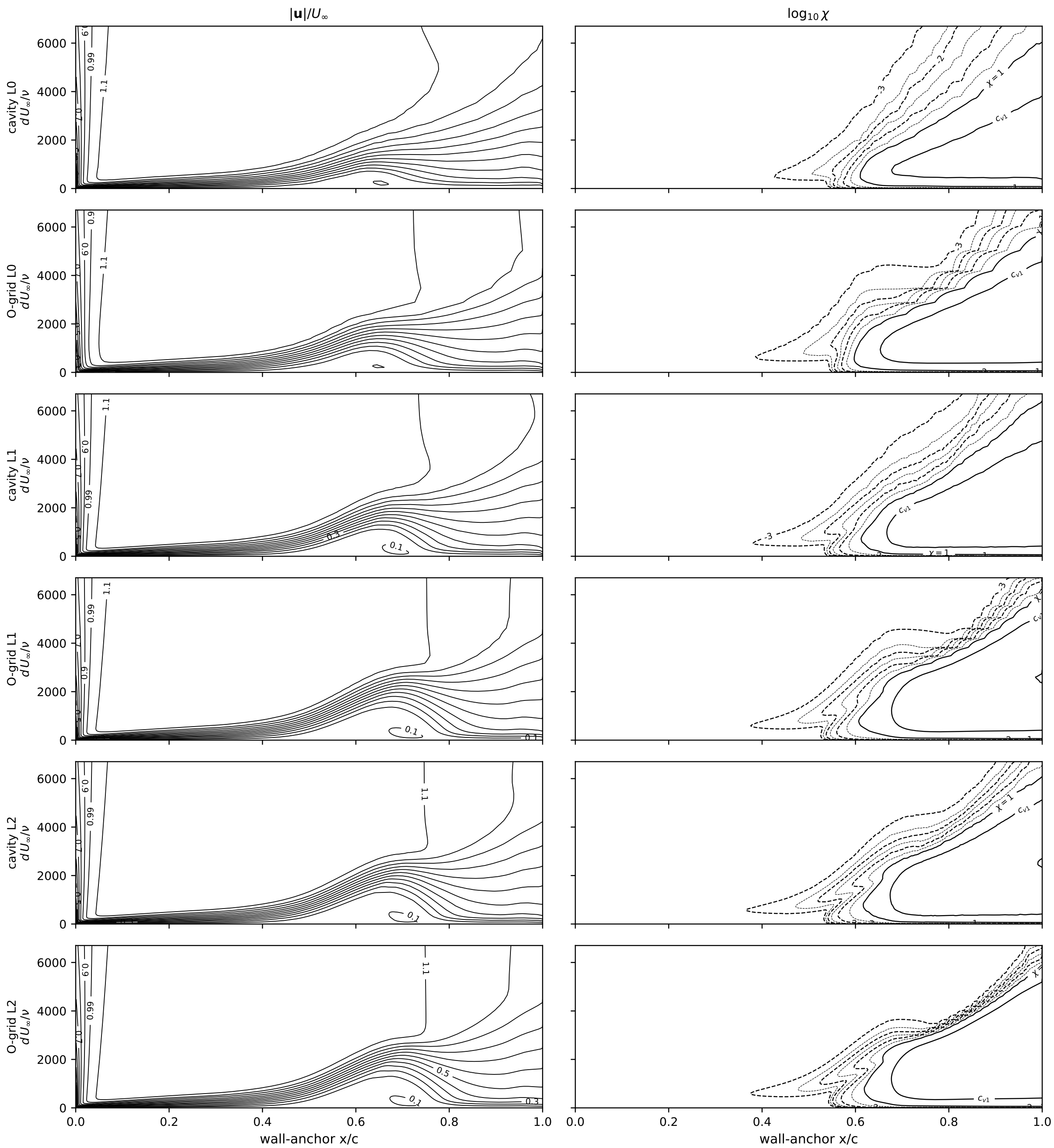
|**u**|/U_∞
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU_∞/ν
wall-anchor x/c



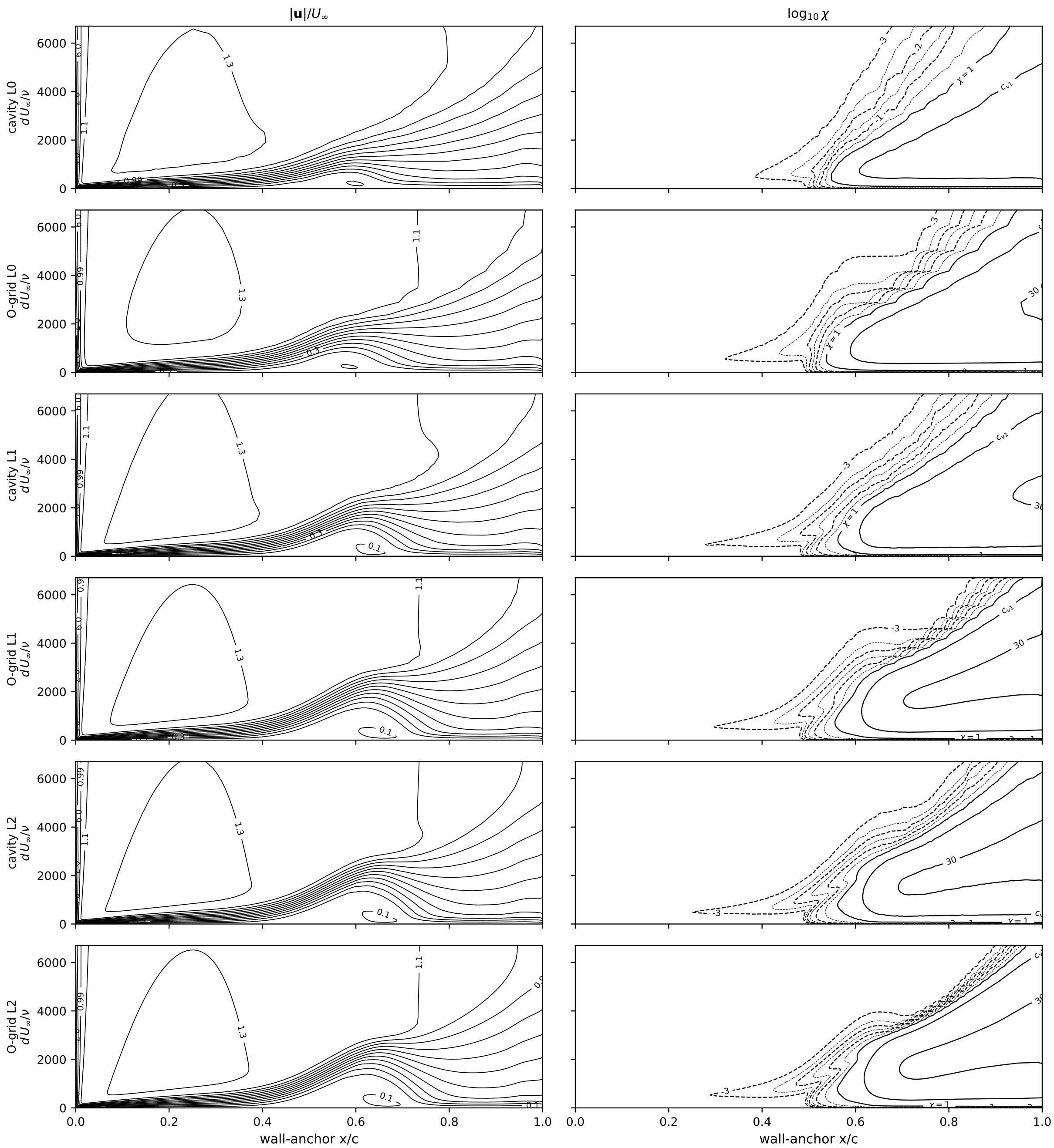
|**u**|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c



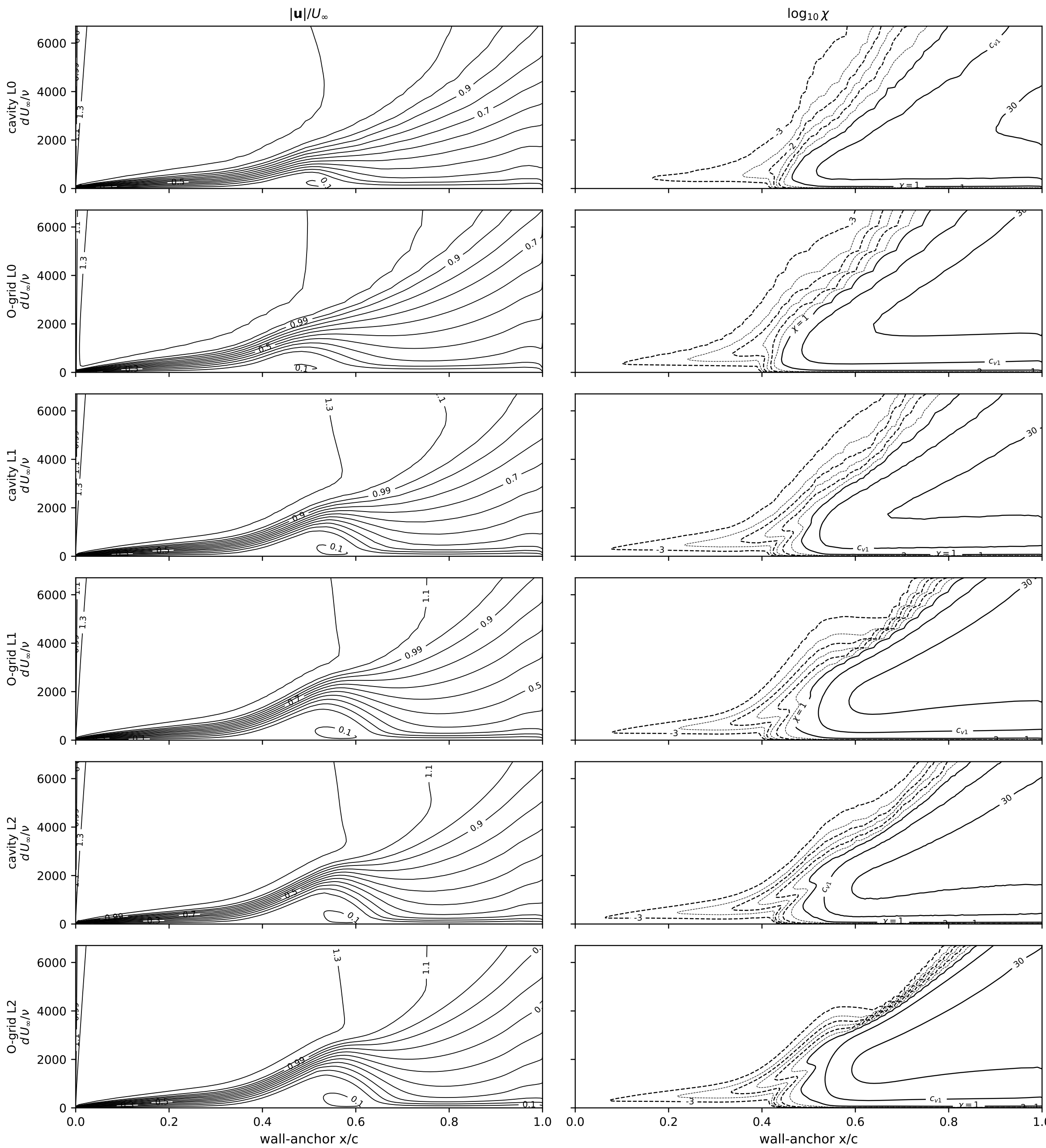
$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L2, $\alpha = -2^\circ$
$d\,U_\infty/\nu$
O-grid L2, $\alpha = -2^\circ$
$d\,U_\infty/\nu$
wall-anchor x/c
wall-anchor x/c
0.9
0.99
1.1
0.1
0.5
$\chi = 1$
c_{v1}
-3



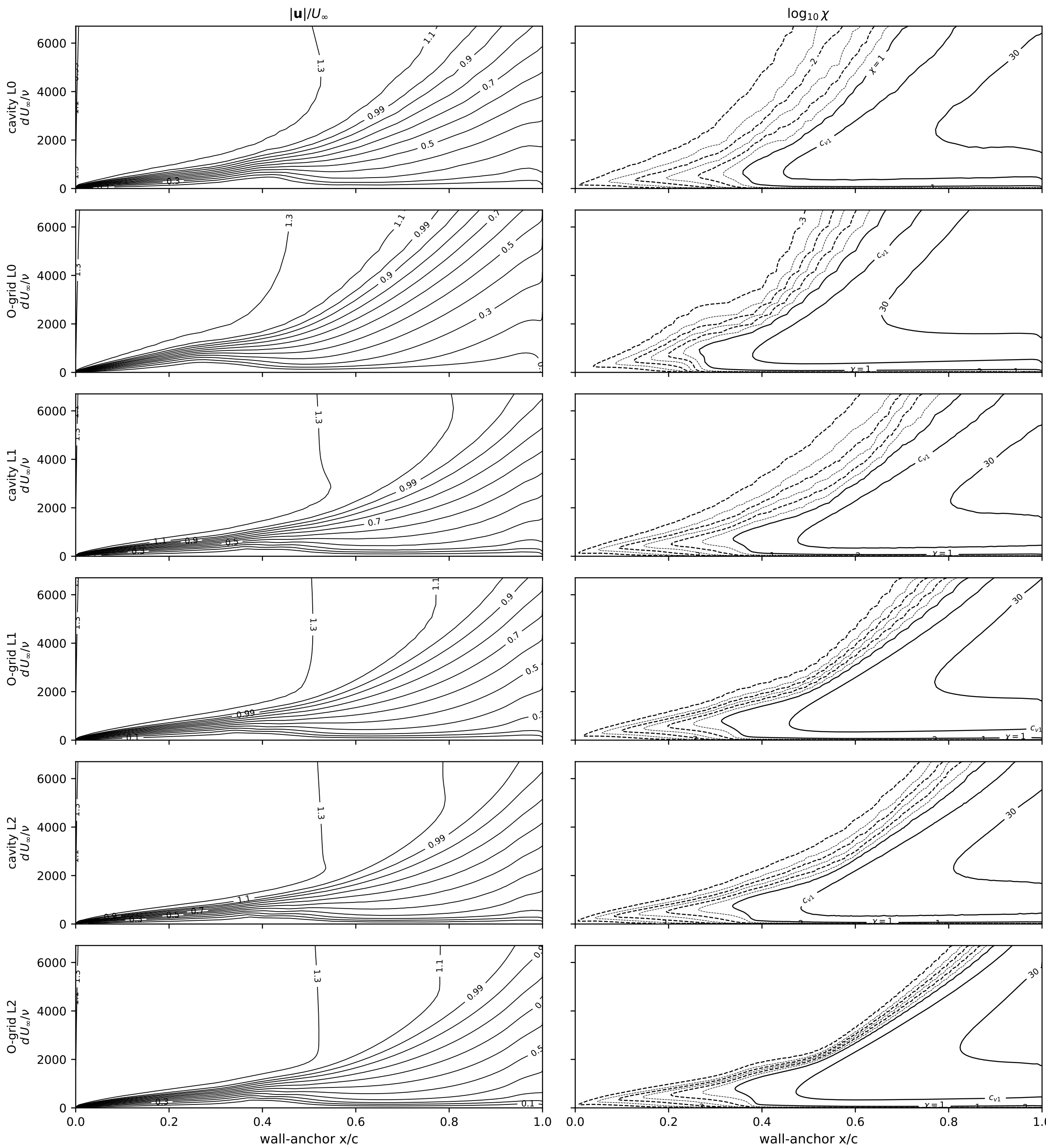
|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
dU∞/ν
wall-anchor x/c
wall-anchor x/c



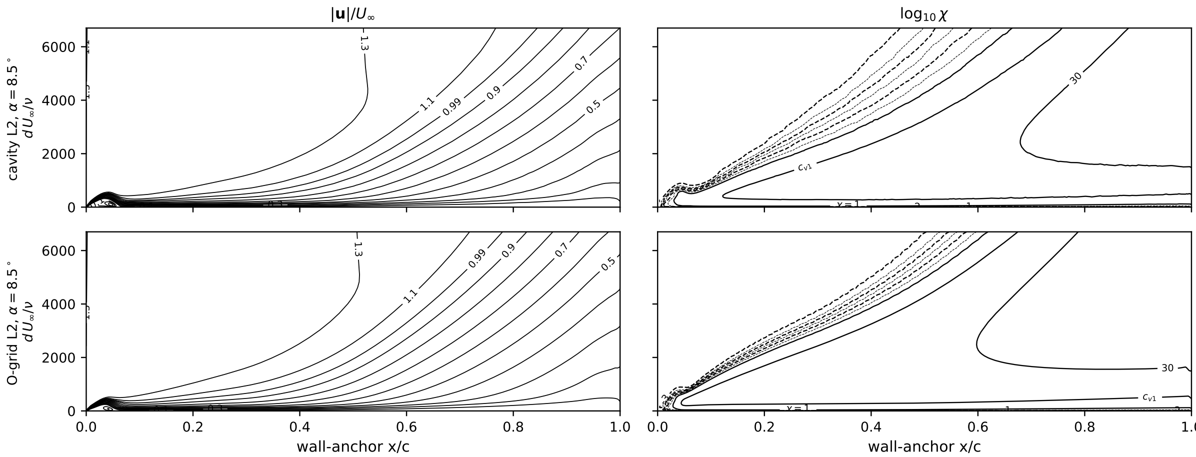
$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c

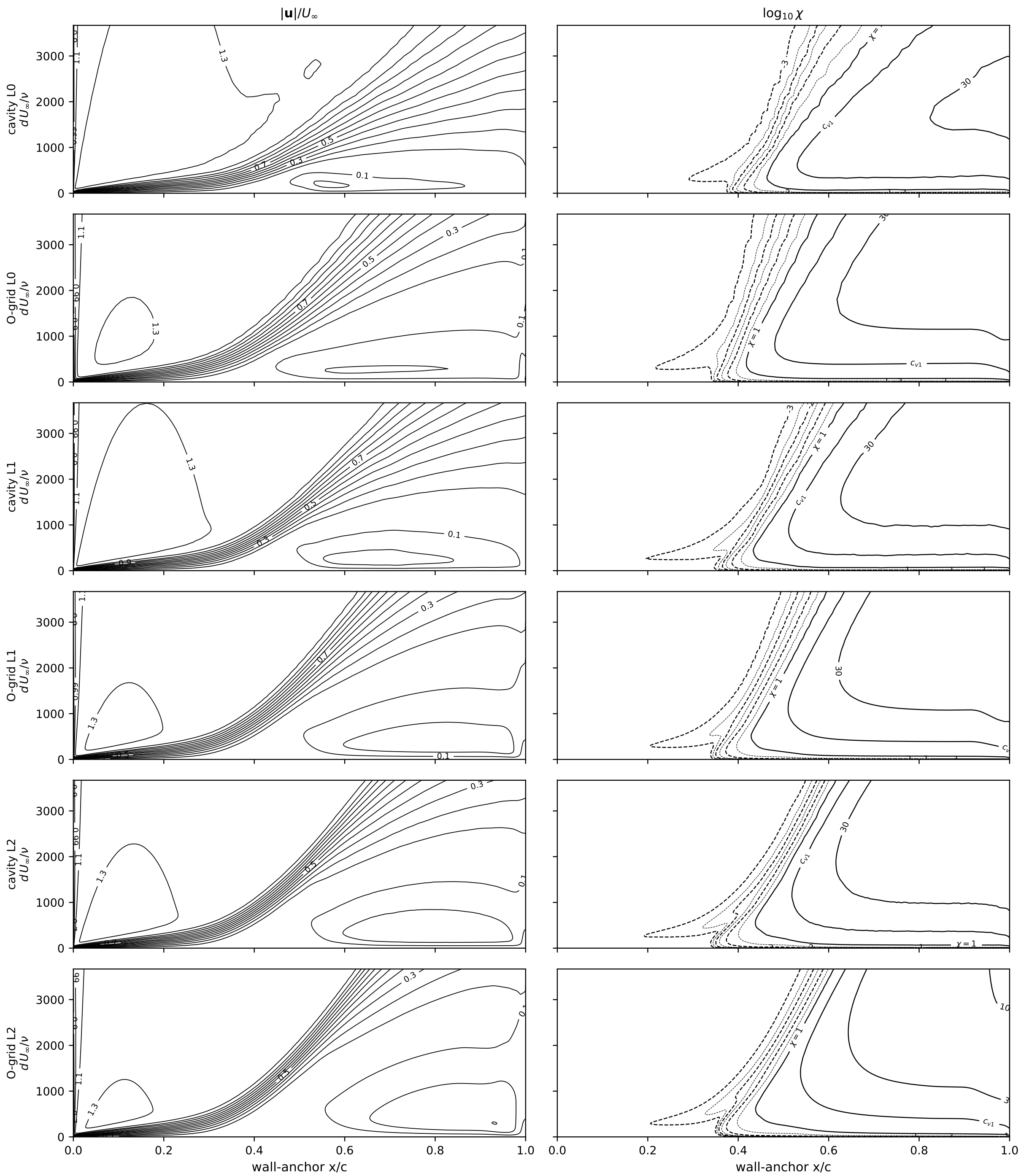


$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c

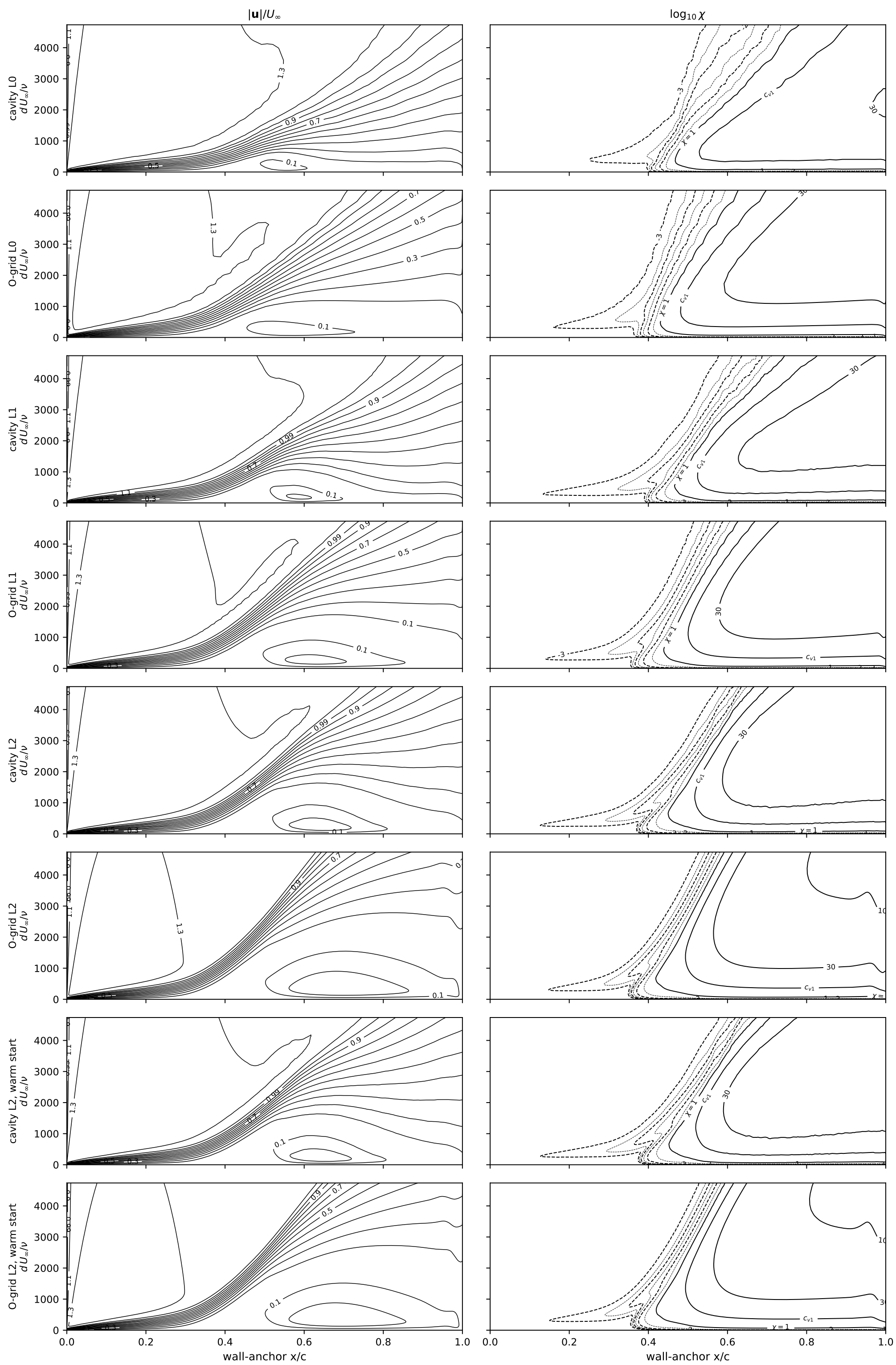


|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c

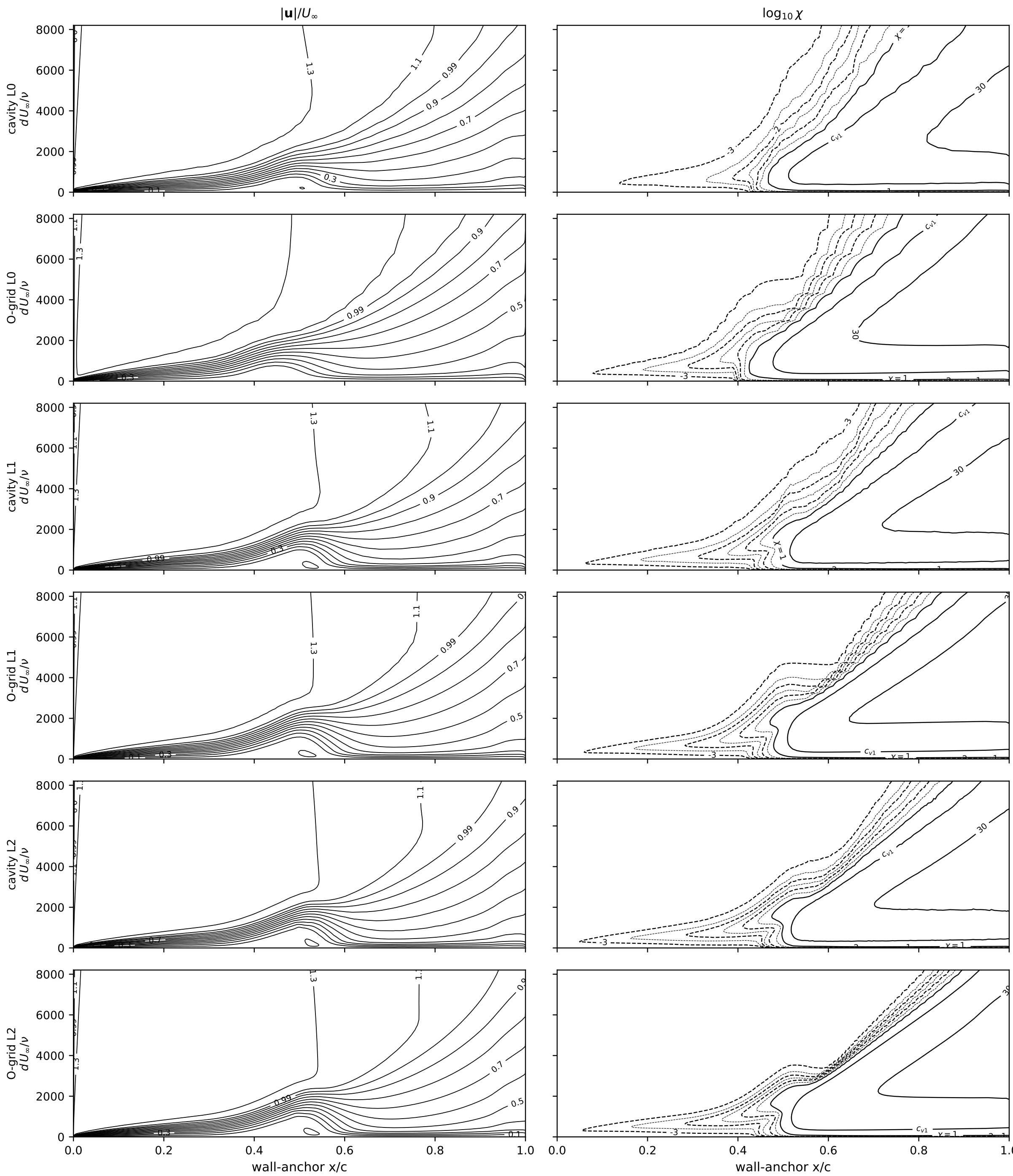

|u|/U∞
log10 χ
cavity L2, α = 8.5°
O-grid L2, α = 8.5°
d U∞/ν
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
wall-anchor x/c
1.3
1.1
0.99
0.9
0.7
0.5
30
cv1



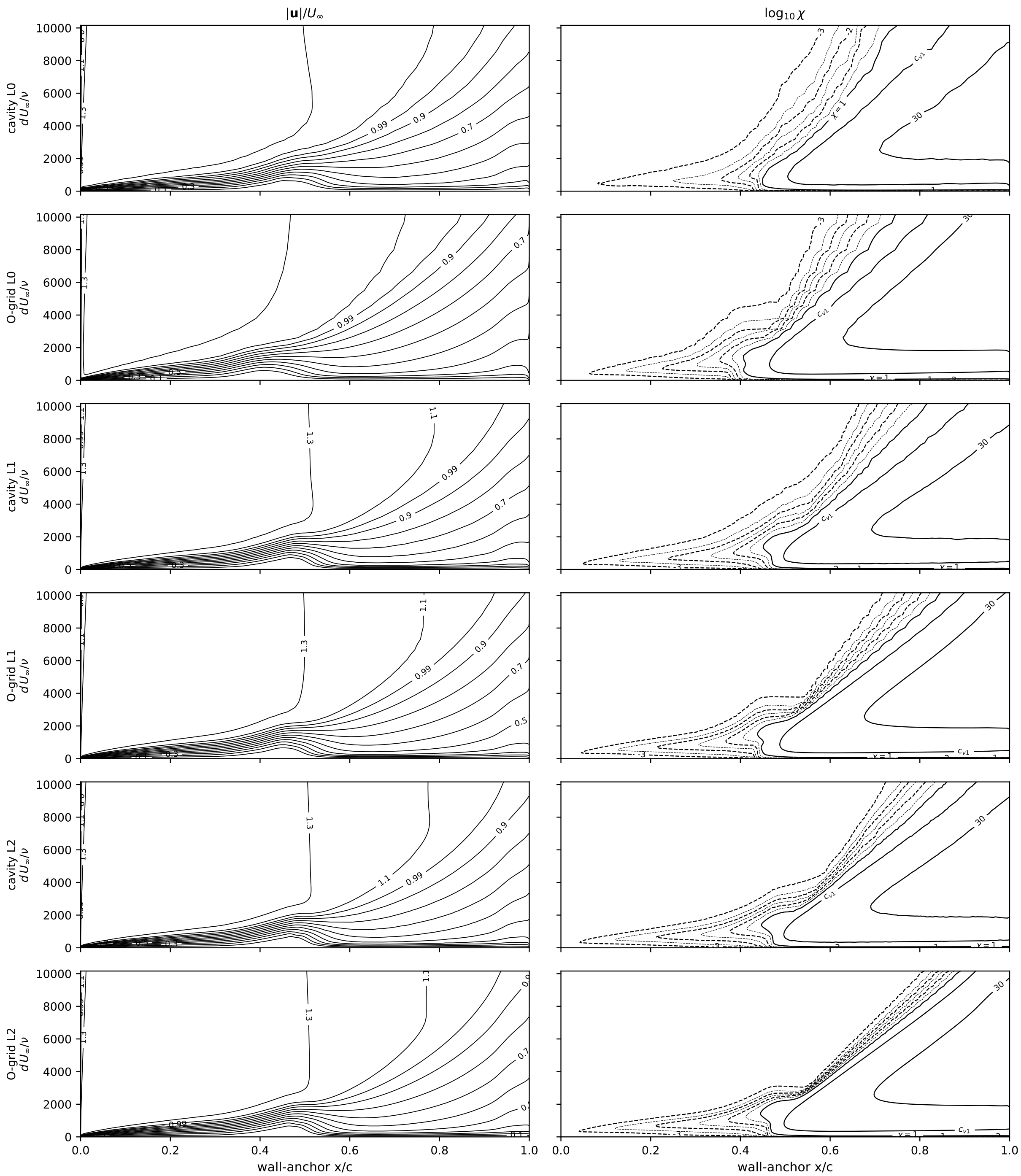
|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
d U∞/ν
wall-anchor x/c
wall-anchor x/c



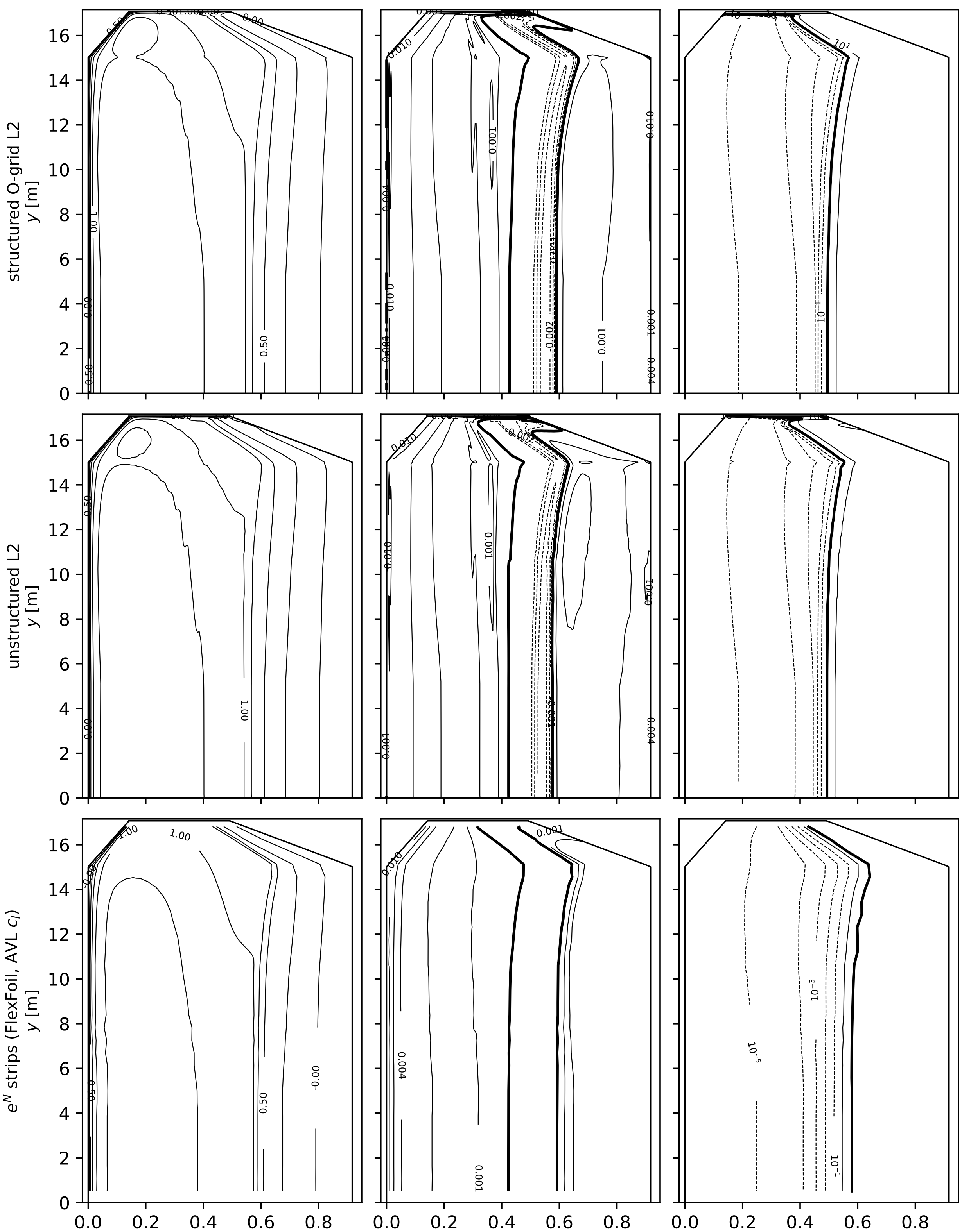
|u|/U∞
log10 χ
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
cavity L2, warm start
O-grid L2, warm start
d U∞/ν
wall-anchor x/c
wall-anchor x/c

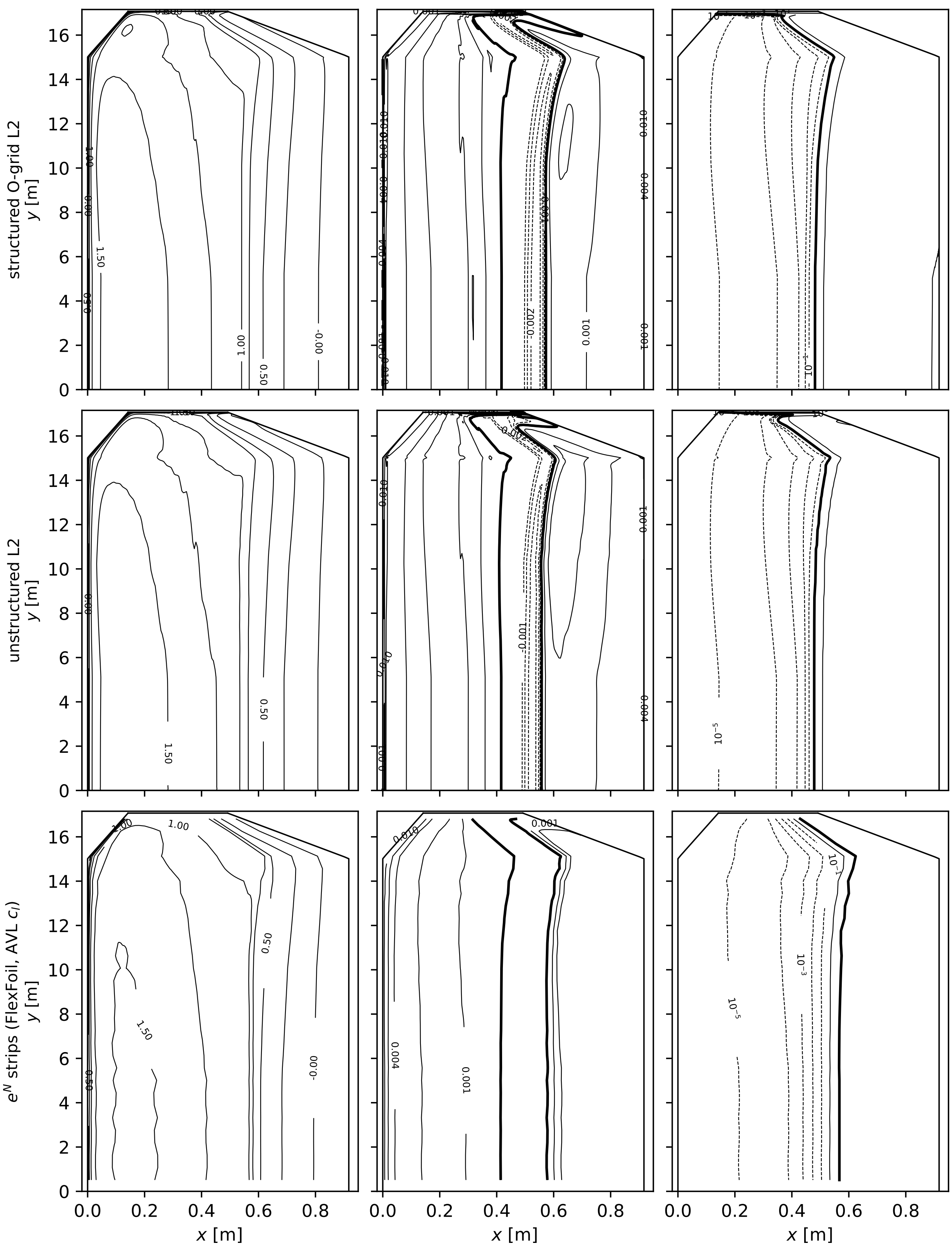


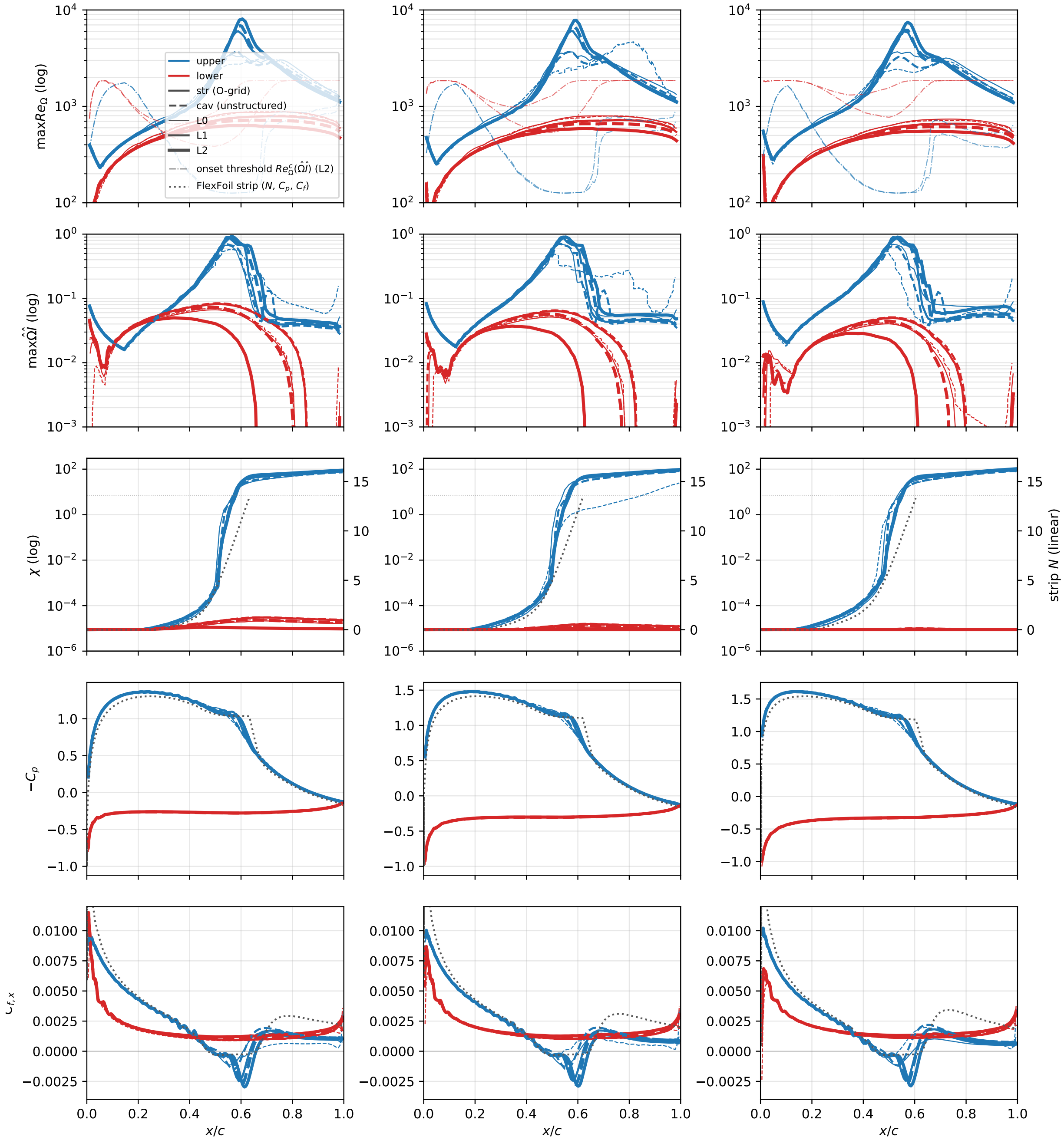
$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0 dU_∞/ν
O-grid L0 dU_∞/ν
cavity L1 dU_∞/ν
O-grid L1 dU_∞/ν
cavity L2 dU_∞/ν
O-grid L2 dU_∞/ν
wall-anchor x/c
wall-anchor x/c

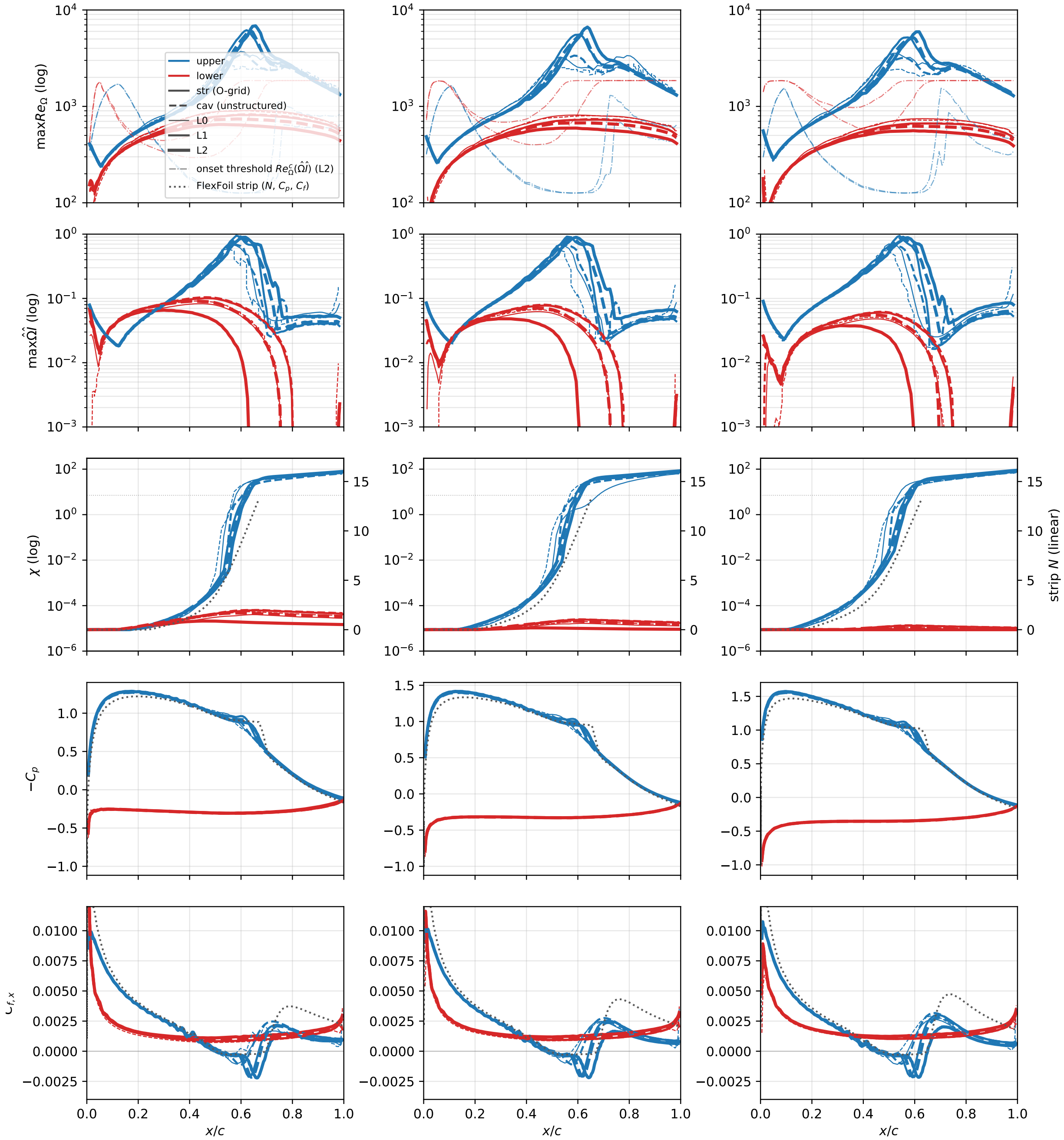


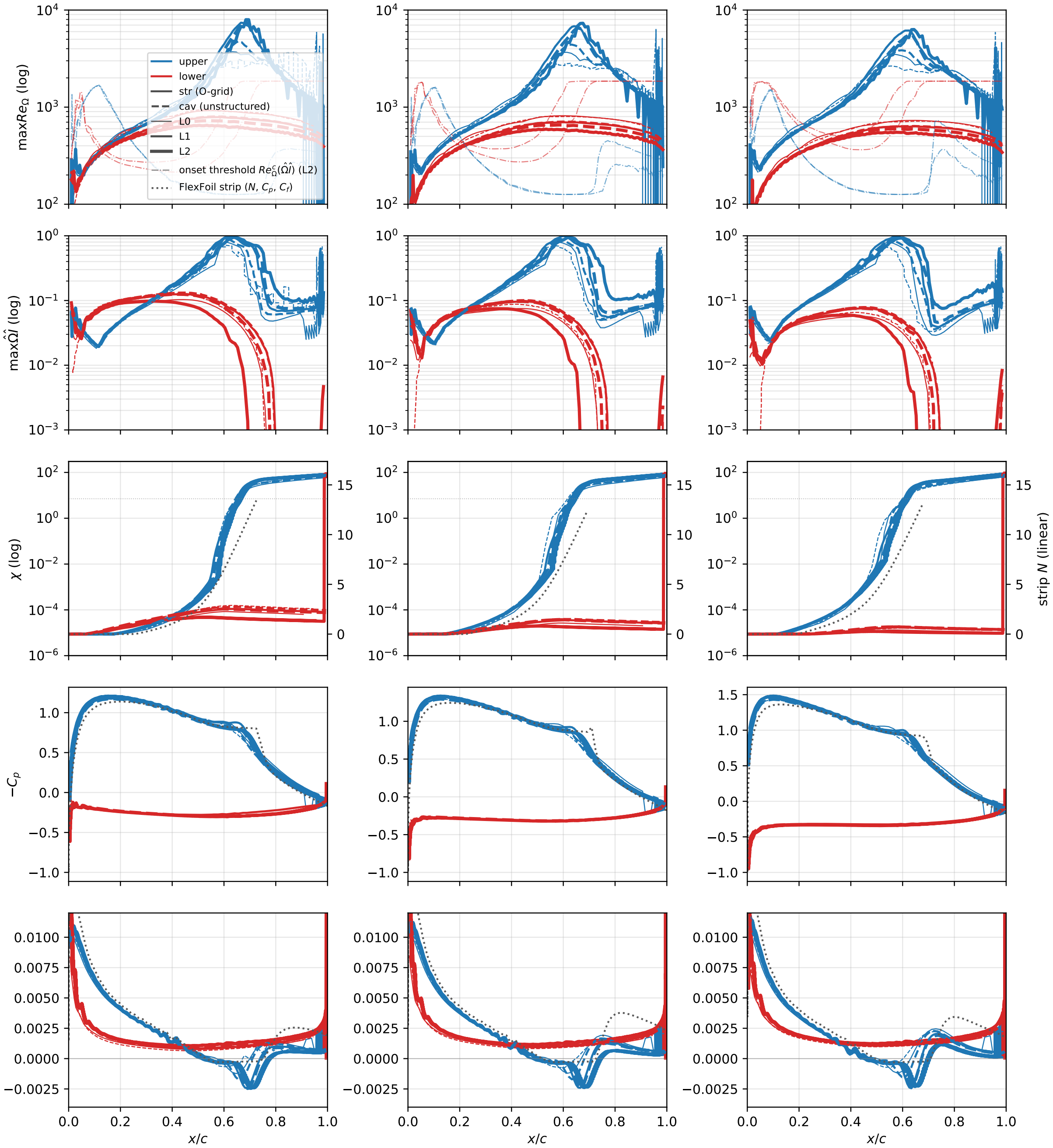
$|\mathbf{u}|/U_\infty$
$\log_{10}\chi$
cavity L0
O-grid L0
cavity L1
O-grid L1
cavity L2
O-grid L2
$d\,U_\infty/\nu$
wall-anchor x/c


structured O-grid L2
y [m]
unstructured L2
y [m]
e^N strips (FlexFoil, AVL c_l)
y [m]
x [m]
x [m]
x [m]


structured O-grid L2
y [m]
unstructured L2
y [m]
e^N strips (FlexFoil, AVL c_l)
y [m]
x [m]
x [m]
x [m]


upper
lower
str (O-grid)
cav (unstructured)
L0
L1
L2
onset threshold $Re^c_\Omega(\hat{\Omega}\hat{l})$ (L2)
FlexFoil strip (N, C_p, C_f)
$\max Re_\Omega$ (log)
$\max \hat{\Omega}\hat{l}$ (log)
χ (log)
strip N (linear)
$-C_p$
$C_{f,x}$
x/c


upper
lower
str (O-grid)
cav (unstructured)
L0
L1
L2
onset threshold $Re_\Omega^c(\hat{\Omega}\hat{l})$ (L2)
FlexFoil strip (N, C_p, C_f)
$\max Re_\Omega$ (log)
$\max \hat{\Omega}\hat{l}$ (log)
χ (log)
strip N (linear)
$-C_p$
$C_{f,x}$
x/c


upper
lower
str (O-grid)
cav (unstructured)
L0
L1
L2
onset threshold $Re_\Omega^c(\hat{\Omega}\hat{l})$ (L2)
FlexFoil strip (N, C_p, C_f)
$\max Re_\Omega$ (log)
$\max \hat{\Omega}\hat{l}$ (log)
χ (log)
strip N (linear)
$-C_p$
$C_{f,x}$
x/c